

PEIEP: See *inverse eigenvalue problem with prescribed entries*.

pencil (defined by $A, B \in \mathbb{C}^{n \times n}$): $A - xB$, or pair (A, B) . **43.1**

pencil (in $\mathbb{D}[x]^{m \times n}$): A matrix $A(x) = A_0 + xA_1$, $A_0, A_1 \in \mathbb{D}^{m \times n}$. **23.4**

perfect elimination ordering: An elimination ordering that does not produce any fill elements during Gaussian elimination. **30.2, 40.5**

perfect matching: A matching M in which each vertex of G is in one edge of M . **30.1**

perfectly-well determined to high relative accuracy (singular values of A): Changing an arbitrary nonzero entry $a_{k\ell}$ to $\theta a_{k\ell}$, with arbitrary $\theta \neq 0$ causes a relative perturbation in σ_i bounded by $|\theta|$ and $1/|\theta|$. **46.3**

period (of access equivalence class J of reducible nonnegative P): The period of the irreducible matrix $P[J]$. **9.3**

period (of irreducible nonnegative matrix P): Greatest common divisor of the lengths of the cycles of the digraph of P . **9.2**

period (of reducible nonnegative matrix P): The least common multiple of the periods of its basic classes. **9.3**

permanent (of $n \times n$ matrix A): $\text{per}(A) = \sum_{\sigma \in S_n} \prod_{i=1}^n a_{i\sigma(i)}$ (also defined for rectangular matrices). **31.1**

permutation: A 1-1 onto function from a set to itself. **Preliminaries**

permutation equivalent (matrices A, B): There exist permutation matrices P and Q such that $B = PAQ$. **31.1**

permutation invariant (vector norm $\|\cdot\|$): $\|P\mathbf{x}\| = \|\mathbf{x}\|$ for all $\mathbf{x}$ and all permutation matrices P . **37.1**

permutation invariant absolute norm: s function $g: \mathbb{R}^n \rightarrow \mathbb{R}_0^+$ that is a norm, and $g(x_1, \dots, x_n) = g(|x_1|, \dots, |x_n|)$ and $g(\mathbf{x}) = g(P\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^n$ and all permutation matrices $P \in \mathbb{R}^{n \times n}$. **17.3**

permutation matrix: A matrix whose rows are a rearrangement of the rows of the identity matrix. **1.2**

permutation pattern: A square sign pattern matrix with entries 0 and +, where the entry + occurs precisely once in each row and in each column. **33.1**

permutation similar (square matrices A, B): There exists a permutation matrix P such that $B = P^{-1}AP (= P^TAP)$. **27.2**

permutational equivalence (of sign pattern A): A product of the form P_1AP_2 , where P_1 and P_2 are permutation patterns. **33.1**

permutational similarity (of sign pattern A) is a product of the form P^TAP , where P is a permutation pattern. **33.1**

Perron branch (at vertex v of a tree): The Perron value of the corresponding bottleneck matrix is maximal among all branches at v . **36.3**

Perron value (of square nonnegative matrix P): the spectral radius of P . **9.1**

perturbation (of matrix A): $A + \Delta A$ **15.1**

perturbation (of scalar β): $\beta + \Delta\beta$ **15.1**

perturbation (of vector $\mathbf{v}$): $\mathbf{v} + \Delta\mathbf{v}$ **15.1**

Petrie matrix: A $(0, 1)$ -matrix with the 1s in each of its columns occurring in consecutive rows. **30.2**

PIEP: Affine parameterized inverse eigenvalue problem. **20.8**

pinching (of a matrix): Defined recursively. See *Section 17.4*.

pivot: An entry of a matrix in a pivot position. **1.3**

pivot column: A column of a matrix that contains a pivot position. **1.3**

pivot position: A position in a matrix in row echelon form that contains a leading entry. **1.3**

pivot row: A row of a matrix that contains a pivot position. **1.3**

planar (graph G): There an embedding of G into $\mathbb{R}^2$. **28.2**

PLDU factorization: A factorization of a row permutation of a given matrix as the product of a unit lower triangular matrix, diagonal matrix, and a unit upper triangular matrix. **1.6**

PLU factorization: A factorization of a row permutation of a given matrix as the product of a unit lower triangular matrix and an upper triangular matrix. **1.6, 38.3**

PO: See *potentially orthogonal*.

pointed (cone K): $K \cap -K = \{0\}$. **8.5, 26.1**

polar decomposition (of matrix $A \in \mathbb{C}^{m \times n}$ with $m \geq n$): A factorization $A = UP$ where $P \in \mathbb{C}^{n \times n}$ is positive semidefinite and $U \in \mathbb{C}^{m \times n}$ satisfies $U^*U = I_n$. **17.1**

polar form: See *polar decomposition*.

polynomial restarting (of the Arnoldi algorithm): Use $\psi(A)\mathbf{v}$ instead of vector $\mathbf{v}$ when restarting. **44.4**

polytope: See *convex polytope*.

positionally symmetric: See *combinatorially symmetric*.

positive (linear map $\phi : \mathbb{C}^{n \times n} \rightarrow \mathbb{C}^{m \times m}$): $\phi(A)$ is positive semidefinite whenever A is positive semidefinite. **18.7**

positive (real matrix A): All of A 's elements are positive. **9.1**

K -positive (matrix $A \in \mathbb{R}^{n \times n}$): $AK \subseteq \text{int } K$. **26.1**

K -positive (vector $\mathbf{v} \in \mathbb{R}^n$): $\mathbf{v} \in \text{int } K$. **26.1**

positive definite (matrix): An $n \times n$ Hermitian matrix satisfying $\mathbf{x}^*A\mathbf{x} > 0$ for all nonzero $\mathbf{x} \in \mathbb{C}^n$. **5.1, 8.4**

positive definite (real symmetric or Hermitian bilinear form f): $f(\mathbf{v}, \mathbf{v}) > 0$ for all nonzero $\mathbf{v} \in V$. **12.2, 12.5**

positive P -matrix: A P -matrix in which every entry is positive. **35.9**

positive semidefinite (function $f : \mathbb{R} \rightarrow \mathbb{C}$): For each $n \in \mathbb{N}$ and all $x_1, x_2, \dots, x_n \in \mathbb{R}$, the $n \times n$ matrix $[f(x_i - x_j)]$ is positive semidefinite. **8.5**

positive semidefinite (matrix): An $n \times n$ Hermitian matrix A satisfying $\mathbf{x}^*A\mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{C}^n$. **8.4**

positive semidefinite (real symmetric or Hermitian bilinear form f): $f(\mathbf{v}, \mathbf{v}) \geq 0$ for all $\mathbf{v} \in V$. **12.2, 12.5**

positive semistable (complex square matrix): See *nonnegative stable*

positive set α of edges (in a signed bipartite graph): $\text{sgn}(\alpha)$ is $+1$. **30.1**

positive stable (complex polynomial): Its roots lie in the open right half plane. **19.2**

positive stable (complex square matrix): Its eigenvalues lie in the open right half plane. **9.5, 19.2**

k -potent (square sign pattern or ray pattern A): k is the smallest positive integer such that $A = A^{k+1}$. **33.9**

potentially orthogonal (PO) (sign pattern): Allows an orthogonal matrix. **33.10**

potentially stable (square sign pattern A): Allows stability, i.e., some matrix $B \in Q(A)$ is stable. **33.4**

Powell–Reid's complete (row and column) pivoting: A particular pivoting strategy for QR-factorization. **46.2**

power method: Method for computing the EVD of A : given starting vector $\mathbf{x}_0$, compute the sequences $\mathbf{v}_k = \mathbf{x}_k^T A \mathbf{x}_k$, $\mathbf{x}_{k+1} = A \mathbf{x}_k / \|A \mathbf{x}_k\|$, $k = 0, 1, 2, \dots$, until convergence. **42.1**

powerful (square sign or ray pattern A): All the powers $A^1, A^2, A^3, \dots$, are unambiguously defined. **33.9**

precision: The number of digits, i.e., p in $x = \pm \left(\frac{m}{b^{p-1}}\right) b^e$. **37.6**

preconditioner: A matrix M designed to improve the performance of an iterative method for solving the linear system $A\mathbf{x} = \mathbf{b}$. **41.1**

preserves: Linear operator T preserves bilinear form f if $f(T\mathbf{u}, T\mathbf{v}) = f(\mathbf{u}, \mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$, **12.1**; Also applied to φ -sesquilinear form. **12.4**

preserves: Linear operator ϕ preserves subset of matrices $\mathcal{M}$ if $\phi(\mathcal{M}) \subseteq \mathcal{M}$. **22.1**

preserves: Linear operator ϕ preserves relation $\sim$ on matrix subspace $\mathcal{V}$ if $\phi(A) \sim \phi(B)$ whenever $A \sim B$, $A, B \in \mathcal{V}$. **22.1**

primary decomposition (of a nonconstant monic polynomial $q(x)$ over F): A factorization $q(x) = (h_1(x))^{m_1} \cdots (h_r(x))^{m_r}$, where the $h_i(x)$, $i = 1, \dots, r$ are distinct monic irreducible polynomials over F . **6.4**

primary factors: The factors in a primary decomposition. **6.4**

primary matrix function: Function of a matrix defined using the Jordan Canonical Form and using the same branch for f and its derivatives for each Jordan block for the same eigenvalue. **11.1**

prime (in domain $\mathbb{D}$): A nonzero, nonunit element p such that for any $a, b \in \mathbb{D}$, $p|ab$ implies $p|a$ or $p|b$. **23.1**

prime (ideal I of a domain): $ab \in I$ implies that either a or b is in I . **23.1**

primitive (digraph): There is a positive integer k such that for every pair of vertices u and v , there is a walk of length k from u to v . **29.6**

primitive (polynomial $p(x) = \sum_{i=0}^m a_i x^{m-i} \in \mathbb{Z}[x]$, $a_0 \neq 0$, $m \geq 1$): 1 is a g.c.d. of $a_0, \dots, a_m$. **23.1**

primitive matrix: A square nonnegative matrix whose digraph is primitive. **29.6**

principal angles (between subspaces X and Y of $\mathbb{C}^r$): See *Section 17.7* (equivalent to *canonical angles*, *Section 15.1*.)

principal ideal: An ideal generated by one element (in a domain). **23.1**

principal ideal domain (PID) $\mathbb{D}$: Any ideal of $\mathbb{D}$ is principal. **23.1**

principal minor: The determinant of a principal submatrix. **4.2**

principal logarithm (of complex square matrix that has no real eigenvalues ≤ 0): The logarithm with all eigenvalues in the strip $\{z : -\pi < \operatorname{im}(z) < \pi\}$. **11.4**

principal square root (of a complex square matrix that has no real eigenvalues ≤ 0): The square root with each eigenvalue having real part > 0 . **11.2**

principal submatrix: A submatrix lying in the same rows as columns. **1.2, 10.1**

principal submatrix at a distinguished eigenvalue λ (of square nonnegative matrix P): The principal submatrix of P corresponding to a set of vertices of $\Gamma(P)$ having no access to a vertex of an access equivalence class C that satisfies $\rho(P[C]) > \lambda$. **9.3**

principal vectors (between subspaces X and Y of $\mathbb{C}^r$): See *Section 17.7*.

product (of simple graphs): See (*strong*) *product*.

profile (of sparse matrix A): The number of elements in the envelope of A . **40.5**

projection (of a vector onto V_1 along V_2 , assuming $V = V_1 \oplus V_2$): The (unique) part of the vector in V_1 ; also, the linear transformation that maps a vector to its projection. **3.6**

proper cone K (in a finite-dimensional real vector space V): A convex cone that is closed, pointed, and full. **26.2**

proper subdigraph (of a digraph $\Gamma = (V, E)$): A subdigraph $\Gamma' = (V', E')$ such that V' is a proper subset of V or E' is a proper subset of E . **29.1**

properly signed nest: See *allows a properly signed nest*.

Property C: An $n \times n$ M -matrix A satisfies property C if there exists a representation of A of the form $A = sI - P$ with $s > 0$, $P \geq 0$ and $\frac{P}{s}$ semiconvergent. **9.4**

Property L: A property that complex square matrices A, B have if their eigenvalues α_k, β_k , ($k = 1, \dots, n$) may be ordered in such a way that the eigenvalues of $xA + yB$ are given by $x\alpha_k + y\beta_k$ for all complex numbers x and y . **7.2, 24.3**

PSD square root (of a positive semidefinite matrix): The square root with each eigenvalue ≥ 0 (same as principal square root for a positive definite matrix). **8.3**

pseudo-code: An informal computer program used for writing algorithms. **37.7**

ε -pseudoeigenvalue (of matrix $A \in \mathbb{C}^{n \times n}$): A number $z \in \mathbb{C}$ such that there exists a nonzero vector $\mathbf{v} \in \mathbb{C}^n$ such that $\|A\mathbf{v} - z\mathbf{v}\| < \varepsilon\|\mathbf{v}\|$. **16.1**

ε -pseudoeigenvector (of matrix $A \in \mathbb{C}^{n \times n}$): A nonzero vector $\mathbf{v} \in \mathbb{C}^n$ such that there exists $z \in \mathbb{C}$ such that $\|A\mathbf{v} - z\mathbf{v}\| < \varepsilon\|\mathbf{v}\|$. **16.1**

ε -pseudospectral abscissa (of a complex square matrix): The rightmost extent of the ε -pseudospectrum. **16.3**

ε -pseudospectral radius (of a complex square matrix): The maximum magnitude of the ε -pseudospectrum. **16.3**

ε -pseudospectrum (of complex square matrix A): The set

$\{z \in \mathbb{C} : z \in \sigma(A + E) \text{ for some } E \in \mathbb{C}^{n \times n} \text{ with } \|E\| < \varepsilon\}$. **16.1**

ε -pseudospectrum of a matrix pencil: See *Section 16.5*.

ε -pseudospectrum of a matrix polynomial: See *Section 16.5*.

ε -pseudospectrum of a rectangular matrix: See *Section 16.5*.

Q

QMR algorithm: Iterative method for solving a linear system using non-Hermitian Lanczos algorithm. **41.3**

QR factorization (of matrix A): $A = QR$ where Q is unitary and R is upper triangular. **5.5**

QR factorization with column pivoting (of $A \in \mathbb{R}^{m \times n}$): The factorization $A\Pi = Q \begin{bmatrix} R \\ 0 \end{bmatrix}$, where Π is a permutation matrix, Q is orthogonal, and R is $n \times n$ upper triangular. **46.2**

QR iteration: Method for computing the EVD of A : starting from the matrix $A_0 = A$, the sequence of matrices $A_k = Q_k R_k$, $A_{k+1} = R_k Q_k$, $k = 0, 1, 2, \dots$ where $Q_k R_k$ is the QR factorization of A_k . **42.1**

quadrangular (bipartite graph): Simple and each pair of vertices with a common neighbor lie on a cycle of length 4. **30.1**

quadratic form (corresponding to symmetric bilinear form f): The map $g : V \rightarrow F$ defined by $g(\mathbf{v}) = f(\mathbf{v}, \mathbf{v})$, $\mathbf{v} \in V$. **12.2**

qualitative class (of complex sign pattern $A = A_1 + i A_2$): $Q(A) = \{B_1 + i B_2 : B_1 \in Q(A_1) \text{ and } B_2 \in Q(A_2)\}$. **33.8**

qualitative class (of real matrix B): The qualitative class of $\text{sgn}(B)$. **33.1**

qualitative class (of sign pattern A): The set of all real matrices B with $\text{sgn}(B) = A$. **33.1**

quotient (of vector space V by subspace W): The set of additive cosets of W with operations $(\mathbf{v}_1 + W) + (\mathbf{v}_2 + W) = (\mathbf{v}_1 + \mathbf{v}_2) + W$ and $c(\mathbf{v} + W) = (c\mathbf{v}) + W$ for $c \in F$. **2.3**

quotient field (of an integral domain): The set of equivalence classes of all quotients $\frac{a}{b}$, $b \neq 0$, constructed the same way $\mathbb{Q}$ is constructed from $\mathbb{Z}$. **23.1**

R

(R, S) -standard map, (R, c, f) -standard map: Alphabetized under *standard*.

radix: The exponentiation base b in floating point number $x = \pm \left(\frac{m}{b^{p-1}}\right) b^e$. **37.6**

range (of a linear transformation $T : V \rightarrow W$): $\{T(\mathbf{v}) : \mathbf{v} \in V\}$. **3.4**

range (of a matrix): The span of the columns. **2.4**

rank (of bilinear form f): The rank of a matrix representing f relative to an ordered basis. **12.1**. Also applied to φ -sesquilinear form. **12.4**

rank (of a matrix or linear transformation): The dimension of the range; for a matrix, equals the number of leading entries in the reduced row echelon form of the matrix. **1.3, 2.4, 3.5**

rank (of a tensor $\mathbf{z}$): The rank of $\mathbf{z}$ is k if $\mathbf{z}$ is the sum of k decomposable tensors, but it cannot be written as sum of l decomposable tensors for any l less than k . **13.3**

rank revealing (QR factorization (with pivoting) of A): Small singular values of A are revealed by correspondingly small diagonal entries of R . **46.2**

rank revealing decomposition (of real matrix A): a two-sided orthogonal decomposition of the form

$$A = U \hat{R} V^T = U \begin{bmatrix} R \\ 0 \end{bmatrix} V^T, \text{ where } U \text{ and } V \text{ are orthogonal, and } R \text{ is upper triangular and reveals the}$$

(numerical) rank of A in the size of its diagonal elements. **39.9**

rational canonical form: See *invariant factors rational canonical form* or *elementary divisors rational canonical form*.

rational function: A quotient of polynomials. **23.1**

ray nonsingular (ray pattern A): The Hadamard product $X \circ A$ is nonsingular for every entry-wise positive $n \times n$ matrix X . **33.8**

ray pattern: A matrix each of whose entries is either 0 or a ray in the complex plane represented by $e^{i\theta}$. **33.8**

ray pattern class (of ray pattern A): $Q(A) = \{B : b_{pq} = 0 \text{ iff } a_{pq} = 0, \text{ and otherwise } \arg b_{pq} = \arg a_{pq}\}$. **33.8**

Rayleigh quotient: $\frac{\mathbf{x}^* A \mathbf{x}}{\mathbf{x}^* \mathbf{x}}$, where A is Hermitian and $\mathbf{x}$ is a nonzero vector. **8.2**

real generalized eigenspace: See *Section 6.3*.

real generalized eigenvector: See *Section 6.3*.

real-Jordan block (of size $2k$ with eigenvalue $\alpha + \beta i$): The $2k \times 2k$ matrix having k copies of $M_2(\alpha, \beta) =$

$$\begin{bmatrix} \alpha & \beta \\ -\beta & \alpha \end{bmatrix} \text{ on the (block matrix) diagonal, } k-1 \text{ copies of } I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ on the first (block matrix)}$$

superdiagonal, and copies of $0_2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ everywhere else. **6.3**

real-Jordan canonical form (of real square matrix A): A real-Jordan matrix that is similar to A . **6.3**

real-Jordan matrix: A block diagonal real matrix having diagonal blocks that are Jordan blocks or real-Jordan blocks. **6.3**

real part (of complex number $a + bi$): a . **Preliminaries**

real structured ε -pseudospectrum (of real square matrix A): See *Section 16.5*.

real vector space: A vector space over the field of real numbers. **1.1**

reduced digraph (of digraph Γ): A digraph whose vertices are the access equivalence classes of Γ (or $\{1, \dots, k\}$ where k is the number of access equivalence classes) and having an arc from the i th vertex to the j th precisely when the i th access class has access to the j th access class. **9.1, 29.5**

reduced QR factorization (of matrix $A \in \mathbb{C}^{m \times n}$, $m \geq n$): $A = \hat{Q}\hat{R}$ where columns of $\hat{Q}$ are orthonormal and R is upper triangular. **5.5**

reduced row echelon form (RREF): A matrix is in RREF if it is in REF, the leading entry in any nonzero row is 1, and all other entries in a column containing a leading entry are 0; matrix B is the RREF of matrix A if B is in RREF and A and B are row equivalent. **1.3**

reduced singular value decomposition (reduced SVD) (of complex matrix A): $A = \hat{U}\hat{\Sigma}\hat{V}^*$, $\hat{\Sigma} = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r) \in \mathbb{R}^{r \times r}$, where $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$ and the columns of $\hat{U} \in \mathbb{C}^{m \times r}$ and the columns of $\hat{V} \in \mathbb{C}^{n \times r}$ are both orthonormal. **5.6, 45.1**

reducible (square matrix A): There is a permutation matrix P such that $PAP^T = \begin{bmatrix} B & C \\ \mathbf{0} & D \end{bmatrix}$, where B, D are square. **9.1, 27.3**

reducing eigenvalue (λ of $A \in \mathbb{C}^{n \times n}$): A is unitarily similar to $[\lambda] \oplus A_2$. **18.2**

REF: See *row echelon form*.

(k -)regular (graph): Every vertex has the same degree k . **28.2**

regular ($m \times n$ (0,1)-design matrix W): W is balanced and $W^T W = t(I + J)$ for some integer t . **32.4**

regular (pencil $A - xB$): There exists a $\lambda \in \mathbb{C}$ such that $A - \lambda B$ is nonsingular. **43.1**

regular (pencil $A_0 + xA_1 \in \mathbb{D}[x]^{n \times n}$): $\det(A_0 + xA_1)$ is not the zero polynomial, or equivalently, there exists a $\lambda \in \mathbb{D}$ such that $A_0 + \lambda A_1$ is nonsingular. **23.4**

relative backward error of the linear system: See *Section 38.1*.

relative condition number: See *condition number*.

relative error (in approximation $\hat{\mathbf{z}}$ to $\mathbf{z}$): $\|\mathbf{z} - \hat{\mathbf{z}}\|/\|\mathbf{z}\|$. **37.4**

requires: If P is a property referring to a real matrix, then a sign pattern A requires P if every real matrix in $Q(A)$ has property P . **33.1**

requires unique inertia (sign pattern A): $\text{in}(B_1) = \text{in}(B_2)$ for all symmetric matrices $B_1, B_2 \in Q(A)$. **33.6**

residual vector (of $\tilde{\mathbf{x}}$ when solving the linear system $A\mathbf{x} = \mathbf{b}$): The vector $\mathbf{r} = \mathbf{b} - A\tilde{\mathbf{x}}$. **38.3**

residual vector (at step k of an iterative method for solving $A\mathbf{x} = \mathbf{b}$): The vector $\mathbf{r}_k = \mathbf{b} - A\mathbf{x}_k$, where $\mathbf{x}_k$ is the approximate solution generated at step k . **41.1**

resolvent (of matrix $A \in \mathbb{C}^{n \times n}$ at a point $z \notin \sigma(A)$): The matrix $(zI - A)^{-1}$. **16.1**

restarted GMRES algorithm, GMRES(j): Restart GMRES every j steps, using the latest iterate as the initial guess for the next GMRES cycle. **41.3**

right singular vector: See *singular vectors*.

right singular space: See *singular spaces*.

ring automorphism of $F^{n \times n}$ induced by f : The map $\phi : F^{n \times n} \rightarrow F^{n \times n}$ defined by $\phi([a_{ij}]) = [f(a_{ij})]$. **22.4**

Ritz pair: $(\theta, \mathbf{v})$ where θ is the Ritz value for Ritz vector $\mathbf{v}$. **43.3**

Ritz value: The scalar θ for a Ritz vector. **43.3**

Ritz vector (of matrix of A from subspace S of $\mathbb{C}^n$): There is a $\theta \in \mathbb{C}$ such that $A\mathbf{v} - \theta\mathbf{v} \perp S$. **43.3**

rook polynomials: Generating functions for the rook numbers defined using permanents. **31.8**

rounding error: In one floating point arithmetic operation, the difference between the exact arithmetic operation and the floating point arithmetic operation; in more extensive calculations, refers to the cumulative effect of the rounding errors in the individual floating point operations. **37.6**

rounding mode: Maps $x \in \mathbb{R}$ to a floating point number $\text{fl}(x)$; default rounding mode in standard-conforming arithmetic is round-to-nearest, ties-to-even. **37.6**

Routh-Hurwitz matrix: See *Section 19.2*.

row (of a matrix): The entries of a matrix lying in a horizontal line in the matrix. **1.2**

row echelon form (REF): A matrix is in REF if every zero row is below all nonzero rows and for two nonzero rows, the leading entry in the upper row is to the left of the leading entry of the lower row; matrix B is a REF of of matrix A if B is in REF and A and B are row equivalent. **1.3**

row equivalent: Matrix A is row equivalent to matrix B if B can be obtained from A by a sequence of elementary row operations (equivalently, $B = QA$ for some invertible matrix Q). **1.3**

row equivalent (matrices $A, B \in \mathbb{D}^{m \times n}$): $B = QA$ for some $\mathbb{D}$ -invertible matrix Q . **23.2**

row operation: See *elementary row operation*.

row signing (of (real or sign pattern) matrix A): DA where D is a signing. **33.3**

row space: The span of the rows of a matrix. **2.4**

row-stochastic (matrix): A square nonnegative matrix having all row sums equal to 1. **9.4**

row sum vector (of a matrix): The vector of row sums. **27.4**

RRD (of real matrix A): A decomposition $A = XDY^T$ with D a diagonal matrix and X and Y full column rank, well-conditioned matrices. **46.3**

RREF: See *reduced row echelon form*.

S

S-matrix: An $n \times (n + 1)$ matrix B such that it is an S^* -matrix and the kernel of every matrix in $Q(B)$ contains a vector all of whose coordinates are positive. **33.3**

S^* -matrix: An $n \times (n + 1)$ matrix B such that each of the $n + 1$ matrices obtained by deleting a column of B is an SNS matrix. **33.3**

SAP: See *spectrally arbitrary pattern*.

scalar: An element of a field. **1.1**

scalar matrix (transformation): A scalar multiple of the identity matrix (transformation). **1.2, 3.2**

scaling (of a real matrix A): A matrix of the form D_1AD_2 where D_1 and D_2 are diagonal matrices with positive diagonal entries. **9.6, 27.6**

Schatten-p norm (of $A \in \mathbb{C}^{m \times n}$): $\|A\|_{S,p} = \left(\sum_{i=1}^q \sigma_i^p(A)\right)^{1/p}$. **17.3**

Schur complement: A matrix defined from a partitioned matrix. **4.2, 10.3**

Schur product: See *Hadamard product*.

score vector: The row sum vector of a tournament matrix. **27.5**

SDR: See *system of distinct representatives*.

Seidel matrix (of simple graph G): The matrix $J - I - 2\mathcal{A}_G$. **28.4**

Seidel switching: An operation on simple graphs. **28.4**

self-adjoint: See *Hermitian*.

self-inverse sign pattern: An $S^2\text{NS}$ -pattern A such that $B^{-1} \in Q(A)$ for every matrix $B \in Q(A)$. **33.2**

semiconvergent (square nonnegative matrix P): $\lim_{m \rightarrow \infty} P^m$ exists. **9.3**

semipositive (real matrix A): A is nonnegative and some element is positive. **9.1**

K -semipositive (matrix $A \in \mathbb{R}^{n \times n}$): A is K -nonnegative and $A \neq 0$. **26.1**

K -semipositive (vector $\mathbf{v} \in \mathbb{R}^n$): $\mathbf{v} \in K$ and $\mathbf{v} \neq 0$. **26.1**

semisimple: An eigenvalue having algebraic multiplicity equal to its geometric multiplicity. **4.3**

semistable matrix: Either positive semistable (i.e., nonnegative stable) or negative semistable, depending on section.

separation (between two square matrices A_1 and A_2): $\inf_{\|X\|_2=1} \|XA_1 - A_2X\|_2$. **15.1**

φ -sesquilinear form (on vector space V over field F with automorphism φ): A map f from $V \times V$ into F that satisfies $f(a\mathbf{u}_1 + b\mathbf{u}_2, \mathbf{v}) = af(\mathbf{u}_1, \mathbf{v}) + bf(\mathbf{u}_2, \mathbf{v})$ and $f(\mathbf{u}, a\mathbf{v}_1 + b\mathbf{v}_2) = \varphi(a)f(\mathbf{u}, \mathbf{v}_1) + \varphi(b)f(\mathbf{u}, \mathbf{v}_2)$. **12.4**

sesquilinear form (on a complex vector space): A φ -sesquilinear form where φ is complex conjugation. **12**

set of Hermitian matrices associated with graph G : $\mathcal{H}(G) = \{B \in \mathcal{H}_n \mid \mathcal{G}(B) = G\}$. **34.1**

set of symmetric matrices associated with graph G : $\mathcal{S}(G) = \{B \in \mathcal{S}_n \mid \mathcal{G}(B) = G\}$. **34.1**

sgn: See *sign*.

Shannon capacity: A parameter of a simple graph. **28.5**

shift: The scalar μ in a shifted matrix $A - \mu I$. **42.1**

shifted matrix (of matrix A): The matrix $A - \mu I$, where μ is the shift. **42.1**

shifted QR iteration: QR iteration of the shifted matrix. **42.1**

sign (denoted sign , of a complex number): A nonzero complex number; $\text{sign}(0) = 1$, otherwise the complex number with the same argument having absolute value 1. **Preliminaries**

sign (denoted sgn , of a complex square matrix): A function defined from the Jordan Canonical Form of the matrix. See *Section 11.6*.

sign (denoted sgn , of a permutation): 1, if the permutation is even and -1 , if the permutation is odd. **Preliminaries**

sign (denoted sgn , of a real number): $+, 0, -$ according as the number is $> 0, = 0, < 0$. **Preliminaries**

sign (denoted sgn , of a set α of edges in a signed bipartite graph): The product of the weights of the edges in α . **30.1**

sign (of a simple cycle in a sign pattern A): The product of the entries in the cycle. **33.1**

sign-central: A sign pattern matrix that requires centrality. **33.11**

sign nonsingular (SNS) (square sign pattern A): Every matrix $B \in Q(A)$ is nonsingular. **33.2**

sign pattern: A matrix whose entries are in $\{+, 0, -\}$. **33.1**

sign pattern class: See *qualitative class*.

sign pattern matrix: See *sign pattern*.

sign pattern of real matrix B : The sign pattern whose entries are the signs of the corresponding entries in B . **33.1**

sign potentially orthogonal SPO (square sign pattern): Does not have a zero row or zero column and every pair of rows and every pair of columns allows orthogonality. **33.10**

sign semistable (square sign pattern A): Every matrix $B \in Q(A)$ is semistable **33.4**

sign singular (square sign pattern A): Every matrix $B \in Q(A)$ is singular. **33.2**

sign-solvable (system of linear equations $Ax = b$): For each $\tilde{A} \in Q(A)$ and for each $\tilde{b} \in Q(b)$, the system $\tilde{A}x = \tilde{b}$ is consistent and $\{\tilde{x} : \text{there exist } \tilde{A} \in Q(A) \text{ and } \tilde{b} \in Q(b) \text{ with } \tilde{A}\tilde{x} = \tilde{b}\}$ is entirely contained in one qualitative class. **33.3**

sign stable (square sign pattern A): Requires stability, i.e., every matrix $B \in Q(A)$ is stable. **33.4**

sign symmetric (square matrix A): $\det A[\alpha, \beta] \det A[\beta, \alpha] \geq 0, \forall \alpha, \beta \subseteq \{1, \dots, n\}, |\alpha| = |\beta|$. **19.2**

signature (of a real symmetric or Hermitian bilinear form): The signature of a matrix representing the form relative to some basis. **12.2, 12.5**

signature (of real symmetric or Hermitian matrix A): The number of positive eigenvalues minus the number of negative eigenvalues. **12.2, 12.5**

signature (of a composite cycle γ in sign pattern A): $(-1)^{\sum_{i=1}^m (l_i - 1)}$ where l_i are the lengths of the simple cycles in γ . **33.1**

signature matrix: A ± 1 diagonal matrix. **32.2**

signature pattern: A diagonal sign pattern matrix, each of whose diagonal entries is $+$ or $-$. **33.1**

signature similarity (of the square sign pattern A): A product of the form SAS , where S is a signature pattern. **33.1**

signed bigraph (of real matrix A): The weighted graph obtained from the bigraph of A weighting the edge $\{i, j\}$ by $+1$ if $a_{ij} > 0$, and by -1 if $a_{ij} < 0$. **30.2**

signed bipartite graph: A weighted bipartite graph with weights $X = \{-1, 1\}$. **30.1**

signed 4-cockade: A 4-cockade whose edges are weighted by ± 1 in such a way that every 4-cycle is negative. **30.2**

signed digraph (of an $n \times n$ sign pattern A): The digraph with vertex set $\{1, 2, \dots, n\}$ where (i, j) is an arc (bearing a_{ij} as its sign) iff $a_{ij} \neq 0$. **33.1**

significand: The (base b) integer m in floating point number $x = \pm \left(\frac{m}{b^{p-1}}\right) b^e$. **37.6**

signing (of order k): A nonzero $(0, 1, -1)$ - or $(0, +, -)$ -diagonal matrix of order k . **33.3**

signless Laplacian matrix (of graph G): The matrix $D + \mathcal{A}_G$, where D is the diagonal matrix of vertex degrees and $\mathcal{A}_G$ is the adjacency matrix of G . **28.4**

similar (to matrix A): Any matrix of the form $C^{-1}AC$. **2.4**

similarity scaling (of square matrix A): A scaling DAD^{-1} of A . **9.6**

simple (cycle) (in a digraph): The induced subdigraph of the vertices in the cycle is the cycle. **35.1**

simple cycle (in sign pattern A): See k -cycle (in sign pattern A).

simple (digraph): A digraph with no loops (a digraph does not have multiple arcs). **29.1**

simple (eigenvalue): An eigenvalue having algebraic multiplicity 1. **4.3**

simple (graph): A graph with no loops where each edge has multiplicity at most one. **28.1**

simple (walk) (in a digraph): A walk in which all vertices, except possibly the first and last, are distinct. **29.1**

simple row operation: A generalization of an elementary row operation used in domains. **23.2**

sine (of a complex square matrix A): The matrix defined by the sine power series,

$$\sin(A) = A - \frac{A^3}{3!} + \cdots + \frac{(-1)^k}{(2k+1)!} A^{2k+1} + \cdots \quad \mathbf{13.5}$$

single precision: Typically, floating point numbers have machine epsilon roughly 10^{-7} and precision roughly 7 decimal digits. **37.6**

singular (matrix or linear operator): Not nonsingular. **1.5, 3.7**

singular (pencil): A pencil that is not regular. **23.4, 43.1**

singular space (right or left) (of complex matrix A): The subspace spanned by the right (or by the left) singular vectors of A . **45.1**

singular-triplet (of $A \in \mathbb{C}^{m \times n}$): A (left singular vector, singular value, right singular vector) triplet. **15.2**

singular value decomposition (SVD) (of matrix $A \in \mathbb{C}^{m \times n}$): $A = U \Sigma V^*$,

$\Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_p) \in \mathbb{R}^{m \times n}$, $p = \min\{m, n\}$, where $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_p \geq 0$ and both $U = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m] \in \mathbb{C}^{m \times m}$ and $V = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n] \in \mathbb{C}^{n \times n}$ are unitary. Also forms with other dimensions. **5.6, 45.1, 17.1**

singular value vector (of complex matrix A): The vector of singular values of A in nonincreasing order. **17.1**

singular values (of a complex matrix): The diagonal entries σ_i of Σ in a singular value decomposition. **5.6, 45.1, 17.1**

singular vectors (of a complex matrix): Left: the columns of U , and right: the columns of V , both in a singular value decomposition. **5.6, 45.1**

size (of matrix A): $m \times n$, where A is an $m \times n$ matrix; also m if $n = m$. **1.2**

Skeel condition number of the linear system $A\hat{\mathbf{x}} = \mathbf{b}$: $\text{cond}(A, \hat{\mathbf{x}}) = \frac{\| |A^{-1}| |A| \|\hat{\mathbf{x}}\|_\infty}{\|\hat{\mathbf{x}}\|_\infty}$. **38.1**

Skeel matrix condition number (of matrix A): $\text{cond}(A) = \| |A^{-1}| |A| \|_\infty$. **38.1**

skew-Hermitian (matrix): A real or complex matrix equal to the negative of its conjugate transpose. **1.2, 7.2**

skew-symmetric (matrix): A matrix equal to the negative of its transpose. **1.2**

Smith invariant factors (of $A \in F^{n \times n}$): The nonconstant polynomials on the diagonal of the Smith normal form of $xI - A$. **6.5**

Smith normal form (of $M \in F[x]^{n \times n}$): The Smith normal matrix obtained from M by elementary row and column operations. **6.5**

Smith normal form (over GCD domain $\mathbb{D}_g$): Matrix $B \in \mathbb{D}_g^{m \times n}$ is in Smith normal form if

$$B = \text{diag}(b_1, \dots, b_r, 0, \dots, 0), b_i \neq 0 \text{ for } i = 1, \dots, r \text{ and } b_{i-1} | b_i \text{ for } i = 2, \dots, r. \quad \mathbf{23.2}$$

Smith normal matrix (in $F[x]^{n \times n}$): A diagonal matrix $\text{diag}(1, \dots, 1, a_1(x), \dots, a_s(x), 0, \dots, 0)$, where the $a_i(x)$ are monic nonconstant polynomials such that $a_i(x)$ divides $a_{i+1}(x)$ for $i = 1, \dots, s - 1$. **6.5**

SNS: See *sign nonsingular*.

S²NS: See *strong sign nonsingular*.

solution: An ordered tuple of scalars that when assigned to the variables satisfies a system of linear equations. **1.4**

solution set: The set of all solutions to a system of linear equations. **1.4**

span, Span (noun): The set of all linear combinations of the vectors in a set. **2.1**

span, spans (verb): A set $S \subseteq V$ spans V if $V = \text{Span}(S)$. **2.1**

spanning subgraph (of a connected graph (V, E)): A connected subgraph (V', E') with $V' = V$. **28.1**

spanning tree (of a connected graph): A spanning subgraph that is a tree. **28.1**

sparse (matrix A): A matrix for which substantial savings in either operations or storage can be achieved when the zero elements of A are exploited during the application of Gaussian elimination to A . **40.2**

sparse approximate inverse (of matrix A): A sparse matrix M^{-1} constructed to approximate A^{-1} . **41.1**

sparsity pattern: See *sparsity structure*.

sparsity structure (of matrix A): The set of indices of nonzero elements of A . **40.3**

special linear group (of order n over F): The subgroup of the general linear group consisting of all matrices that have determinant 1. **67.1**

spectral absolute value (of complex matrix A): $(A^*A)^{1/2}$. **17.1**

spectral norm (of complex matrix A): $\|A\|_2 = \sqrt{\rho(A^*A)} = \max\{\|A\mathbf{v}\|_2 : \|\mathbf{v}\|_2 = 1\} =$ the largest singular value of A . **7.1, 37.3**

spectral radius (of $A \in \mathbb{C}^{n \times n}$): $\max\{|\lambda| : \lambda \text{ an eigenvalue of } A\}$. **4.3**

spectrally arbitrary pattern (SAP): A sign pattern A of size n such that every monic real polynomial of degree n can be achieved as the characteristic polynomial of some matrix $B \in Q(A)$. **33.6**

spectrum (of a matrix): The multiset of all eigenvalues of a matrix, each eigenvalue appearing with its algebraic multiplicity. **4.3**

spectrum (of a graph): The spectrum of its adjacency matrix. **28.3**

speed (of a processor): The number of floating-point operations per second (flops). **42.9**

splitting: See *preconditioner*.

SPO: See *sign potentially orthogonal*.

square matrix: The number of columns is the same as the number of rows. **1.2**

square root (of a square complex matrix A): A matrix B such that $B^2 = A$. **11.2** (See also *principal square root*, *PSD square root*)

stable (algorithm): See *numerically stable*.

stable matrix: Either positive stable or negative stable, depending on section; positive in *Section 9.5* and *Section 19.2*.

standard basis (for F^n or $F^{m \times n}$): The set of vectors or matrices having one entry equal to 1 and all other entries equal to 0. **2.1**

standard-conforming (floating point system): conforming to IEEE standard. **37.6**

standard inner product (in $\mathbb{R}^n$ (or $\mathbb{C}^n$)): $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{v}$ ($\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{v}^* \mathbf{u}$). **5.1**

standard linear preserver problem: A linear preserver problem where the preservers take one of the standard forms. **22.2**

(R, c, f)-standard map: For $A \in F^{n \times n}$, $\phi(A) = cRAR^{-1} + f(A)I$, or $\phi(A) = cRA^T R^{-1} + f(A)I$. R invertible, c nonzero, f a linear functional. **22.2**

(R, S)-standard map: For $A \in F^{m \times n}$, $\phi(A) = RAS$, or $m = n$ and $\phi(A) = RA^T S$. R and S must be invertible and may be required satisfy some additional assumptions. **22.2**

standard matrix of transformation $T : F^n \rightarrow F^m$: The matrix of T with respect to the standard bases for F^n, F^m . **3.3**

star on n vertices: a tree in which there is a vertex of degree $n - 1$. **34.6**

state (of $n \times n$ stochastic matrix P): An index $i \in \{1, \dots, n\}$. **9.4**

stationary distribution (of stochastic matrix P): Nonnegative vector π that satisfies $\pi^T \mathbf{1} = \mathbf{1}$ and $\pi^T P = \pi^T$. **9.4**

k -step Arnoldi factorization: See *Arnoldi factorization*.

stochastic (square nonnegative matrix P): Same as row-stochastic; the sum of the entries in each row is 1. **9.4**

stopping (matrix): A transient substochastic matrix. **9.4**

strict column signing (of (real or sign pattern) matrix A): AD where D is a strict signing. **33.3**

strict row signing (of (real or sign pattern) matrix A): DA where D is a signing. **33.3**

strict signing: A signing where all the diagonal entries are nonzero. **33.3**

strictly block lower triangular (matrix A): A^T is strictly block upper triangular. **10.3**

strictly block upper triangular (matrix): A block upper triangular matrix in which all the diagonal blocks are 0. **10.3**

strictly copositive (matrix A): $\mathbf{x}^T A \mathbf{x} > 0$ for all $\mathbf{x} \geq 0$ and $\mathbf{x} \neq 0$. **35.5**

strictly diagonally dominant ($n \times n$ complex matrix A): $|a_{ii}| > \sum_{j=1, j \neq i}^n |a_{ij}|$ for $i = 1, \dots, n$. **9.5**

strictly equivalent (pencils $A - xB$ and $C - xD$): If there exist nonsingular matrices S_1 and S_2 such that $C - \lambda D = S_1(A - \lambda B)S_2$ for all $\lambda \in \mathbb{C}$. **43.1**

strictly equivalent (pencils $A(x), C(x) \in \mathbb{D}[x]^{m \times n}$): $C(x) = QA(x)P$ for some $\mathbb{D}$ -invertible matrices P, Q . **23.4**

strictly lower triangular (matrix A): A^T is strictly upper triangular **10.2**

strictly upper triangular (matrix): A matrix such that every entry having row number greater than or equal to column number is zero. **10.2**

strictly unitarily equivalent (pencils $A - xB$ and $C - xD$): strictly equivalent pencils in which S_1, S_2 can be taken unitary. **43.1**

Strong Arnold Hypothesis: Satisfied by a real symmetric matrix M provided there does not exist a real symmetric nonzero matrix X such that $MX = 0, M \circ X = 0, I \circ X = 0$. **28.5**

strong combinatorial invariant (of a matrix): A quantity or property that does not change when the rows and/or columns are permuted. **27.1**

strong Parter vertex: a Parter–Wiener vertex j for an eigenvalue λ of an $A \in \mathcal{H}(G)$ such that λ occurs as an eigenvalue of at least three direct summands of $A(j)$. **34.1**

(strong) product (of simple graphs $G = (V, E), G' = (V', E')$): The simple graph with vertex set $V \times V'$, where two distinct vertices are adjacent whenever in both coordinate places the vertices are adjacent or equal in the corresponding graph. **28.1**

strong sign nonsingular (S^2NS) (square sign pattern A): A is an SNS-pattern such that the matrix B^{-1} is in the same sign pattern class for all $B \in Q(A)$. **33.2**

strongly connected: A digraph whose vertices all lie in a single access equivalence class. **9.1, 29.5**

strongly connected components (of a digraph): The subdigraphs induced by the access equivalence classes. **29.5**

strongly inertia preserving (real square matrix A): The inertia of AD is equal to the inertia of D for every real diagonal matrix D . **19.5**

strongly nonsingular (square matrix): All its principal submatrices are nonsingular. **47.6**

ϕ **strongly preserves** (subset of matrices $\mathcal{M}$): $\phi(\mathcal{M}) = \mathcal{M}$. **22.1**

ϕ **strongly preserves** (relation $\sim$ on matrix subspace $\mathcal{V}$): For every pair $A, B \in \mathcal{V}$, we have $\phi(A) \sim \phi(B)$ if and only if $A \sim B$. **22.1**

strongly regular: A simple graph with parameters (n, k, λ, μ) that has n vertices, is k -regular with $1 \leq k \leq n - 2$, every two adjacent vertices have exactly λ common neighbors, and every two distinct nonadjacent vertices have exactly μ common neighbors. **28.2**

strongly stable: See *additive D-stable*.

subdigraph (of a digraph $\Gamma = (V, E)$): A digraph $\Gamma' = (V', E')$ with $V' \subseteq V$ and $E' \subseteq E$. **29.1**

subgraph (of a graph (V, E)): A graph $G' = (V', E')$ with $V' \subseteq V$ and $E' \subseteq E$. **28.1**

submatrix: A matrix lying in certain rows and columns of a given matrix. **1.2, 10.1**

$\mathbb{D}$ -**submodule**: A nonempty set $\mathbf{N} \subseteq \mathbf{M}$ that is closed under addition and multiplication by ($\mathbb{D}$) scalars. **23.3**

submultiplicative (norm $\|\cdot\|$ on $\mathbb{C}^{n \times n}$): A vector norm satisfying $\|AB\| \leq \|A\|\|B\|$ for all $A, B \in \mathbb{C}^{n \times n}$ (satisfies the conditions of a matrix norm except not required to be part of a family). **18.4**

subordinate (matrix norm): See *induced norm*.

(k -th)-subpermanent sum (of matrix A): The sum of the permanents of all order k submatrices of A . **31.7**

subpattern (of a sign pattern): A sign pattern obtained by replacing some (possibly none) of the nonzero entries with 0. **33.1**

subspace: Subset of a vector space V that is itself a vector space under the operations of V . **1.1**

subspace iteration: See *orthogonal iteration*.

substochastic (square nonnegative matrix P): The sum of the entries in each row is ≤ 1 . **9.4**

subtractive cancellation of significant digits: Occurs in floating point sums when the relative error in the rounding-error-corrupted approximate sum is substantially greater than the relative error in the summands. **37.6**

sum (of subspaces): $W_1 + \cdots + W_k = \{\mathbf{w}_1 + \cdots + \mathbf{w}_k : \mathbf{w}_i \in W_i\}$. **2.3**

sum norm: See *1-norm*.

sup-norm: See *∞ -norm*.

support (of rectangular matrix A or vector): The set of indices ij with $a_{ij} \neq 0$. **9.6**

support line (of convex set S): A line ℓ that intersects ∂S such that S lies entirely within one of the closed half-planes determined by ℓ . **18.2**

G supports rank decompositions: Each matrix $A \in \mathcal{M}(G)$ is the sum of $\text{rank}(A)$ matrices in $\mathcal{M}(G)$ each having rank 1 (also a variant for signed bipartite graphs). **30.3**

surjective: See *onto*.

SVD: See *singular value decomposition*.

switching equivalent: A simple graph that can be obtained from another simple graph by a Seidel switching. **28.4**

Sym: The map $\text{Sym}(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_k) = \frac{1}{m!} \sum_{\pi \in S_m} \mathbf{v}_{\pi(1)} \otimes \cdots \otimes \mathbf{v}_{\pi(k)}$. **13.6**

sym multiplication: $(\mathbf{v}_1 \vee \cdots \vee \mathbf{v}_p) \vee (\mathbf{v}_{p+1} \vee \cdots \vee \mathbf{v}_{p+q}) = \mathbf{v}_1 \vee \cdots \vee \mathbf{v}_{p+q}$. **13.7**

symbol (of a family of Toeplitz matrices): The function $a(z) = \sum a_k z^k$ where the a_k are the constants of the Toeplitz matrices. **16.2**

symbol curve (of a Toeplitz family): The image of the complex unit circle under the symbol. **16.2**

symmetric (bilinear form f): $f(\mathbf{u}, \mathbf{v}) = f(\mathbf{v}, \mathbf{u})$ for all $\mathbf{u}, \mathbf{v} \in V$. **12.2**

symmetric (digraph $\Gamma = (V, E)$): $(i, j) \in E$ implies $(j, i) \in E$ for all $i, j \in V$. **35.1**

symmetric (form): A multilinear form that is a symmetric map. **13.4**

symmetric (map $\psi \in L^m(V; U)$): $\psi(\mathbf{v}_{\pi(1)}, \dots, \mathbf{v}_{\pi(m)}) = \psi(\mathbf{v}_1, \dots, \mathbf{v}_m)$, for all permutations π . **13.4**

symmetric (matrix): A matrix equal to its transpose. **1.2**

symmetric algebra (on vector space V): $\bigvee V = \bigoplus_{k \in \mathbb{N}} (\bigvee^k V)$ with sym multiplication. **13.9**

symmetric inertia set (of a symmetric sign pattern A): $\text{in}(A) = \{\text{in}(B) : B = B^T \in Q(A)\}$; **33.6**

symmetric matrices associated with graph G : See *set of symmetric matrices associated with graph G* .

symmetric maximal rank (of symmetric sign pattern A): $\max\{\text{rank } B : B^T = B \text{ and } B \in Q(A)\}$. **33.6**

symmetric minimal rank (of symmetric sign pattern A): $\min\{\text{rank } B : B^T = B \text{ and } B \in Q(A)\}$. **33.6**

symmetric power (of a vector space): See *symmetric space*.

symmetric product (of vectors $\mathbf{v}_1, \dots, \mathbf{v}_m$): $\mathbf{v}_1 \vee \cdots \vee \mathbf{v}_m = m! \text{Sym}(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_k)$. **13.6**

symmetric rank revealing decomposition (SRRD) (of symmetric real matrix H): A decomposition $H = XDX^T$, where D is diagonal and X is a full column rank well-conditioned matrix. **46.5**

symmetric space (of vector space V): $\bigvee^m V = \text{Sym}(\bigotimes^m V)$. **13.6**

symmetric scaling (of a matrix): A scaling where $D_1 = D_2$. **9.6, 27.6**

system of distinct representatives (SDR): For the finite sets $S_1, S_2, \dots, S_n$, a choice of $x_1, x_2, \dots, x_n$ with the properties that $x_i \in S_i$ for each i and $x_i \neq x_j$ whenever $i \neq j$. **31.3**

system of linear equations: A set of one or more linear equations in the same variables. **1.4**

T

tensor: An element of a tensor product. **13.2**

tensor algebra (on vector space V): $\bigotimes V = \bigoplus_{k \in \mathbb{N}} (\bigotimes^k V)$. **13.9**

tensor multiplication: $(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_p) \otimes (\mathbf{v}_{p+1} \otimes \cdots \otimes \mathbf{v}_{p+q}) = \mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_{p+q}$. **13.7**

tensor power: A tensor product of copies of one vector space. **13.2**

tensor product (of matrices): See *Kronecker product*.

tensor product (of vector spaces $V_1, \dots, V_m$): a multilinear map satisfying a universal factorization property, See Section 13.2.

term rank (of a $(0, 1)$ -matrix A): The largest size of a collection of 1s of A with no two 1s in the same line. 27.1

term rank (of sign pattern A): The maximum number of nonzero entries of A no two of which are in the same row or same column. 33.6

tight sign-central: A sign-central pattern A for which the Hadamard product of any two columns contains a negative element. 33.11

TN: See *totally nonnegative*.

Toeplitz (matrix): A matrix whose entries are constant along each sub- and super- diagonal, i.e., the value of the i, j -entry is the constant a_k whenever $i - j = k$. 16.2, 48.1

Toeplitz-block matrix: A block matrix $A = [A_{ij}]$ where each block A_{ij} is an $n \times n$ Toeplitz matrix. 48.1

Toeplitz inverse eigenvalue problem (ToIEP): Given $\lambda_1, \dots, \lambda_n \in \mathbb{R}$, find $\mathbf{c} = [c_1, \dots, c_n]^T \in \mathbb{R}^n$ such that $[t_{ij}]_{i,j=1}^n$ with $t_{ij} = c_{|i-j|+1}$ has spectrum $\{\lambda_1, \dots, \lambda_n\}$. 20.9

ToIEP: See *Toeplitz inverse eigenvalue problem*.

total least squares problem: See Section 39.1.

total signed compound (TSC) (sign patterns): Every square submatrix of every matrix $A \in Q(S)$ is either sign nonsingular or sign singular. 46.3

total support: An $n \times n$ $(0, 1)$ -matrix A has total support if $A \neq \mathbf{0}$ and each 1 of A is on a diagonal of A . 27.2

totally nonnegative (real matrix A): Every minor is nonnegative. 21.1

totally positive (real matrix A): Every minor is positive. 21.1

totally unimodular (real matrix A): All minors of A are from $\{-1, 0, 1\}$. 46.3

tournament: Digraph of a tournament matrix. 27.5

tournament matrix: A $(0, 1)$ -matrix with 0 diagonal and exactly one of a_{ij}, a_{ji} equal to 1 for all $i \neq j$. 27.5

TP: See *totally positive*.

trace: The sum of all the diagonal entries of the matrix. 1.2

trace-minimal (graph G in a family $\mathcal{F}$ of graphs): The trace-sequence of the adjacency matrix of G is least in lexicographic order among all graphs in $\mathcal{F}$. 32.7

trace norm (of $A \in \mathbb{C}^{m \times n}$): The sum of the singular values of A . 17.3

trace-sequence (of $n \times n$ matrix A): $(\text{tr}(A), \text{tr}(A^2), \dots, \text{tr}(A^n))$. 32.7

transient class (of a stochastic matrix P): an access equivalence class of P that is not ergodic. 9.4

transient matrix: See convergent.

transient state: A state in a transient class. 9.4

transition matrix: See *change-of-basis matrix*.

transitive tournament matrix A : $a_{ij} = a_{jk} = 1$ implies $a_{ik} = 1$. 27.5

transpose (of a linear transformation): A specific linear transformation from the dual of the codomain to the dual of the domain. 3.8

transpose (of $m \times n$ matrix A): The $n \times m$ matrix $B = [b_{ij}]$ where $b_{ij} = a_{ji}$. 1.2

transposition: A 2-cycle. Preliminaries

tree (digraph): A digraph whose associated graph is a tree. 29.1

tree (graph): A connected graph with no cycles. 28.1

tree sign pattern (t.s.p.): A combinatorially symmetric sign pattern matrix whose graph (suppressing loops) is a tree. 33.5

triangular (matrix): Upper or lower triangular. 10.2

triangular property (of class of matrices X): Whenever A is a block triangular matrix and every diagonal block is an X -matrix, then A is an X matrix. 35.1

triangular system: A linear system $T\mathbf{x} = \mathbf{b}$ where T is a triangular matrix. 38.2

triangular totally positive (Δ TP): A triangular matrix all of whose nontrivial minors are positive. 21.2

tridiagonal matrix: A square matrix A such that $a_{ij} = 0$ if $|i - j| > 1$.

trivial face (F of cone K): $F = \{0\}$ or $F = K$. 26.1

trivial linear combination: A linear combination in which all the scalar coefficients are 0 (or over the empty set). 2.1

truncation error: The error made by replacing an infinite process by a finite process. **37.6**

TSC: See *total signed compound*.

t.s.p. See *tree sign pattern*.

two-sided Lanczos algorithm: See *Section 41.3*.

type I (tree): Has exactly one characteristic vertex. **36.3**

type II (tree): Has two characteristic vertices. **36.3**

U

UFD: See *unique factorization domain*.

u.i.: See *unitarily invariant*.

underflow: $\text{fl}(x) = 0$ for $x \neq 0$. **37.6**

unicyclic: A graph containing precisely one cycle. **36.2**

unimodular (matrix over a domain): See $\mathbb{D}$ -*invertible*.

union (of two graphs $G = (V, E)$ and $G' = (V', E')$): The graph with vertex set $V \cup V'$, and edge (multi)set $E \cup E'$. **28.1**

unique factorization domain (UFD): Any nonzero, nonunit element a can be factored as a product of irreducible elements $a = p_1 \cdots p_r$, and this factorization is unique within order and unit factors. **23.1**

unisigned (real or sign pattern vector): Not balanced. **33.3**

unit (in a domain): An element a such that a divides 1. **23.1**

unit lower triangular matrix: A lower triangular matrix such that all diagonal entries are equal to one.

unit round: $u = \inf \{\delta > 0 \mid \text{fl}(1 + \delta) > 1\}$. **37.6**

unit upper triangular matrix: An upper triangular matrix such that all diagonal entries are equal to one. **1.2**

unit vector: A vector of length 1. **5.1**

unital (linear map $\phi : \mathbb{C}^{n \times n} \rightarrow \mathbb{C}^{m \times m}$): $\phi(I_n) = I_m$. **18.7**

unitarily equivalent: See *unitarily similar*.

unitarily invariant (vector norm $\|\cdot\|$ on $\mathbb{C}^{m \times n}$): $\|A\| = \|UAV\|$ for any unitary $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$ and any $A \in \mathbb{C}^{m \times n}$. **17.3**

unitarily similar (matrices A and B): There exists a unitary matrix U such that $B = U^*AU$. **7.1**

unitarily similarity invariant (vector norm $\|\cdot\|$ on $\mathbb{C}^{n \times n}$): $\|U^*AU\| = \|A\|$ for all $A \in \mathbb{C}^{n \times n}$ and unitary $U, V \in \mathbb{C}^{n \times n}$. **18.4**

unitary matrix: A matrix U such that $U^*U = I$. **5.2, 7.1**

unknown vector: The vector of variables of a system of linear equations. **1.4**

unreduced upper Hessenberg matrix: An upper Hessenberg matrix A such that $a_{j+1,j} \neq 0$ for $j = 1, \dots, n-1$. **43.2**

unstable (algorithm): See *numerically unstable*.

updating (QR factorization): See *Section 39.7*.

upper Hessenberg (matrix A): $a_{ij} = 0$, for all $i \geq j+2$, $1 \leq i, j \leq n$. **10.2, 43.2**

upper triangular (pencil $A - xB$): Both A and B are upper triangular. **43.1**

upper triangular matrix: A matrix such that every entry having row number greater than column number is zero. **1.2, 10.2**

V

(v, k, λ) -design: See *2-design*.

valency: See *degree*.

Vandermonde matrix: A matrix having each row equal to successive powers of a number. **48.1**

variables (of a linear equation): The unknowns. **1.4**

vec-function (of matrix A): The vector formed by stacking the columns of A on top of each other in their natural order. **10.4**

vector: An element of a vector space. **1.1**

vector norm: A real-valued function $\|\cdot\|$ on $\mathbb{R}^n$ or $\mathbb{C}^n$ such that for all vectors $\mathbf{x}, \mathbf{y}$ and all scalars α : (1) $\|\mathbf{x}\| \geq 0$, and $\|\mathbf{x}\| = 0$ implies $\mathbf{0}$; (2) $\|\alpha\mathbf{x}\| = |\alpha|\|\mathbf{x}\|$; (3) $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$. **37.1**

vector seminorm: A real-valued function $\|\cdot\|$ on $\mathbb{R}^n$ or $\mathbb{C}^n$ such that for all vectors $\mathbf{x}, \mathbf{y}$ and all scalars α : (1) $\|\mathbf{x}\| \geq 0$; (2) $\|\alpha\mathbf{x}\| = |\alpha|\|\mathbf{x}\|$; (3) $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$. **37.2**

vector space (over field F): A nonempty set V with two operations, addition and scalar multiplication, such that V is an abelian group under addition, scalar multiplication distributes over addition, scalar multiplication is associative, and the multiplicative identity of F acts as the identity on V . **1.1**

vertex: An element of the vertex set of a graph or digraph **28.1, 29.1**

vertex coloring: A partition of the vertex set of a graph into cliques. **28.5**

vertex-edge incidence matrix: See *incidence matrix*. **30.1**

vertex independence number: The largest order of a coclique in G . **28.5**

vertices: Plural of *vertex*.

Volterra–Lyapunov stable: See *Lyapunov diagonally stable*.

W

walk (in a digraph): A sequence of arcs $(v_0, v_1), (v_1, v_2), \dots, (v_{k-1}, v_k)$ (the vertices need not be distinct). **29.1**

walk (in a graph): An alternating sequence $(v_{i_0}, e_{i_1}, v_{i_1}, e_{i_2}, \dots, e_{i_\ell}, v_{i_\ell})$ of vertices and edges, not necessarily distinct, such that $v_{i_{j-1}}$ and v_{i_j} are endpoints of e_{i_j} for $j = 1, \dots, \ell$. **28.1**

walk product: $\prod_{j=1}^k a_{v_{j-1}, v_j}$ where A is a square matrix and $(v_0, v_1), (v_1, v_2), \dots, (v_{k-1}, v_k)$ is a walk in the digraph of A . **29.3**

walk-regular (graph): For every vertex v the number of walks from v to v of length ℓ , depends only on ℓ (not on v). **28.2**

weak combinatorial invariant (of a matrix): A quantity or property that does not change when the rows and columns are simultaneously permuted (by the same permutation). **27.1**

weakly majorizes: Real sequence α weakly majorizes β if for all k , the sum of the first k entries of α in decreasing order is $\geq$ the sum of the first k entries of β in decreasing order. **Preliminaries**

weakly numerically stable (algorithm): The magnitude of the forward error is roughly the same as the magnitude of the error induced by perturbing the data by small multiple of the unit round. **37.8**

weakly sign symmetric (square matrix A): $\det A[\alpha, \beta] \det A[\beta, \alpha] \geq 0$, $\forall \alpha, \beta \subseteq \{1, \dots, n\}$, $|\alpha| = |\beta| = |\alpha \cap \beta| + 1$. **19.2**

weakly unitarily invariant: See *unitarily similarity invariant*.

weighted bigraph (of matrix A): The bigraph of A with the weight of $\{i, j'\} = a_{ij}$. **30.2**

weighted bipartite graph: A simple bipartite graph with a weight function on the edges. **30.1**

weighted digraph: A digraph with a weight function on the arcs. **29.1**

weight function: A function from the edges or arcs of a graph, bipartite graph, or digraph to $\mathbb{R}^+$ **36.4, 30.1, 29.1**

weighted graph: A simple graph with a weight function. **36.4**

weighted least squares problem: See *Section 39.1*.

Wiener vertex: See *Parter–Wiener vertex*.

well-conditioned (input $\mathbf{z}$ for P): Small relative perturbations of $\mathbf{z}$ cause small relative perturbations of $P(\mathbf{z})$. **37.4**

Wilkinson's shift: The shift μ is the eigenvalue of the bottom right 2×2 submatrix of T which is closer to $t_{n,n}$. **42.3**

Z

Z-matrix (square real matrix): All off-diagonal elements are nonpositive. **9.5, 19.2**

zero completion (of a partial matrix): Obtained by setting all unspecified (off-diagonal) entries to 0 (partial matrix must have all diagonal entries specified). **35.6**

zero divisor (in a ring R): A nonzero element $a \in R$ such that there exists a nonzero $b \in R$ with $ab = 0$ or $ba = 0$. **Preliminaries**

zero-free diagonal (property of matrix A): All the diagonal elements of A are nonzero. **40.3**

zero line (in a matrix): A line of all zeros. **27.1**

zero matrix: A matrix with all entries equal to zero. **1.2**

(zero) pattern (of a matrix): See *pattern*.

zero pattern (of sign pattern A): The $(0, +)$ -pattern obtained by replacing each $-$ entry in A by a $+$. **33.2**

zero transformation: A linear transformation that maps every vector to zero. **3.1**

Notation Index

This notation index covers most of the terms defined in Chapters 1 to 49. It does not cover some terminology used in a single section (including most of the terminology that is specific to a particular application (Chapters 50 to 70)), nor does it cover most of the terminology used in computer software (Chapters 71 to 77).

Notation is in “alphabetical” order. If you are looking for something done to a matrix, like the transpose, look under A . If you are looking for something done to a field, look under F . If you are looking for something done to a linear transformation, look under T . If you are looking for something done to a vector or vector space, look under V .

The meaning of a symbol depends on what it is applied to; e.g., $\rho(A)$, where A is a matrix, is the spectral radius of A , whereas $\rho(G)$, where G is a group, is a representation of the group.

Warning: Tilde and hat, as in $\tilde{A}$, $\hat{Q}$, frequently have the meanings *perturbation* and *reduced*, respectively, but are also redefined in some chapters to mean other things.

For most symbols, the section where the symbol is introduced is listed at the end of the definition.

$\mathbf{0}_{mn}$	the zero matrix (in $F^{m \times n}$), can be shortened to $\mathbf{0}$. 1.2
$\mathbf{1}_n$	all 1s vector in F^n can be shortened to $\mathbf{1}$.
$A = [a_{ij}]$	matrix A and its elements. 1.2
A^T	transpose of matrix A . 1.2
$\overline{A}$	complex conjugate of matrix A . 1.2
A^*	Hermitian adjoint (conjugate-transpose) of matrix A . 1.2
A^{-1}	inverse of square matrix A . 1.5
$A^\dagger$	Moore–Penrose pseudo-inverse of matrix A . 5.7
$A^\#$	group inverse of square matrix A . 9.1
$A[\alpha, \beta]$	submatrix of A with row indices in α and column indices in β . 1.2, 10.1
$A[\alpha]$	$= A[\alpha, \alpha]$, principal submatrix, also denoted $A[1, 2, 3]$ for $A[\{1, 2, 3\}]$. 1.2, 10.1
$A(\alpha, \beta)$	submatrix of A with row indices not in α and column indices not in β . 1.2, 10.1
$A(\alpha)$	$= A(\alpha, \alpha)$, principal submatrix with row and column indices not in α . 1.2, 10.1
$A_{i:k,j:l}$	submatrix $A[\{i, i+1, \dots, k\}, \{j, j+1, \dots, l\}]$, analogously $A_{i,j:l}$, $A_{i:k,j}$.
$(A)_{ij}$	i, j -entry of A . 1.2
A_{ij}	i, j -block in a block matrix of A . 10.1
$A/A[\alpha]$	Schur complement of $A[\alpha]$. 4.2, 10.3
$ A $	matrix having as entries the absolute values of the entries of matrix A . 9.1
$ A _{pd}$	spectral absolute value $(A^*A)^{1/2}$. 17.1
$ A $	zero pattern of sign pattern A . 33.2
$\sqrt{A}$	a square root of a square complex matrix A . 11.2
$A^{1/2}$	the principal square root of a square complex matrix A . 11.2
$\tilde{A}^{1/2}$	the positive semidefinite square root of a positive semidefinite matrix A . 8.3
$\tilde{A}$	a perturbation $A + \Delta A$ of matrix A . 15.1 (also other meanings)
$\ A\ _k$	operator norm of matrix A (induced by $\ \mathbf{v}\ _k$). 37.3
$\ A\ _F$	Frobenius (Euclidean) norm of matrix A . 7.1, 37.3
$\ A\ _{K,k}$	Ky Fan k norm. 17.3
$\ A\ _{S,p}$	Schatten- p norm. 17.3
$\ A\ _{tr}$	trace norm (not same as Frobenius norm). 17.3

$\ A\ _{UI}$	unitarily invariant matrix norm. 17.3
$A > 0$	matrix A is positive. 9.1
$A \geq 0$	matrix A is nonnegative. 9.1
$A \succeq 0$	matrix A is semipositive. 9.1
$A > B$	$A - B$ is positive.
$A \geq B$	$A - B$ is nonnegative.
$A \succeq B$	$A - B$ is semipositive.
$A \succeq^K 0$	A is K -nonnegative. 26.1
$A >^K 0$	A is K -positive. 26.1
$A \succeq^K 0$	A is K -semipositive. 26.1
$A \succeq^K B$	$A - B$ is K -nonnegative. 26.1
$A >^K B$	$A - B$ is K -positive. 26.1
$A \succeq^K B$	$A - B$ is K -semipositive. 26.1
$A \succ 0$	matrix A is positive definite. 8.4
$A \succeq 0$	matrix A is positive semidefinite. 8.4
$A \succ B$	$A - B$ is positive definite. 8.4
$A \succeq B$	$A - B$ is positive semidefinite. 8.4
$\hat{A} \preceq A$	sign pattern $\hat{A}$ is a subpattern of sign pattern A . 33.1
$A \stackrel{c}{\sim} B$	A and B are *-congruent. 8.3
$A \sim_c B$	A and B are column equivalent (in a domain). 23.2
$A \sim_r B$	A and B are row equivalent (in a domain). 33.2
$A \sim B$	A and B are equivalent (in a domain). 23.2
$[A b]$	augmented matrix. 1.4
$A + B$	sum of matrices A and B . 1.2
AB	matrix product of matrices A and B . 1.2
$A \oplus B$	direct sum (block diagonal matrix) of matrices A and B . 2.4, 10.2
$A \circ B$	Hadamard product of A and B . 8.5
$A \otimes B$	Kronecker product of A and B . 10.4
$\mathcal{A}_G$	adjacency matrix of graph G , can be shortened to $\mathcal{A}$. 28.3
$\mathcal{A}_\Gamma$	adjacency matrix of digraph Γ , can be shortened to $\mathcal{A}$. 29.2
$\mathcal{A}(R, S)$	the class of all $(0, 1)$ -matrices with row sum vector R and column sum vector S . 27.4
$A^m(V; U)$	subset of $L^m(V; U)$ consisting of the antisymmetric maps. 13.5
$a b$	a divides b (in a domain). 23.1
$a \equiv b$	a, b are associates (in a domain). 23.1
$\{\{a\}\}$	the set of associates (equivalence class) of a (in a domain). 23.1
$(a_1, \dots, a_n)$	any g.c.d. of $a_1, \dots, a_n$ (in a domain). 23.1
$\{(a_1, \dots, a_n)\}$	the equivalence class of all g.c.d.s of $a_1, \dots, a_n$ (in a domain). 23.1
α^c	set complement of α in $1, \dots, n$. Preliminaries
$\alpha \downarrow$	permutation of real number sequence α , with entries in nonincreasing order. Preliminaries
$\alpha \uparrow$	permutation of real number sequence α , with entries in nondecreasing order. Preliminaries
$\alpha \succeq \beta$	real number sequence α majorizes β . Preliminaries
$\alpha \succeq_w \beta$	natural number sequence α weakly majorizes β . Preliminaries
$\alpha_\varepsilon(A)$	ε -pseudospectral abscissa of complex matrix A . 16.3
$\alpha(G)$	algebraic connectivity of graph G . 36.1
$\hat{\alpha}(G)$	absolute algebraic connectivity of G . 36.1
$\alpha(\lambda)$ or $\alpha_A(\lambda)$	algebraic multiplicity of eigenvalue λ (of A). 4.3
$\alpha(m, n)$	$\max\{\det W^T W W \in \{\pm 1\}^{m \times n}\}$. 32.1
$\alpha(n)$	$= \alpha(n, n)$. 32.1

$\alpha \triangleleft \beta$	partial order on partitions of n . 31.10
$\text{adj } A$	adjugate (classical adjoint) of matrix A , also denoted $\text{adj}(A)$. 4.2
$\text{Alt}(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_k)$	$= \frac{1}{k!} \sum_{\pi \in S_m} \text{sgn}(\pi) \mathbf{v}_{\pi(1)} \otimes \cdots \otimes \mathbf{v}_{\pi(k)}$. 13.6
B_0	zero completion of partial matrix B . 35.6
$\mathcal{B}$	basis $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ or ordered basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ for a vector space. 2.2
$\mathcal{B}^*$	dual basis (for dual space V^*) determined by basis $\mathcal{B}$ of V . 3.8
$\mathcal{B}_G$	biadjacency matrix of a bipartite graph G , cannot be shortened. 30.1
b_λ	number of Jordan blocks with eigenvalue λ . 6.2
$B(V, V, F)$	vector space of bilinear forms on vector space V over field F . 12.1
$B(V, V, F, \varphi)$	vector space of φ -sesquilinear forms (φ is an automorphism of F). 12.4
$B(X, Y)$	biclique of between $X, Y \subset V$ in graph $G = (V, E)$. 30.3
$\beta(m, n)$	$\max\{\det W^T W \mid W \in \{0, 1\}^{m \times n}\}$. 32.1
$\beta(n)$	$= \beta(n, n)$. 32.1
$\text{bc}(G)$	biclique cover number of graph G . 30.3
$\text{bp}(G)$	biclique partition number of graph G . 30.3
$\mathbb{C}$	complex numbers. Preliminaries
$\mathbb{C}^+$	the open right half plane. Preliminaries
$\mathbb{C}_0^+$	the closed right half plane. Preliminaries
$\mathbb{C}^-$	the open left half plane. Preliminaries
$\mathbb{C}_0^-$	the closed left half plane. Preliminaries
$\bar{c}$	complex conjugate of the complex number c . Preliminaries
$ c $	the absolute value of real or complex number c . Preliminaries
c_{ij}	i, j th cofactor of a matrix. 4.1
C_n	cycle on n vertices. 28.1
C_n	n -cycle matrix. 48.1
$C_{n,g}$	graph formed by appending C_g to a pendent vertex of P_{n-g} . 36.2
$c(A)$	the number of lines in a minimum line cover of A . 27.1
$c^*(A)$	the number of 1s in a minimum co-cover of A . 27.1
$c_i(A)$	Euclidean length of a column of A (the i th length in nonincreasing order). 17.1
$C_k(A)$	k th compound matrix of A . 4.2
$C(\mathbf{x}, \mathbf{y})$	Cauchy matrix. 48.1
$C(p(x))$	companion matrix of polynomial $p(x)$. 4.3
$\text{Con}(S)$	convex hull of set S of vectors. Preliminaries
$\text{cond}(\lambda)$	individual condition number of eigenvalue λ . 15.1
$\cos(A)$	cosine of matrix A . 11.5
$\chi(G)$	chromatic number of graph G . 28.5
$\chi(a, b)$	measure of relative separation of real numbers a, b . 17.4
$\mathbb{D}$	an integral domain. 23.1
$\mathbb{D}_b$	a Bezout domain. 23.1
$\mathbb{D}_g$	a greatest common divisor domain. 23.1
$\mathbb{D}_e$	a Euclidean domain. 23.1
$\mathbb{D}_{ed}$	an elementary divisor domain. 23.1
$\mathbb{D}_p$	a principal ideal domain. 23.1
$\mathbb{D}_u$	a unique factorization domain. 23.1
$\mathbb{D}^{m \times n}$	$m \times n$ matrices over domain $\mathbb{D}$. 23.2
$D_{g,n-g}$	graph formed by $n - g$ isolated vertices to a single vertex of C_g . 36.2
$\mathbb{D}[\mathbf{x}] = \mathbb{D}[x_1, \dots, x_n]$	ring of polynomials $p(\mathbf{x}) = p(x_1, \dots, x_n)$ with coefficients in $\mathbb{D}$. 23.1
$D(A)$	signed digraph of sign pattern A . 33.1
$\mathcal{D}(B)$	digraph of partial matrix B . 35.1
$\Delta(G)$	$\max\{p - q : q \text{ vertices may be deleted from graph } G \text{ leaving } p \text{ paths}\}$. 34.2

$\frac{\partial f}{\partial x}$	the partial derivative of F with respect to x .
$\partial(S)$	the boundary of a set S of real or complex numbers. Preliminaries
$\partial(V')$	number of edges of a graph having one endpoint inside vertex subset V' and other outside. 28.5
δ_{ij}	Kronecker delta, i.e., 1 if $i = j$ and 0 otherwise.
$\delta(A)$	the zero part of the inertia of complex square matrix A . 8.3, 19.1
$\delta_G(v)$	degree of vertex v in graph G , also denoted $\delta(v)$. 28.1
Δ_n^k	the set of $n \times n$ matrices of nonnegative integers with each row and column sum = k . 31.4
$d(\mathbf{v}, \mathbf{u})$	distance from vector $\mathbf{v}$ to vector $\mathbf{u}$ in a normed vector space. 5.1
$d_G(v, u)$	distance from vertex v to vertex u in graph G , also denoted $d(v, u)$. 28.1
$d(T)$	diameter of tree T , same as $\text{diam}(T)$. 34.3
$\deg(p)$	degree of polynomial $p(x)$. 23.1
$\det A$	determinant of matrix A , also denoted $\det(A)$. 4.1
$\text{diag}(d_1, \dots, d_n)$	$n \times n$ diagonal matrix with listed diagonal entries. 1.2
$\text{diam}(G)$	diameter of graph. 28.1
$\dim \mathcal{P}$	dimension of convex polytope $\mathcal{P}$, also denoted $\dim(\mathcal{P})$. 27.6
$\dim V$	dimension of vector space V , also denoted $\dim(V)$. 2.2
ϵ	machine precision (i.e., machine epsilon). 37.6
$\mathbf{e}_i$	i th standard basis vector (1 in i th coordinate, 0s elsewhere). 2.1
E_{ij}	standard basis matrix (1 in i, j th coordinate, 0s elsewhere). 2.1
$\mathcal{E}_n$ or $\mathcal{E}$	standard basis $\mathbf{e}_1, \dots, \mathbf{e}_n$ for F^n . 2.1
ε_n	the identity in the symmetric group S_n . Preliminaries
E_λ or $E_\lambda(A)$	the eigenspace for eigenvalue λ (of A). 4.3
$E(A, \alpha + \beta i)$	the real generalized eigenspace of real matrix A for eigenvalue $\alpha + \beta i$. 6.3
$E_k(\mu)$	a particular lower elementary bidiagonal matrix. 21.2
e^A	exponential function of matrix A . 11.3
$\eta(G)$	a graph parameter involving minimum rank. 28.5
$\eta_p(A, \mathbf{b}; \tilde{\mathbf{x}})$	relative backward error. 38.1
End V	algebra of linear operators on vector space V . 69.1
$\exp(A)$	exponential function of matrix A , same as e^A . 11.3
$f g$	polynomial $f(x)$ divides $g(x)$ in $F[x]$. 20.2
$\mathbb{F}_q$	finite field with q elements. 61
$F^{m \times n}$	$m \times n$ matrices over the field F . 1.2
F^n	F -vector space of column n -tuples of elements of F . 1.1
$F[x]$	polynomials over F . 1.1
$F[\mathbf{x}] = F[x_1, \dots, x_n]$	ring of polynomials $p(\mathbf{x}) = p(x_1, \dots, x_n)$ with coefficients in F . 23.1
$F(\mathbf{x})$	field of rational functions in $x_1, \dots, x_n$ over F . 23.1
$F[x; n]$	polynomials of degree $\leq n$ over F .
$\mathcal{F}(A)$	all doubly stochastic matrices that have 0's at least wherever the $(0,1)$ -matrix A has 0s. 27.6
$F \vee G$	the join $\Phi(F \cup G)$ (where (F, G) are cone faces). 26.1
$F \wedge G$	the meet $F \cap G$ (where (F, G) are cone faces). 26.1
$\Phi(G)$	conductance (isoperimetric number) of a graph. 28.5
$\Phi(S)$	face generated by S . 26.1
$\Phi(\mathbf{x})$	face generated by $\{\mathbf{x}\}$. 26.1
fix T	fixed space of linear transformation T . 3.6
G	group. Preliminaries
G or $G = (V_G, E_G)$	graph = (vertices, edges), also denoted $G = (V, E)$. 28.1
Γ or $\Gamma = (V_\Gamma, E_\Gamma)$	digraph = (vertices, arcs), also denoted $\Gamma = (V, E)$. 29.1
$\overline{G}$	graph complement. 28.1

$G - X$	subgraph induced by $V \setminus X$. 28.1
$G \cup H$	union of graphs G, H . 28.1
$G \cap H$	intersection of graphs G, H . 28.1
$G + H$	join of graphs G, H . 28.1
$G \cdot H$	(strong) product of graphs G, H . 28.1
G^ℓ	the strong product of ℓ copies of graph G . 28.1
$G - v$	the result of deleting vertex v from graph G . 28.1
G_w	a weighted graph. 36.4
$G(A)$	(simple) graph of square matrix A . 19.3, 34.1
$\mathcal{G}(B)$	graph of partial matrix B . 35.1
$\mathcal{G}(S)$	bipartite graph of sparsity pattern of S . 46.3
G^F	the fill graph, digraph, or bigraph of real square matrix A . 40.5
$G^+(A)$	bipartite fill-graph of square matrix A . 30.2
$G(\mathcal{P})$	simple graph of convex polytope $\mathcal{P}$. 27.6
$\Gamma(A)$	digraph of square matrix A (may have loops). 9.1, 29.2
$\mathcal{G}(v, \delta)$	the set of all δ -regular graphs on v vertices. 32.7
$G(i, j, \theta, \vartheta)$	Givens transformation. 38.4
$\gcd(a_1, \dots, a_k)$	greatest common divisor of $(a_1, \dots, a_k)$.
$\text{GL}(n, F)$	general linear group of order n over F . 67.1
$\text{GL}(n, \mathbb{D})$	the group of $\mathbb{D}$ -invertible matrices in $\mathbb{D}^{n \times n}$. 23.2
$\gamma(\lambda)$ or $\gamma_A(\lambda)$	geometric multiplicity of eigenvalue λ (of A). 4.3
H_n	Hadamard matrix of size n . 32.2
$\mathcal{H}_n$	the set of Hermitian matrices of size n . 8.1
$\mathcal{H}(G)$	the set of Hermitian matrices A such that $G(A) = G$. 34.1
$H(\Omega)$	the ring of analytic functions over a nonempty path-connected set $\Omega \subset \mathbb{C}^n$. 23.1
H_ζ	$= H(\{\zeta\})$. 23.1
I_n or I	identity matrix or transformation (in $F^{n \times n}$ or $L(V, V)$). 1.2, 3.1
$B'[I]_B$	change-of-basis (transition) matrix from basis B to basis B' . 2.6
$i \mapsto j$	vertex i has access to vertex j (in a digraph). 9.1
$\iota(G)$	the vertex independence number of graph G . 28.5
$\text{im}(c)$	the imaginary part of complex number c . Preliminaries
$\text{im}(f)$	image of the function $f (= \text{range}(f))$ if f is a linear transformation)
$\text{in}(A) = (\pi, \nu, \delta)$	inertia of complex square matrix A . 8.3, 19.1
$\text{int}K$	interior of K . 26.1
$ip(A)$	the invariant polynomials (i.e., invariant factors) of matrix A . 20.2
J_A	Jordan canonical form of matrix A . 6.2
$J_A^{\mathbb{R}}$	real-Jordan canonical form of real matrix A . 6.3
J_{mn}	all 1s matrix in $F^{m \times n}$, can be shortened to J_n or J when $m = n$.
$J_k(\lambda)$	Jordan block of size k for λ . 6.2
$J_{2k}^{\mathbb{R}}(\lambda)$	real-Jordan block of size $2k$ for λ . 6.3
$\text{JCF}(A)$	Jordan canonical form of matrix A . 6.2
$\text{JCF}^{\mathbb{R}}(A)$	real-Jordan canonical form of matrix A . 6.3
K^*	dual space (dual cone) of cone K . 8.5, 26.1
K_n	complete graph on n vertices. 28.1
$K_{n,m}$	complete bipartite graph on n and m vertices. 28.1
$\mathcal{K}_k(A, \mathbf{v})$	Krylov subspace of dimension k for matrix A and vector $\mathbf{v}$. 44.1
$\kappa(A), \kappa_p(A)$	condition number of matrix A for linear system $\mathbf{Ax} = \mathbf{b}$ (in p norm). 37.5
$\kappa_v(G)$	vertex connectivity of graph G . 36.1
$\kappa_e(G)$	edge connectivity of graph G . 36.1
$\ker A$	kernel of matrix A , also denoted $\ker(A)$. 2.4

$\ker T$	kernel of linear transformation T , also denoted $\ker(T)$. 3.5
λ	eigenvalue of a matrix or transformation (can also use other Greek). 4.3
Λ	diagonal matrix of eigenvalues. 15.1
Λ_n^k	the set of $n \times n$ binary matrices in which each row and column sum is k . 31.3
L_G	the Laplacian matrix of graph G . 28.4
$L(G_w)$	Laplacian matrix of weighted graph G_w . 36.4
$ L_G $	the signless Laplacian matrix of graph G . 28.4
$L(G)$	the line graph of graph G . 28.2
$L(G; a_1, \dots, a_n)$	a generalized line graph. 28.2
$\ell_k(x)$	Laguerre polynomial. 31.8
$L(V, W)$	the set of all linear transformations of V into W . 3.1
$L(V_1, \dots, V_m; U)$	the set of all multilinear maps from $V_1 \times \dots \times V_m$ into U with operations. 13.1
$L^m(V; U)$	$L(V_1, \dots, V_m; U)$ with $V_i = V$ for $i = 1, \dots, m$. 13.5
$\text{lcm}(a_1, \dots, a_n)$	least common multiple of $a_1, \dots, a_n$.
$\lim_{m \rightarrow \infty} a_m = a(C, 0)$	$\lim_{m \rightarrow \infty} a_m = a$. 9.1
$\lim_{m \rightarrow \infty} a_m = a(C, k)$	(C, k) limit. 9.1
$\log(A)$	principal logarithm of matrix A . 11.4
$M(A)$	the comparison matrix of A . 19.5
$m(G)$	the maximum rank deficiency of graph G . 34.2
μ_i	i th eigenvalue of the Laplacian matrix of a graph (nondecreasing order). 28.4
$\mu(G)$	Colin de Verdière parameter of graph G . 28.5
$M(G)$	maximum (eigenvalue) multiplicity of graph G . 34.2
$\mathcal{M}(G)$	set of matrices A such that bigraph of A is a subgraph of G . 30.3
$\mathcal{M}(\Omega)$	the quotient field of $H(\Omega)$. 23.1
$\mathcal{M}_\zeta$	$= \mathcal{M}(\{\zeta\})$. 23.1
$\text{MR}(A)$	maximal rank of sign pattern A . 33.6
$\text{mr}(A)$	minimal rank of sign pattern A . 33.6
$\text{mr}(G)$	minimum rank of graph G . 34.2
$\mathbb{N}$	natural numbers, i.e., $\{0, 1, 2, \dots\}$.
$\langle n \rangle$	the set of integers $\{1, 2, \dots, n\}$. 9.1
$N_\lambda^k(A)$	the k th generalized eigenspace of square matrix A . 6.1
$N_\lambda^v(A)$	the generalized eigenspace of matrix A , can be abbreviated $N_\lambda(A)$. 6.1
$\mathcal{N}(T)$	the minimum number of distinct eigenvalues of a matrix in $\mathcal{S}(T)$ where T is a tree. 34.3
$\nu(A)$	the negative part of the inertia of complex square matrix A . 8.3, 19.1
$\nu_A(\lambda)$	the index of matrix A at eigenvalue λ . 6.1
ν_P	the index of P , i.e. $\nu_P(\rho)$. 9.3
$\bar{\nu}_P$	the co-index of P . 9.3
N_G	the (vertex-edge) incidence matrix of graph G . 28.4
$\text{nnz}(A)$	the number of nonzero entries in sparse matrix A . 40.2
null A	nullity of a matrix, also $\text{null}(A)$. 2.4
null T	nullity of a linear transformation or of a matrix, also $\text{null}(T)$. 3.5
$O(f)$	big-oh of function f . Preliminaries
$o(f)$	little-oh of function f . Preliminaries
Ω_n	set of $n \times n$ doubly stochastic matrices. 9.4
$\omega(G)$	clique number of graph G . 28.4
P_n	path on n vertices. 28.1
$P[\lambda]$	principal submatrix at a distinguished eigenvalue. 9.3
$P(G)$	path cover number of graph G . 34.2
$P(n, d)$	tree constructed in a specific way. 36.3
$\pi(A)$	the positive part of the inertia of complex square matrix A . 8.3, 19.1

Π_n	regular n —sided unit polygon. 20.3
$p_A(x)$	characteristic polynomial of matrix A , or linear transformation. 4.3
$p_G(x)$	characteristic polynomial of graph G . 28.3
PD_n	the set of $n \times n$ positive definite matrices, can be shortened to PD. 8.4
per A	permanent of matrix A , also denoted $\text{per}(A)$. 31.1
$\text{per}_k A$	the sum of the subpermanents of order k , also denoted $\text{per}_k(A)$. 31.1
$\text{proj}_W(\mathbf{v})$	the orthogonal projection of $\mathbf{v}$ onto W (along $W^\perp$). 5.4
$\text{proj}_{W,Z}(\mathbf{v})$	the projection of $\mathbf{v}$ onto W along Z . 3.6
PSD_n	the set of $n \times n$ positive semidefinite matrices, can be shortened to PSD. 8.4
$\mathbb{Q}$	rational numbers.
$Q(A)$	qualitative class of sign pattern A (or of $\text{sgn}(A)$ if A is a real matrix). 33.1
$q_A(x)$	minimum polynomial of matrix A , or linear transformation. 4.3
$Q_{r,n}$	the set of all strictly increasing sequences of r integers chosen from the set $\{1, 2, \dots, n\}$. 23.2
QR	QR-factorization of a real or complex matrix. 5.5
$\hat{Q}\hat{R}$	reduced QR-factorization of a real or complex matrix. 5.5
$\rho(A)$	spectral radius of real or complex square matrix A . 4.3
$\rho_\varepsilon(A)$	ε -pseudospectral radius of complex matrix A . 16.3
$q(A)$	term rank of matrix A . 27.1
$q^*(A)$	largest size of a zero submatrix of A . 27.1
$\rho(G)$	representation of group G . 68.1
$\rho_1(A, x), \rho_2(A, x)$	rook polynomials. 31.8
$\mathbb{R}$	real numbers.
$\mathbb{R}^+$	positive reals (subset of $\mathbb{R}$ or $\mathbb{C}$).
$\mathbb{R}_0^+$	nonnegative reals (subset of $\mathbb{R}$ or $\mathbb{C}$).
$\mathbb{R}^-$	negative reals (subset of $\mathbb{R}$ or $\mathbb{C}$).
$\mathbb{R}_0^-$	nonpositive reals (subset of $\mathbb{R}$ or $\mathbb{C}$).
$\mathbb{R}_{\max}$	max-plus semiring. 25.1
$\overline{\mathbb{R}}_{\max}$	completed max-plus semiring. 25.1
$R(\Gamma)$	reduced digraph of Γ . 9.1, 29.5
$R^*(P)$	basic reduced digraph of matrix P . 9.3
$\mathbb{R}[x; n]$	polynomials of degree $\leq n$ over $\mathbb{R}$.
$r_i(A)$	Euclidean length of a row of A (the i th length in nonincreasing order). 17.1
$\text{RCF}_{ED}(A)$	elementary divisors rational canonical form of A . 6.4
$\text{RCF}_{IF}(A)$	invariant factors rational canonical form of A . 6.6
$\text{re}(c)$	the real part of complex number. c
range A	range of matrix A , = column space of A , also denoted $\text{range}(A)$. 2.4
range T	range of linear transformation T , also denoted $\text{range}(T)$. 3.5
rank A	rank of matrix A , also denoted $\text{rank}(A)$. 1.3, 2.4
$\text{rank}(f)$	rank of bilinear form f . 12.1
rank T	rank of linear transformation T , also denoted $\text{rank}(T)$. 3.5
REF(A)	a row echelon form of matrix A . 1.3
RREF(A)	the reduced row echelon form of matrix A . 1.3
$\text{reldist}(H, \tilde{H})$	component-wise relative distance between $H, \tilde{H}$. 46.4
$RS(A)$	row space of matrix A . 2.4
S^a	annihilator (in the dual space V^*) of subspace S of vector space V . 3.8
$S \setminus X$	set complement of X in S . Preliminaries
S_n	the symmetric group on $\{1, \dots, n\}$. Preliminaries
$\mathcal{S}_n$	the set of real symmetric matrices of size n . 8.1
σ_i	i th singular value (in nonincreasing order). 5.6
$\sigma(A)$	spectrum of square matrix A . 4.3

$\sigma(G)$	spectrum of graph G . 28.3
$\sigma_\varepsilon(A)$	ε -pseudospectrum of complex square matrix A . 16.1
$\sigma_\varepsilon^{\mathbb{R}}(A)$	real structured ε -pseudospectrum of real square matrix A . 16.5
$\sigma_\varepsilon(A, B)$	ε -pseudospectrum of the matrix pencil $A - xB$. 16.5
$\sigma_\varepsilon(A, B, C)$	spectral value set of the matrix triplet A, B, C . 16.5
$\sigma_\varepsilon(P)$	ε -pseudospectrum of matrix polynomial P . 16.5
$ S $	cardinality of set S .
$S_k(A)$	sum of all principal minors of size k of matrix A . 4.2
$\mathcal{S}(G)$	the set of real symmetric matrices A such that $G(A) = G$. 34.1
$S^m(V; U)$	subset of $L^m(V; U)$ consisting of the symmetric maps. 13.5
$S_k(\alpha_1, \dots, \alpha_n)$	k th elementary symmetric function of $\alpha_i, i = 1, \dots, n$. Preliminaries
$S(u)$	$\sum_{k \in V} (d(u, k))^2$ where $G = (V, E)$ is a simple graph. 36.5
$\text{sep}(A_1, A_2)$	separation between matrices A_1, A_2 . 15.1
$\text{sgn}(\pi)$	sign of the permutation π . Preliminaries
$\text{sgn}(a)$	sign of the real number a as used in sign patterns, one of $+, 0, -$. Preliminaries
$\text{sgn}(A)$	sign pattern matrix of the real matrix A , with entries in $+, 0, -$. 33.1
$\text{sgn}(\alpha)$	sign of a set α of edges in a signed bipartite graph. 30.1
$\text{sign}(A)$	matrix sign function of the complex square matrix A . 11.6
$\text{sign}(z)$	sign of the complex number z as used in numerical linear algebra, always nonzero. Preliminaries
$\text{SL}(n, F)$	special linear group of order n over F . 67.1
$\text{SMR}(A)$	symmetric maximal rank of sign pattern A . 33.6
$\text{smr}(A)$	symmetric minimal rank of sign pattern A . 33.6
$\text{Span}(S)$	span of the set S of vectors. 2.1
$\text{Struct}(A)$	the sparsity structure (support) of A . 9.6, 40.3
$\text{sv}(A)$	vector of singular values of complex matrix A . 15.2, 17.1
$\text{sv}_{\text{ext}}(A)$	extended vector of singular values of complex matrix A . 15.2
$\text{Sym}(\mathbf{v}_1 \otimes \dots \otimes \mathbf{v}_k)$	$= \frac{1}{m!} \sum_{\pi \in S_m} \mathbf{v}_{\pi(1)} \otimes \dots \otimes \mathbf{v}_{\pi(k)}$. 16.6
T	linear transformation T . 3.1
T^{-1}	inverse of linear transformation T . 3.7
T^T	transpose of linear transformation T . 3.8
${}_{\mathcal{B}'}[T]_{\mathcal{B}}$	matrix of T with respect to bases $\mathcal{B}$ (input) and $\mathcal{B}'$ (output). 3.3
$[T]_{\mathcal{B}}$	matrix of T with respect to bases $\mathcal{B}$ and $\mathcal{B}$ (same as ${}_{\mathcal{B}}[T]_{\mathcal{B}}$). 3.3
$[T]$	for $T: F^n \rightarrow F^m$, matrix of T with respect to the standard bases, $[T] = {}_{\mathcal{E}_m}[T]_{\mathcal{E}_n}$. 3.3
T_A	linear transformation associated to matrix A , $T_A(\mathbf{v}) = A\mathbf{v}$. 3.3
T_r^n	triples of subsets of $\{1, \dots, n\}$ of cardinality r . 17.6
$T(k, l, d)$	tree constructed from P_d by appending k and l isolated vertices to the ends. 36.3
$\tau(A)$	the ergodicity coefficient of matrix A . 9.1
$\mathcal{T}(R)$	the class of all tournament matrices with score vector R . 27.5
$\vartheta(G)$	Lovász parameter of graph G . 28.5
$\Theta(G)$	Shannon capacity of graph G . 28.5
$\Theta(X, Y)$	canonical angle matrix between X and Y . 15.1
$\text{tr } A$	trace of matrix A , also denoted $\text{tr}(A)$. 1.2
$\mathcal{U}(T)$	the minimum number of simple eigenvalues of a matrix in $\mathcal{S}(T)$ where T is a tree. 34.4
$U\Sigma V^*$	singular value decomposition of a real or complex matrix. 5.6
$\hat{U}\hat{\Sigma}\hat{V}^*$	reduced singular value decomposition of a real or complex matrix. 5.6
V	vector space V . 1.1
V^*	dual space of V . 3.8

V^{**}	bidual space of V (dual of the dual). 3.8
$V_1 \times \cdots \times V_k$	external direct sum of vector spaces V_i . 2.3
$V_1 \otimes \cdots \otimes V_k$	tensor product of vector spaces V_i . 13.2
$\bigotimes^m V$	$V \otimes \cdots \otimes V$ (m copies of V). 13.2
$\bigotimes V$	the tensor algebra on V . 13.9
$\bigvee^m V$	the symmetric space of degree m . 13.6
$\bigvee V$	the symmetric algebra on V . 13.9
$\bigwedge^m V$	the Grassman (exterior) space of degree m . 13.6
$\bigwedge V$	the Grassman algebra on V . 13.9
V_W	an $(n-1) \times (n-1)$ $(0, 1)$ -matrix constructed from an $n \times n$ ± 1 -matrix. 32.2
V/W	quotient space of V by subspace W . 2.3
$\mathbf{v}$	vector $\mathbf{v}$. 1.1
$\tilde{\mathbf{v}}$	a perturbation $\mathbf{v} + \Delta\mathbf{v}$ of vector $\mathbf{v}$. 15.1 (Also other meanings.)
$\langle \mathbf{v}, \mathbf{u} \rangle$	inner product of vectors $\mathbf{v}$ and $\mathbf{u}$. 5.1
$\ \mathbf{v}\ $	norm of vector $\mathbf{v}$ (which norm depends on context). 5.1, 37.1
$\ \mathbf{v}\ _2$	Euclidean norm of vector $\mathbf{v}$ in $\mathbb{R}^n$ or $\mathbb{C}^n$, = standard inner product norm. 5.1, 37.1
$\ \mathbf{v}\ _\infty$	∞ -norm (maximum of absolute values) of vector $\mathbf{v}$ in $\mathbb{R}^n$ or $\mathbb{C}^n$. 37.1
$\ \mathbf{v}\ _1$	1-norm (absolute column sum) of vector $\mathbf{v}$ in $\mathbb{R}^n$ or $\mathbb{C}^n$. 37.1
$\ \mathbf{v}\ _{\text{UI}}$	unitarily invariant norm. 15
$[\mathbf{v}]_{\mathcal{B}}$	coordinate vector of $\mathbf{v}$ with respect to basis $\mathcal{B}$. 2.6
$\mathbf{v} \perp \mathbf{w}$	$\mathbf{v}$ is orthogonal to $\mathbf{w}$. 5.2
$\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_k$	tensor product of vectors $\mathbf{v}_i$. 13.2
$\mathbf{v}_1 \vee \cdots \vee \mathbf{v}_k$	symmetric product of vectors $\mathbf{v}_i$. 13.6
$\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k$	exterior product of vectors $\mathbf{v}_i$. 13.6
$\mathbf{v} \geq \mathbf{0}$	$\mathbf{v}$ is nonnegative. 9.1
$\mathbf{v} > \mathbf{0}$	$\mathbf{v}$ is positive. 9.1
$\mathbf{v} \succeq \mathbf{0}$	$\mathbf{v}$ is semi-positive. 9.1
$\mathbf{v} \succeq \mathbf{w}$	$\mathbf{v} - \mathbf{w}$ is nonnegative.
$\mathbf{v} > \mathbf{w}$	$\mathbf{v} - \mathbf{w}$ is positive.
$\mathbf{v} \succeq \mathbf{w}$	$\mathbf{v} - \mathbf{w}$ is semi-positive.
$\mathbf{v} \succeq^K \mathbf{0}$	$\mathbf{v}$ is K -nonnegative. 26.1
$\mathbf{v} >^K \mathbf{0}$	$\mathbf{v}$ is K -positive. 26.1
$\mathbf{v} \succeq^K \mathbf{0}$	$\mathbf{v}$ is K -semipositive. 26.1
$\mathbf{v} \succeq^K \mathbf{w}$	$\mathbf{v} - \mathbf{w}$ is K -nonnegative. 26.1
$\mathbf{v} >^K \mathbf{w}$	$\mathbf{v} - \mathbf{w}$ is K -positive. 26.1
$\mathbf{v} \succeq^K \mathbf{w}$	$\mathbf{v} - \mathbf{w}$ is K -semipositive. 26.1
$\text{vec } A$	the vector of columns of A . 10.4
$W^\perp$	orthogonal complement of subspace W . 5.2
$W_1 \oplus \cdots \oplus W_k$	direct sum of subspaces W_i . 2.3
$W_1 + \cdots + W_k$	sum of subspaces W_i . 2.3
W_V	an $(n+1) \times (n+1)$ ± 1 -matrix constructed from an $n \times n$ $(0, 1)$ -matrix. 32.2
$W(A)$	numerical range of A . 7.1, 18.1
$w(A)$	numerical radius of complex square matrix A . 18.1
$\tilde{w}(A)$	distance of $W(A)$ to the origin. 18.1
$\vec{X}$	characteristic vector of $X \subseteq V$ where $G = (V, E)$. 30.3
$\mathbf{x}_{\text{LS}}$	least squares solution. 39.1
$\angle(\mathbf{x}, \mathbf{y})$	the canonical angle between the two vectors $\mathbf{x}, \mathbf{y} = \Theta(\{\mathbf{x}\}, \{\mathbf{y}\})$. 15.1
$\mathbb{Z}$	integers.
$\mathbb{Z}^+$	positive integers.
$\mathbb{Z}_n$	integers mod n . 23.1
Z_n	$n \times n$ lower shift matrix. 48.1

Index

A

- Abelian, Lie algebras, 70–2
- Abs, Mathematica software, 73–26
- Absolute, simple graphs, 36–9 to 36–10
- Absolute bound, 28–11
- Absolute errors
 - conditioning and condition numbers, 37–7
 - floating point numbers, 37–13, 37–16
- Absolute irreducibility, 67–1
- Absolute matrix norms, 37–4
- Absolute values, 17–1
- Absolute vector norm, 37–2
- Absorbing
 - irreducible classes, 54–5
 - vector norms, 37–3
 - vector seminorms, 37–4
- Access equivalence
 - irreducible classes, 54–5
 - irreducible matrices, 29–6, 29–7
 - max-plus eigenproblem, 25–6
 - nonnegative and stochastic matrices, 9–2
- Action, group representations, 68–2
- Active branch, algebraic connectivity, 36–5
- Active constraints, 51–1
- Acyclic matrices
 - multiplicative D -stability, 19–6
 - rank revealing decompositions, 46–9
- Adaptive filtering, signal processing, 64–12 to 64–13
- Addition, 1–1, 1–3
- Additive coset, 2–5
- Additive D -stability, 19–7 to 19–8
- Additive identity axiom, 1–1
- Additive IEPs (AIEPs), 20–10
- Additive inverse axiom, 1–1
- Additive preservers, 22–7 to 22–8
- Adjacency
 - convex set points, 50–13
 - digraphs, 29–3 to 29–4
 - graphs, 28–5 to 28–7
 - Hermitian matrices, 8–2
 - linear preservers, 22–7
 - vertices, 28–2
- Adjoint linear transformation, 51–3
- Adjoint map, Lie algebras, 70–2
- Adjoint, inner product spaces, 13–22
- Adjoint, inner product spaces
 - inner product spaces, 5–5 to 5–6
 - semidefinite programming, 51–3
- Adjugates, determinants, 4–3
- Adjusting random vectors, 52–4
- Admissibility, control theory, 57–2
- Admittance matrix, 28–7
- Ad-nilpotency, 70–4
- Ad-semisimple linear transformation, 70–4
- Advanced linear algebra
 - bilinear forms, 12–1 to 12–9
 - cone invariant departure, matrices, 26–1 to 26–14
 - equalities, matrices, 14–1 to 14–17
 - functions of matrices, 11–1 to 11–12
 - inequalities, matrices, 14–1 to 14–17
 - inertia, matrices, 19–1 to 19–10
 - integral domains, matrices over, 23–1 to 23–10
 - inverse eigenvalue problems, 20–1 to 20–12
 - linear preserver problems, 22–1 to 22–8
 - matrix equalities and inequalities, 14–1 to 14–17
 - matrix perturbation theory, 15–1 to 15–16
 - matrix stability and inertia, 19–1 to 19–10
 - max-plus algebra, 25–1 to 25–14
 - multilinear algebra, 13–1 to 13–26
 - numerical range, 18–1 to 18–11
 - perturbation theory, matrices, 15–1 to 15–16
 - pseudospectra, 16–1 to 16–15
 - quadratic forms, 12–1 to 12–9
 - sesquilinear forms, 12–1 to 12–9
 - similarity of matrix families, 23–1 to 23–10
 - singular values and singular value inequalities, 17–1 to 17–15
 - stability, matrices, 19–1 to 19–10
 - total negativity, matrices, 21–1 to 21–12
 - total positivity, matrices, 21–1 to 21–12

- Affine algebraic variety, 24–8
- Affine function, 50–1
- Affine-independence, 65–2
- Affine parameterized IEPs (PIEPs), 20–10 to 20–12
- Affine subspace determination, 65–2
- AIIEPs (additive IEPs), 20–10
- Aissen studies, 21–12
- Aitken estimator, 52–8
- Akian, Marianne, 25–1 to 25–14
- Akivis algebra, 69–16 to 69–17
- Albert algebra, 69–4, 69–14
- Alexandroff inequality, 25–10
- Alexandrov's inequality, 31–2
- Algebra, P–1
- Algebra applications, *see also* Nonassociative algebra
 - group representations, 68–1 to 68–11
 - Lie algebras, 70–1 to 70–10
 - matrix groups, 67–1 to 67–7
 - nonassociative algebra, 69–1 to 69–25
- Algebraic aspects, least squares solutions, 39–4 to 39–5
- Algebraic connectivity
 - absolute, simple graphs, 36–9 to 36–10
 - Fiedler vectors, 36–7 to 36–9
 - generalized Laplacian, 36–10 to 36–11
 - matrix representations, 28–7
 - multiplicity, 36–10 to 36–11
 - simple graphs, 36–1 to 36–4, 36–9 to 36–10
 - trees, 36–4 to 36–6
 - weighted graphs, 36–7 to 36–9
- Algebraic eigenvalues, 25–9, *see also* Eigenvalues and eigenvectors
- Algebraic function, matrix similarities, 24–1
- Algebraic geometric Goppa (AG) code, 61–10
- Algebraic multigrid, preconditioners, 41–11
- Algebraic multiplicity, 4–6
- Algebraic Riccati equation (ARE), 57–10, 57–12
- Algorithms, *see also specific algorithm*
 - Arbitrary Precision Approximating (APA), 47–6
 - Arnoldi, 41–7
 - BiCGSTAB, 41–8
 - biconjugate gradient (BCG/BiCG), 41–7, 49–13
 - bilinear noncommutative, 47–2
 - bit flipping algorithm, 61–11
 - Conjugate Gradient (CG), 41–4, 41–6
 - conjugate gradient squared (CGS), 41–8
 - Denardo, 25–8
 - error analysis, 37–16 to 37–17
 - ESPRIT, 64–17
 - Euclid's, 23–2
 - fast matrix multiplication, 47–2 to 47–7
 - Full Orthogonalization Method (FOM), 41–7
 - Generalized Minimal Residual (GMRES), 41–7, 49–13
 - Lanczos algorithm, 41–4 to 41–5
 - least squares solutions, 39–6 to 39–7
 - left-preconditioned BiCGSTAB algorithm, 41–12
 - Levinson-Durbin algorithm, 64–8
 - Minimal Residual (MINRES), 41–4, 41–6
 - MUSIC, 64–17
 - noncommutative, 47–2
 - non-Hermitian Lanczos algorithm, 41–7
 - policy iteration, 25–7
 - power algorithm, 25–7
 - preconditioned conjugate gradient (PCG), 41–13
 - quasi-minimal residual (QMR), 41–8, 49–13
 - restarted GMRES algorithm, 41–7
 - singular value decomposition, 45–4 to 45–12
 - transpose-free quasi-minimal residual (TFQMR), 49–14
 - two-sided Lanczos algorithm, 41–7
- All-ones matrix, 52–4
- Allowing characteristics
 - random vectors, 52–4
 - sign-pattern matrices, 33–9 to 33–11
- Alphabet, coding theory, 61–1
- Alternate path, single arc, 35–14
- Alternating bilinear forms, 12–5 to 12–6
- Alternative algebras, 69–2, 69–10 to 69–12
- Alternative bimodule, 69–10
- Alternator, 13–12
- Alt multiplication, 13–17 to 13–19
- AMG code, 41–12
- Analysis applications
 - control theory, 57–1 to 57–17
 - differential equations, 55–1 to 55–16
 - dynamical systems, 56–1 to 56–21
 - Fourier analysis, 58–1 to 58–20
 - LTI systems, 57–7 to 57–10
 - stability, 55–1 to 55–16
- Analytical similarity, 24–1
- Analyzing fill, 40–10 to 40–13
- Angles, inner product spaces, 5–1
- Angular momentum, 59–9 to 59–10
- Annihilator, 3–8 to 3–9
- Anticommutative algebra, 69–2
- Anticommutativity, 70–2
- Anticommutator, 69–3
- Anti-identity matrices, 48–2
- Antisymmetric maps, 13–10 to 13–12
- Antisymmetry, 12–5
- Aperiodicity
 - characterizing, 9–3
 - irreducible classes, 54–5
 - irreducible matrices, 9–3
 - reducible matrices, 9–7
- Append, Mathematica software
 - linear systems, 73–23
 - matrices manipulation, 73–13
 - vectors, 73–3
- AppendColumns, Mathematica software, 73–13
- Appending, vertices, 36–3
- AppendRows, Mathematica software, 73–13
- Applications
 - fast matrix multiplication, 47–9 to 47–10
 - semidefinite programming, 51–9 to 51–11
- Applications, algebra
 - group representations, 68–1 to 68–11
 - Lie algebras, 70–1 to 70–10
 - matrix groups, 67–1 to 67–7
 - nonassociative algebra, 69–1 to 69–25
- Applications, analysis
 - control theory, 57–1 to 57–17
 - differential equations, 55–1 to 55–16

- dynamical systems, 56–1 to 56–21
- Fourier analysis, 58–1 to 58–20
- stability, 55–1 to 55–16
- Applications, biological sciences, 60–1 to 60–13
- Applications, computer science
 - coding theory, 61–1 to 61–13
 - information retrieval, 63–1 to 63–14
 - quantum computation, 62–1 to 62–19
 - signal processing, 64–1 to 64–18
 - Web searches, 63–1 to 63–14
- Applications, geometry
 - Euclidean geometry, 66–1 to 66–15
 - geometry, 65–1 to 65–9
- Applications, optimization
 - linear programming, 50–1 to 50–24
 - semidefinite programming, 51–1 to 51–11
- Applications, physical sciences, 59–1 to 59–11
- Applications, probability and statistics
 - linear statistical models, 52–1 to 52–15
 - Markov chains, 54–1 to 54–14
 - multivariate statistical analysis, 53–1 to 53–14
 - random vectors, 52–1 to 52–15
- Apply, Mathematica software, 73–3, 73–5, 73–27
- Approximate Jordan form, Maple software, 72–15
- Approximate prescribed-line-sum scalings, 9–21 to 9–22
- Approximation
 - fast matrix multiplication, 47–6 to 47–7
 - linear programming, 50–20 to 50–23
 - orthogonal projection, 5–7
- Arbitrary Precision Approximating (APA) algorithms, 47–6
- ArcSin, Mathematica software, 73–26
- ARE, *see* Algebraic Riccati equation (ARE)
- Arithmetic Euclidean vector space, 66–1
- Arm, stars, 34–10
- Arnold Hypothesis, Strong, 28–9, 28–10
- Arnoldi algorithm, 41–7, 41–8
- Arnoldi decomposition, 16–11
- Arnoldi factorization
 - implicitly restarted Arnoldi method, 44–2 to 44–4
 - pseudospectra, 16–3
- Arnoldi matrices, 49–11
- Arnoldi method, *see also* Implicitly restarted Arnoldi method (IRAM)
 - eigenvalue computations, 49–12
 - implicit restarting, 44–6
 - large-scale matrix computations, 49–10 to 49–11
 - sparse matrices, 43–9, 43–11
- Arnoldi vectors
 - Arnoldi factorization, 44–3
 - Arnoldi process, 49–11
- ARPACK subroutine package
 - computational modes, 76–8 to 76–9
 - directory structure and contents, 76–3
 - fundamentals, 76–1 to 76–2, 76–4 to 76–5
 - Lanczos methods, 42–21
 - Matlab's EIGS, 76–9 to 76–10
 - naming conventions, 76–3 to 76–4
 - precisions, 76–3 to 76–4
 - pseudospectra computation, 16–12
 - reverse communication, 76–2
 - setup of problem, 76–5 to 76–7
 - sparse matrices, 43–9
 - types, 76–3 to 76–4
 - use, 76–7 to 76–8
- Array, Maple software, 72–1, 72–2
- array, Maple software, 72–1
- Array, Mathematica software
 - matrices, 73–6
 - vectors, 73–3, 73–4
- Array arithmetic, Maple software, 72–8
- ArrayPlot, Mathematica software
 - fundamentals, 73–27
 - matrices, 73–7, 73–9
- ArrayRules, Mathematica software, 73–6
- Arrays
 - Fourier analysis, 58–2
 - manifold, 64–16
 - Maple software, 72–8 to 72–9
 - response vector, 64–16
- Arrays, Maple software, 72–8, 72–9
- Arrival estimation direction, 64–15 to 64–18
- Arrow, Mathematica software, 73–5
- Artin's theorem, 69–3, 69–11
- AspectRatio, Mathematica software, 73–27
- Assignment polytope, 27–10
- Associated divisor, 61–10
- Associated linear programming, 50–14
- Associated maps, 13–19 to 13–20
- Associations, generalized stars, 34–11
- Association schemes, graphs, 28–11 to 28–12
- Associative algebra, 69–2
- Associative center, 69–5
- Associative enveloping algebra, 69–13
- Associative nucleus, 69–5
- Associativity axiom, 1–1, 1–2
- Associator, nonassociative algebra, 69–2
- Associator ideal, 69–5
- Asymmetric digraphs, 35–2, *see also* Digraphs
- Asymptotics, matrix powers, 25–8 to 25–9
- Asymptotic spectrum, 47–8
- Asymptotic stability
 - control theory, 57–2
 - linear differential-algebraic equations, 55–14
 - linear ordinary differential equations, 55–10
 - LTI systems, 57–7
- ATLAST M-file collection, 71–20
- Attractor-repeller decompositions, 56–7
- Augmented matrices
 - Bezout domains, 23–9
 - systems of linear equations, 1–9, 1–11
- Augmented systems, least squares solution, 39–4
- Autocorrelation matrix, 64–5
- Autocorrelation sequence, 64–4
- Automatic, Mathematica software
 - fundamentals, 73–27
 - singular values, 73–17
- Automorphism
 - Lie algebras, 70–1
 - ring, linear preservers, 22–7
- Autonomy, control theory, 57–2

B

Backward errors

- analysis, 37–20
- numerical stability and instability, 37–18, 37–20 to 37–21

- one-sided Jacobi SVD algorithm, 46–2

BackwardsSubstitute, Maple software, 72–9

Backward stability

- numerical stability and instability, 37–18
- one-sided Jacobi SVD algorithm, 46–2

Bad columns, 50–8

Badly-conditioned data, 37–7

Bad rows, 50–8

Bai, Zhaojun, 75–1 to 75–23

Bailey, D., 47–5

Balanced Boolean function

- Deutsch-Jozsa problem, 62–9
- Deutsch's problem, 62–8

Balanced column signing, 33–5

Balanced matrices

- D-optimal matrices, 32–12
- nonsquare case, 32–5

Balanced row signing, 33–5

Balanced vectors, 33–5

Banded matrices, 41–2

Banding, Toeplitz matrices, 16–6

Bandlimited random signals, 64–5

Bapat, Ravindra, 25–1 to 25–14

Barely L -matrices, 33–5

Barioli, Francesco, 3–1 to 3–9

Barker and Schneider studies, 26–3

Barrett, Wayne, 8–1 to 8–12

Barvinok rank, 25–13

Barycentric coordinates, 66–8

Bases

- Bezout domains, 23–8
- complex sign and ray patterns, 33–14
- component, linear skew product flows, 56–11
- coordinates, 2–10 to 2–12
- induced, symmetric and Grassmann tensors, 13–15
- LTI systems, 57–7
- orthogonality, 5–3
- semisimple and simple algebras, 70–4
- similarity, 3–4
- vector spaces, 2–3 to 2–4

Basic class of P , 9–7

Basic variable, 1–10

Bauer-Fike theorem

- eigenvalue problems, 15–2
- pseudospectra, 16–2 to 16–3

BCG/BiCG (biconjugate gradient) algorithm

- Krylov space methods, 41–7, 41–10
- linear systems of equations, 49–13
- preconditioners, 41–12

BCH (Bose-Chadhuri-Hocquenghem) code, 61–8, 61–9 to 61–10

Beattie, Christopher, 38–1 to 38–17

Belitskii reduction, 24–10

Benner, Peter, 57–1 to 57–17

Bernoulli random variables, 52–2

Bernstein-Vazirani problem, 62–11 to 62–13

Bessel's Inequality, 5–4

Best Approximation Theorem, 5–7

Best linear approximation, 50–20

Best linear unbiased estimate (BLUE), 39–2

Best linear unbiased estimator, 52–9

Best PSD approximation, 17–13

Best rank k approximation, 17–12 to 17–13

Best unitary approximation, 17–13

Between-groups matrix, 53–6

Bezout domains

- certain integral domains, 23–2
- matrices over integral domains, 23–8 to 23–9
- matrix equivalence, 23–6

Bhatia studies, 17–13

Biacyclic matrices, 46–8

Biadjacency matrix, 30–1

Biclique, bipartite graphs, 30–8

Biclique cover, 30–8

Biclique cover number, 30–8

Biclique partition, 30–8

Biclique partition number, 30–8

Biconjugate gradient (BCG/BiCG) algorithm

- Krylov space methods, 41–7, 41–10
- linear systems of equations, 49–13
- preconditioners, 41–12

Bideterminants, 25–13

Bidiagonal singular values by bisection, 45–9

Bidual space, 3–8

 Bigraphs, 40–10, 40–11 to 40–12, *see also* Bipartite graphs

Bilinear forms

- alternating forms, 12–5 to 12–6
- fundamentals, 12–1 to 12–3
- Jordan algebras, 69–13
- symmetric form, 12–1 to 12–5

Bilinear maps, 13–1

Bilinear noncommutative algorithms, 47–2

Bimodule algebras, 69–6

Binary even weight codes, 61–4

Binary Golay code, 61–8, 61–9

Binary linear block code (BLBC), 61–3

Binary matrices, permanents, 31–5 to 31–7

Binary symmetric channel, 61–3 to 61–4

Binary trees, 34–15

Binet-Cauchy identity, 25–13

Binet-Cauchy theorem, 31–2

Bini, Dario A., 47–1 to 47–10

Binomial, Mathematica software, 73–26

Binomial distribution, 52–4

Biological sciences applications, 60–1 to 60–13

Biomolecular modeling

- flux balancing equation, 60–10 to 60–13
- fundamentals, 60–1, 60–13
- Karle-Hauptman matrix, 60–7 to 60–9
- mapping, 60–2 to 60–4
- metabolic network simulation, 60–10 to 60–13
- NMR protein structure determination, 60–2 to 60–4
- protein motion modes, 60–9 to 60–10
- protein structure comparison, 60–4 to 60–7
- x-ray crystallography, 60–7 to 60–9

Biorthogonalization singular value decomposition, 45–11

Biorthogonal pair of bases, 66–5

Bipartite graphs

- factorizations, 30–8 to 30–10
- fill-graph, 30–4
- fundamentals, 30–1 to 30–3
- graphs, 28–2
- matrices, 30–4 to 30–7
- modeling and analyzing fill, 40–10
- rank revealing decomposition, 46–8

Bipartite sign pattern matrices, 33–9

Birepresentations, 69–6

Birkhoff's theorem, 27–11

Bisection method, 42–14 to 42–15

Bit flipping algorithm, 61–11

Bit quantum gate, 62–2

Black-box, 66–13

Black-box matrix, 66–13 to 66–15

Bland's rule, 50–12, 50–13

BLAS subroutine package

- fundamentals, 74–1 to 74–7
- method comparison, 42–21

BLBC (binary linear block code), 61–3

Block, graphs and digraphs, 35–2

Block-clique, 35–2

Block code of length, 61–1

Block diagonal matrices, 10–4 to 10–6

Block lower triangular matrices, 10–4

Block matrices

- partitioned matrices, 10–1 to 10–3
- structured matrices, 48–3

BlockMatrix, Mathematica software, 73–13

Block positive semidefinite matrices, 17–9

Blocks, square case, 32–2

Block-Toeplitz matrices, 48–3

Block-Toeplitz-Toeplitz-Block (BTTB) matrices, 48–3

Block triangular matrices

- inequalities, 17–9
- partitioned matrices, 10–4 to 10–6

Block upper triangular matrices, 10–4

Bloomfield-Watson efficiency, 52–9, 52–10, 52–13 to 52–14

Bloomfield-Watson-Knott Inequality, 52–10, 52–13 to 52–14

Bloomfield-Watson Trace Inequality, 52–10, 52–13 to 52–14

BLUE (best linear unbiased estimate), 39–2

BN structure, matrix groups, 67–4 to 67–5

Bochner's theorem, 8–10, 8–11, 8–12

Boolean properties

- algebra, 30–8
- bipartite graphs, 30–8, 30–9
- Deutsch-Jozsa problem, 62–9
- Deutsch's problem, 62–8
- fast matrix multiplication, 47–10
- matrices, 30–8, 47–10
- rank, 30–8, 30–9

Borel subgroup, 67–4, 67–5

Borobia, Alberto, 20–1 to 20–12

Bose-Chadhuri-Hocquenghem (BCH) code, 61–8

Bose-Mesner algebra, 28–11

Bosons, 59–10

Bottleneck matrices, 36–4

Boundaries

- fundamentals, P–1
- interior point methods, 50–23
- numerical range, 18–3 to 18–4

Bounded properties, 9–11

Box, Euclidean spaces, 66–10

Branches

- matrix similarities, 24–1
- multiplicities and parter vertices, 34–2

Brauer theorem, 14–6

Brègman's bound, 31–7

Bremmer, Murray R., 69–1 to 69–25

Brent, R., 47–5

Brin, Sergey, 54–4, 63–9, 63–10

Browne's theorem, 14–2

Brualdi, Richard A., 27–1 to 27–12

BTTB (Block-Toeplitz-Toeplitz-Block) matrices, 48–3

bucky command, Matlab software, 71–11

Built-in functions, Matlab software, 71–4 to 71–5

Bulge, 42–10, 43–5

Bunch-Parlett factorization, 46–14

Burnside's Vanishing theorem, 68–6

Businger-Golub pivoting

- preconditioned Jacobi SVD algorithm, 46–5
- rank revealing decompositions, 46–9

Butterfly relations, 58–18, 58–20

Byers, Ralph, 37–1 to 37–21

C

Cameron, Peter J., 67–1 to 67–7

Canonical angle and canonical angle matrix, 15–2

Canonical angles, 17–15

Canonical correlations and variates

- multivariate statistical analysis, 53–7
- singular values, 17–15

Canonical forms

- eigenvectors, generalized, 6–2 to 6–3
- elementary divisors, 6–8 to 6–11
- fundamentals, 6–1 to 6–2
- invariant factors, 6–12 to 6–14
- Jordan canonical form, 6–3 to 6–6
- linear programming, 50–7, 50–7 to 50–8
- Maple software, 72–15 to 72–16
- rational canonical forms, 6–8 to 6–11, 6–12 to 6–14
- real-Jordan canonical form, 6–6 to 6–8
- Smith normal form, 6–11 to 6–12

Canonical variates, 53–6

Cartan matrix, 70–4

Cartan's Criterion for Semisimplicity, 70–4

Cartesian coordinates, 55–3

Cartesian decomposition, 17–11

Cartesian product, 13–11 to 13–12

Cassini, ovals of, 14–6, 14–6 to 14–7

- cat command, Matlab software, 71–2
- Cauchy-Binet formula
 - determinants, 4–4, 4–5
 - matrix equivalence, 23–6
- Cauchy-Binet Identity, 21–2
- Cauchy-Binet inequalities, 14–10
- Cauchy boundary conditions, 55–3
- Cauchy integral, 11–2
- Cauchy interlace property, 42–20
- Cauchy matrices
 - rank revealing decompositions, 46–9
 - structured matrices, 48–2
 - symmetric indefinite matrices, 46–16
 - totally positive and negative matrices, 21–4
- Cauchy-Schwartz inequality
 - inner product spaces, 5–1, 5–2
 - vector norms, 37–3
- Causal part, Wiener filtering, 64–10
- Causal signal processing, 64–2
- Cayley-Dickson algebra
 - alternative algebra, 69–11, 69–12
 - Jordan algebras, 69–16
 - Malcev algebras, 69–17
 - nonassociative algebra, 69–4
 - power associative algebras, 69–15
- Cayley-Dickson doubling process, 69–4
- Cayley-Dickson matrix algebra, 69–9
- Cayley-Hamilton Theorem, 4–8
- Cayley numbers
 - nonassociative algebra, 69–4
 - standard forms, 22–4
- Cayley's Formula, 7–6
- Cayley's Transform, 7–6
- CCS (compressed column storage) scheme, 40–3
- Cell bracket, Mathematica software, 73–2
- Cells, Mathematica software, 73–2
- Censoring, Markov chains, 54–11
- Center, 69–5
- Centering matrix, 52–4
- Center subspaces, 56–3
- Central algebra, 69–5
- Central controller, 57–15
- Central distribution, 53–3
- Central force motion, 59–4 to 59–5
- Central matrices, 33–17
- Central Moufang identity, 69–10
- Central path, 51–8
- Central vertex, 34–10
- Centroid, 69–5
- Certain integral domains, 23–1 to 23–4
- CG (Conjugate Gradient) algorithm
 - convergence rates, 41–14 to 41–15
 - Krylov space methods, 41–4, 41–6
- CGS (classical Gram-Schmidt) scheme, 44–3 to 44–4, *see also* Gram-Schmidt methods
- CGS (Conjugate Gradient Squared) algorithm, 41–8
- Chain exponent, 56–16
- Chain recurrence, 56–7 to 56–9
- Chain recurrent component, 56–7
- Chain recurrent set, 56–7
- Chain transitive, 56–7
- Change of basis
 - coordinates, 2–10 to 2–12
 - LTI systems, 57–7
 - similarity, 3–4
- Change-of-basis matrix, 2–10
- Channels, coding theory, 61–2
- Characteristic equation, 43–2
- Characteristic polynomial
 - adjacency matrix, 28–5
 - fast matrix multiplication, 47–10
 - generalized eigenvalue problem, 43–2
- CharacteristicPolynomial, Maple software
 - eigenvalues and eigenvectors, 72–11, 72–12
 - matrix stability, 72–21
 - nonlinear algebra, 72–14
- Characteristic polynomial function, 25–9
- Characteristic polynomials, 4–6
- CharacteristicPolynomials, Mathematica software, 73–14
- Characteristic vector, 30–8
- Characteristic vertex, 36–4
- Characterizations, singular values, 17–1 to 17–3
- Characters
 - grading, 70–9
 - group representations, 68–5 to 68–6
 - restriction, 68–8 to 68–10
 - table, 68–6 to 68–8
- Character table, 68–7
- Chebyshev polynomial
 - convergence rates, 41–15
 - polynomial restarting, 44–6
 - rook polynomials, 31–11
- Checkerboard partial order, 21–9
- Chemical flux, 60–10
- Cholesky algorithm, 46–11
- Cholesky decomposition
 - preconditioners, 41–12, 41–13
 - symmetric factorizations, 38–15, 38–16
- CholeskyDecomposition, Mathematica software, 73–18, 73–27
- Cholesky factor
 - modeling and analyzing fill, 40–13
 - positive definite matrices, 46–12
 - QR factorization, 39–9
 - reordering effect, 40–16, 40–18
- Cholesky factorization
 - extensions, 16–13
 - least squares algorithms, 39–7
 - linear prediction, 64–8
 - positive definite matrices, 8–7, 46–13
 - sparse matrices, 49–3 to 49–5
 - sparse matrix factorizations, 49–3
 - symmetric factorizations, 38–15
- Cholesky factorization with pivoting, 46–10
- Cholesky-like factorization, 8–9
- Chop, Mathematica software, 73–25
- Chordal bipartite graph
 - bipartite graphs, 30–1
- Chordal distance
 - eigenvalue problems, 15–10

- Chordal graph
 - bipartite graphs, 30–1
- Chordal graphs, 35–2
- Chordal symmetric Hamiltonian, minimally, 35–15
- Chromatic index, 27–10
- Chromatic number, 28–9
- Chu, Ka Lok, 53–14
- Circulant matrices
 - Maple software, 72–18
 - nonnegative inverse eigenvalue problems, 20–5
 - structured matrices, 48–2
- circul function, Matlab software, 71–6
- Circumcenter, 66–8
- Circumscribed hypersphere, 66–8
- Class functions, orthogonality relations, 68–6
- Classical Gram-Schmidt (CGS) scheme, 44–3 to 44–4
- Classical groups, 67–5 to 67–7
- Classical Turing machine, 62–2
- Classification, states, 54–7 to 54–9
- Classifications I and II, 24–7 to 24–11
- Classification theorem, 24–9, 24–11
- Clifford algebra, 69–13
- Clifford's theorem
 - characters, 68–9
 - matrix groups, 67–2
- Cline, Alan Kaylor, 45–1 to 45–12
- Clique
 - graph parameters, 28–9
 - graphs, 35–2
- Clique number, 28–9
- Close cones, 8–10
- Closed ball of radius, 61–2
- Closed halfspace, 66–2
- Closed-loop ODE, 57–2
- Closed-loop system, 57–13
- Closed subset, 50–13
- Closed under matrix direct sums, 35–2
- Closed under permutation similarity, 35–2
- Closed under taking principal submatrices, 35–2
- Closed walk, 54–5
- Closure under addition, 1–2
- Closure under scalar multiplication, 1–2
- Cockades, 30–4
- Cocktail party graph, 28–4
- Coclique, 28–9
- Co-cover, 27–2
- Codewords, 61–1
- Codimension, 66–2
- Coding theory
 - convolutional codes, 61–11 to 61–13
 - distance bounds, 61–5 to 61–6
 - fundamentals, 61–1 to 61–2
 - linear block codes, 61–3 to 61–4
 - linear code classes, 61–6 to 61–11
 - main linear coding problem, 61–5 to 61–6
- Codomain, 3–1
- Coefficient matrices
 - Bezout domains, 23–9
 - linear differential equations, 55–1
- Coefficients
 - determination, 52–8
 - multiple determination, 52–8
 - systems of linear equations, 1–9
- Cofactors, determinants, 4–1
- Cogame utility, ATLAST, 71–20, 71–22
- Co-index, 9–7
- Colin de Verdière parameter, 28–9, 28–10
- Collatz-Wielandt sets
 - cone invariant departure, matrices, 26–3 to 26–5
 - irreducible matrices, 9–5
- Collineation, projective spaces, 65–7
- Colonus, Fritz, 56–1 to 56–21
- Color class, 28–9
- ColorFunction, Mathematica software, 73–27
- colormap command, Matlab software, 71–15
- colspace command, Matlab software, 71–17
- Columns
 - equivalence, 23–5
 - feasibility, 50–14
 - indices, 23–9
 - linear independence, span, and bases, 2–6
 - matrices, 1–3
 - pivoting, 46–5
 - rank, 25–13
 - signing, balanced, 33–5
 - sign solvability, 33–5
 - sum vector, 27–7
 - vectors, 1–3
- Column-stochastic matrices, 9–15
- Combinatorial matrix theory
 - algebraic connectivity, 36–1 to 36–11
 - bipartite graphs and matrices, 30–1 to 30–10
 - classes of (0,1)-matrices, 27–7 to 27–10
 - completion problems, 35–1 to 35–20
 - convex polytopes, 27–10 to 27–12
 - digraphs, 29–1 to 29–13
 - D-optimal matrices, 32–1 to 32–12
 - doubly stochastic matrices, 27–10 to 27–12
 - fundamentals, 27–1 to 27–12
 - graphs, 28–1 to 28–12
 - monotone class, 27–7 to 27–8
 - multiplicity lists, 33–1 to 33–17
 - permanents, 31–1 to 31–13
 - sign-pattern matrices, 33–1 to 33–17
 - square matrices, 27–3 to 27–6
 - strong combinatorial invariants, 27–3 to 27–5
 - structure and invariants, 27–1 to 27–3
 - tournament matrices, 27–8 to 27–10
 - weak combinatorial invariants, 27–5 to 27–6
- Combinatorial orthogonality, 33–16
- Combinatorial problems, 47–10
- Combinatorial symmetric partial matrix, 35–2
- Combinatorial symmetric sign pattern matrices, 33–9
- Communication
 - irreducible classes, 54–5
 - irreducible matrices, 29–6
 - nonnegative and stochastic matrices, 9–2
- Commutative algebras, 69–2

- Commutative ring
 - certain integral domains, 23–1
 - permanents, 31–1
- Commutativity and Jordan identity, 69–12
- Commutativity axiom, 1–1
- Commutators
 - nonassociative algebra, 69–2
 - Schrödinger's equation, 59–7
- Commute
 - matrices, 1–4
 - vector spaces, 3–2
- Companion matrices
 - eigenvalues and eigenvectors, 4–6
 - rational canonical forms, 6–8
- CompanionMatrix, Maple software, 72–17
- Comparison matrices, 19–9
- Compatibility, 33–2
- Complement
 - binary matrices, 31–5
 - fundamentals, P–1
 - graphs, 28–2
- Complementary parameters, 64–8
- Complementary projections, 3–6
- Complementary slackness
 - duality, 50–14
 - duality and optimality conditions, 51–6
 - max-plus eigenproblem, 25–7
- Complete bipartite graphs, 28–2, 30–1, *see also* Bipartite graphs
- Completed max-plus semiring, 25–1
- Complete flag, 56–9
- Complete graphs, 28–2, *see also* Graphs
- Completely positive matrices
 - completion problems, 35–10 to 35–11
 - nonnegative factorization, 9–22
- Complete min-plus semiring, 25–1
- Complete orthogonal decomposition, 39–11
- Complete orthogonal set, 5–3
- Complete pivoting, 38–10, *see also* Pivoting
- Complete reducibility
 - matrix group, 67–1
 - modules, 70–7
 - square matrices, weak combinatorial invariants, 27–5
- Completion problems, matrices
 - completely positive matrices, 35–10 to 35–11
 - copositive matrices, 35–11 to 35–12
 - doubly nonnegative matrices, 35–10 to 35–11
 - entry sign symmetric P -, $P_{0,1}$ - and P_0 -matrices, 35–19 to 35–20
 - entry weakly sign symmetric P -, $P_{0,1}$ - and P_0 -matrices, 35–19 to 35–20
 - Euclidean distance matrices, 35–9 to 35–10
 - fundamentals, 35–1 to 35–8
 - inverse M -matrices, 35–14 to 35–15
 - M - and M_0 - matrices, 35–12 to 35–13
 - nonnegative P -, $P_{0,1}$ - and P_0 -matrices, 35–17 to 35–18
 - P -, $P_{0,1}$ - and P_0 -matrices, 35–15 to 35–17
 - positive definite matrices, 35–8 to 35–9
 - positive P -matrices, 35–17 to 35–18
 - positive semidefinite matrices, 35–8 to 35–9
 - strictly copositive matrices, 35–11 to 35–12
- Completion property, 35–2
- Complex conjugate, 68–5
- Complex elements, Maple software, 72–7
- Complexity, convolutional codes, 61–12
- Complex numbers
 - fundamentals, P–1 to P–2
 - nonassociative algebra, 69–4
- Complex polynomial, 19–3
- Complex sign patterns, 33–14 to 33–15
- Component backward stable, 38–6
- Component-wise perturbation theory, 39–7
- Component-wise relative backward errors, linear system, 38–2
- Component-wise relative distance, 46–10
- Composite cycle, 33–2
- Composition, symmetric and Grassmann tensors, 13–13
- Composition algebras, 69–8 to 69–10
- Compound matrix, 4–3
- Compressed column storage (CCS) scheme, 40–3
- Compressed row storage (CRS) scheme, 40–4
- Computational linear algebra
 - fast matrix multiplication, 47–1 to 47–10
 - large-scale matrix computations, 49–1 to 49–15
 - structured matrix computations, 48–1 to 48–9
- Computational methods, 69–20 to 69–25
- Computational modes, 76–8 to 76–9
- Computational software
 - freeware, 77–1 to 77–3
 - Maple, 72–1 to 72–21
 - Mathematica, 73–1 to 73–27
 - Matlab, 71–1 to 71–22
- Computational software, subroutine packages
 - ARPACK, 76–1 to 76–10
 - BLAS, 74–1 to 74–7
 - EIGS, 76–1 to 76–10
 - LAPACK, 75–1 to 75–23
- Computations
 - pseudospectra, 16–11 to 16–12
 - singular value decomposition, 45–1 to 45–12
- Computations, large-scale matrices
 - Arnoldi process, 49–10 to 49–11
 - dimension reduction, 49–14 to 49–15
 - eigenvalue computations, 49–12
 - fundamentals, 49–1 to 49–2
 - Krylov subspaces, 49–5 to 49–6
 - linear dynamical systems, 49–14 to 49–15
 - linear systems, equations, 49–12 to 49–14
 - nonsymmetric Lanczos process, 49–8 to 49–10
 - sparse matrix factorizations, 49–2 to 49–5
 - symmetric Lanczos process, 49–6 to 49–7
- Computations, structured matrices
 - direct Toeplitz solvers, 48–4 to 48–5
 - fundamentals, 48–1 to 48–4
 - iterative Toeplitz solvers, 48–5
 - linear systems, 48–5 to 48–8
 - total least squares problems, 48–8 to 48–9
- Computer science applications
 - coding theory, 61–1 to 61–13
 - information retrieval, 63–1 to 63–14
 - quantum computation, 62–1 to 62–19

- signal processing, 64–1 to 64–18
 - Web searches, 63–1 to 63–14
- Concentration, metabolites, 60–10
- Conceptual PageRank, 63–11, *see also* PageRank
- cond, Mathematica software, 73–18
- Condensation digraph, 29–7
- Condensation method, 4–4
- Conditioning
 - condition numbers, 37–7 to 37–9
 - linear systems, 37–9 to 37–11
- Conditioning of eigenvalues, Maple software, 72–15
- ConditionNumber, Maple software, 72–9
- Condition numbers
 - linear system perturbations, 38–2
 - polar decomposition, 15–8
 - sensitivity, 39–7
- Conductance, 28–9
- Conductivity, 66–13
- Cone invariant departure, matrices
 - Collatz-Wielandt sets, 26–3 to 26–5
 - core, 26–5 to 26–7
 - distinguished eigenvalues, 26–3 to 26–5
 - elementary analytic results, 26–12 to 26–13
 - fundamentals, 26–1
 - K -reducible matrices, 26–8 to 26–10
 - linear equations over cones, 26–11 to 26–12
 - peripheral spectrum, 26–5 to 26–7
 - Perron-Frobenius theorem, 26–1 to 26–3
 - Perron-Schaefer condition, 26–5 to 26–7
 - spectral theory, 26–8 to 26–10
 - splitting theorems, 26–13 to 26–14
 - stability, 26–13 to 26–14
- Cones
 - Perron-Frobenius theorem, 26–1 to 26–3
 - positive definite matrices, 8–10
 - programming, 51–2
 - reducible matrices, 26–8
- Conformal and conformable partitions, 10–1
- Conformal partitions, 10–1
- Congruence
 - bilinear forms, 12–2
 - Hermitian and positive definite matrices, 8–5 to 8–6
 - Hermitian forms, 12–8
 - linear inequalities and projections, 25–10
 - sesquilinear forms, 12–6
- Conjugacy, 56–5
- Conjugate character, 68–9
- Conjugate Gradient (CG) algorithm
 - convergence rates, 41–14 to 41–15
 - Krylov space methods, 41–4, 41–6
- Conjugate Gradient Squared (CGS) algorithm, 41–8
- Conjugate partition, P–2
- Conjugates, linear dynamical systems, 56–5
- ConjugateTranspose, Mathematica software
 - fundamentals, 73–27
 - matrix algebra, 73–9, 73–11
 - singular values, 73–17, 73–18
- Connected properties and components
 - digraphs, 29–2
 - graphs, 28–1, 28–2
- Consecutive ones property, 30–4
- Consistency
 - least squares solution, 39–1
 - linear differential equations, 55–2
 - linear statistical models, 52–8
 - matrix norms, 37–4
 - systems of linear equations, 1–9
- Consistent sign pattern matrices, 33–9
- Constant Boolean function
 - Deutsch-Jozsa problem, 62–9
 - Deutsch's problem, 62–8
- Constant coefficients
 - differential equations, 55–1 to 55–5
 - linear differential equations, 55–1
- Constant term, 1–9
- Constant vector, 1–9
- Constituents, characters, 68–5
- ConstrainedMax, Mathematica software, 73–27
- ConstrainedMin, Mathematica software, 73–27
- Constraint length, 61–12
- Constraint qualification
 - semidefinite programming, 51–7
 - strong duality, 51–7
- Contact matrix, 60–9
- Containment gap, 44–9
- Content index, Web search, 63–9
- Continuation character backslash, Maple software, 72–2
- Continuous dynamical systems, 56–5
- Continuous invariants, 24–11
- Continuous real-valued functions, 2–2
- contour command, Matlab software, 71–15
- Contour integration, 11–10
- Contraction matrices, 18–9
- Contractions, graphs, 28–4
- Contragredient, 68–3
- Control design, LTI systems, 57–13 to 57–17
- Controllability Hessenberg form, 57–9
- Controllability matrix, 57–7
- Controlled-NOT gate
 - quantum computation, 62–4
 - universal quantum gates, 62–8
- Controller, LTI systems, 57–13
- Controlling random vectors, 52–4
- Control system, 57–2
- Control theory
 - analysis, LTI systems, 57–7 to 57–10
 - control design, LTI systems, 57–13 to 57–17
 - frequency-domain analysis, 57–5 to 57–6
 - fundamentals, 57–1 to 57–5
 - LTI systems, 57–7 to 57–10, 57–13 to 57–17
 - matrix equations, 57–10 to 57–11
 - state estimation, 57–11 to 57–13
- Control vector, 57–2
- Convergence
 - implicitly restarted Arnoldi method, 44–9 to 44–10
 - nonnegative and stochastic matrices, 9–2
 - reducible matrices, 9–8, 9–11
 - Toeplitz matrices, 16–6
- Convergence rates
 - CG, 41–14 to 41–15
 - GMRES, 41–15 to 41–16
 - MINRES, 41–14 to 41–15

Convergent regular splitting, 9–17
 Convex hull, 66–2
 Convexity
 affine spaces, 65–2
 Euclidean point space, 66–2
 fundamentals, P–2
 phase 2 geometric interpretation, 50–13
 vector norms, 37–3
 vector seminorms, 37–4
 Convex linear combination, 50–13
 Convex polytopes, 27–10 to 27–12
 Convolution
 Fourier analysis, 58–3
 signal processing, 64–2
 Convolutional codes, 61–11 to 61–13
 Convolution identities
 discrete theory, 58–10
 Fourier analysis, 58–5
 Coordinate matrix, 60–2
 Coordinates
 change of basis, 2–10 to 2–12
 Euclidean point space, 66–1
 NMR protein structure determination, 60–2
 Coordinate vectors, 60–2
 Coordinatization theorem, 69–13
 Copositive matrices, 35–11 to 35–12
 Coppersmith and Winograd studies, 47–9
 Coprime elements, 23–2
 Core, 26–5 to 26–7
 Corless, Robert M., 72–1 to 72–21
 Corner minor, 21–7, *see also* Principal minors
 Corrected seminormal equations, 39–6
 Correlation, random vectors, 52–3
 Correlation coefficient, 52–3
 Correlation matrix
 positive definite matrices, 8–6
 random vectors, 52–4
 Correlations and variates, 53–7 to 53–8
 Cosine and sine, 11–11
 Cospectral graphs, 28–5
 costs, Mathematica software, 73–24
 Coupling time, 25–9
 Courant-Fischer inequalities, 14–4
 Courant-Fischer theorem
 eigenvalues, 14–4
 Hermitian matrices, 8–3
 Covariance, random vectors, 52–3
 Covariance matrix
 positive definite matrices, 8–9
 random vectors, 52–3
 Cover, combinatorial matrix theory, 27–2
 Cramer's Rule, 4–3, 37–17
 Craven and Csordas studies, 21–11 to 21–12
 Critical digraphs, 25–6, 25–7, *see also* Digraphs
 Critical vertices, 25–6
 Cross, Mathematica software, 73–3, 73–5
 Cross correlation, 64–4
 Cross-covariance matrix, 52–4
 Cross-positives, 26–13
 CrossProduct, Maple software, 72–3
 CRS (compressed row storage) scheme, 40–4

Csordas, Craven and, studies, 21–11 to 21–12
 Cubics, Mathematica software, 73–14, 73–15
 Cui, Feng, 60–13
 Cumulative distribution function, 52–2
 Cuthill-McKee algorithm, 40–16
 Cut lattice, 30–2
 Cut space, 30–2
 Cut vertices, 36–3
 Cycle-clique, 35–14
 Cycle conditions, 35–9
 Cycle of length, 28–1, 28–2
 Cycle products, digraphs, 29–4 to 29–6
 Cycles
 digraphs, 29–2
 Jacobi method, 42–18
 matrices, 48–2
 pattern, 33–9
 simplex method, 50–12
 time, 25–8
 Cycles of length, 33–2
 Cyclically real ray pattern, 33–14
 Cyclic code, 61–6
 Cyclicity, matrix power asymptotics, 25–8
 Cyclicity theorem, 25–8
 Cyclic normal form
 digraphs, 29–9 to 29–11
 imprimitive matrices, 29–10
 Cyclic simplexes, 66–12

D

Damped least squares, 39–9 to 39–10
 Dangling node, 63–11
 Daniel studies, 44–4
 Data Encryption Standard (DES) cryptography system, 62–17
 Data fitting, 39–3 to 39–4
 Data matrix, 53–2 to 53–3
 Data perturbations, 38–2
 Datta, Biswa Nath, 37–1 to 37–21
 Davidson's method, 43–10
 Day, Jane, 1–1 to 1–15
 DCT, *see* Discrete Fourier transform (DFT)
 DeAlba, Luz M., 4–1 to 4–11
 Decoder, 61–2
 Decoding, 61–2
 Decomposable tensors, 13–3
 Decomposition
 direct solution of linear systems, 38–7 to 38–15
 high relative accuracy, 46–7 to 46–10
 least squares solutions, 39–11 to 39–12
 Mathematica software, 73–18 to 73–19
 matrix group, 67–1
 Morse, dynamical systems, 56–7 to 56–9
 rank revealing decomposition, 39–11 to 39–12
 semisimple and simple algebras, 70–4
 singular values, 17–15
 symmetric and Grassmann tensors, 13–13
 symmetric factorizations, 38–15
 tensors, multilinear algebra, 13–7

- Deconvolution
 - Fourier analysis, 58–8
 - functional and discrete theories, 58–16
- Decoupling Principle, 59–1, 59–2
- Deeper properties, 21–9 to 21–12
- Defective matrices, 4–6
- Definite matrices, *see* Positive definite matrices (PSD)
- Definite pencils, 15–10
- Deflation, 42–2
- Degenerate characteristics
 - sesquilinear forms, 12–6
 - simplex method, 50–12
- Degree
 - certain integral domains, 23–2
 - characters, 68–5
 - control theory, 57–2
 - convolutional codes, 61–11
 - frequency-domain analysis, 57–6
 - general properties, 69–5
 - graphs, 28–2
 - group representations, 68–1
 - matrix group, 67–1
 - matrix representations, 68–3
 - max-plus permanent, 25–9
- Deletion, edges and vertices, 28–4
- Delsarte's Linear Programming Bound, 28–12
- Delta function, 58–2
- demand, Mathematica software, 73–24
- Demmel, James, 75–1 to 75–23
- Demmel-Kahan singular value decomposition, 45–7 to 45–8
- Denardo algorithm, 25–8
- Denman-Beavers iteration, 11–11
- Dense matrices
 - fundamentals, 43–1
 - large-scale matrix computations, 49–2
 - software, 77–2
 - techniques, 43–3 to 43–9
- de Oliveria studies, 20–3
- Depth, Jordan canonical form, 6–3
- Derangements, 31–6
- De Rijk's row-cyclic pivoting, 46–4
- Derivation, Lie algebras, 70–1
- Derived algebra, 70–3
- Derogatory matrices, 4–6
- Desargues' theorem, 65–8, 65–9
- DES (Data Encryption Standard) cryptography system, 62–17
- Design, square case, 32–2
- Design matrices
 - D-optimal matrices, 32–1
 - linear statistical models, 52–8
- Det, Mathematica software, 73–10, 73–11
- det command, Matlab software, 71–17
- Detection, control theory, 57–2
- Determinant, Maple software, 72–5
- Determinantal region, 33–14
- Determinantal relations, 14–10 to 14–12
- Determinants
 - advanced results, 4–3 to 4–6
 - connections, 31–12 to 31–13
 - fast matrix multiplication, 47–10
 - fundamentals, 4–1 to 4–3
 - invariants, 23–5
- Deterministic Markov decision process, 25–3
- Deterministic spectral estimation, 64–14
- det function, Matlab software, 71–3
- Deutsch-Jozsa problem, 62–9 to 62–11
- Deutsch's problem, 62–8 to 62–9
- Developer'HessenbergDecomposition', Mathematica software, 73–27
- DGEMM BLAS subroutine package, 42–21
- DGKS mechanism, 44–4
- Dhillon, Inderjit S., 45–1 to 45–12
- diag, Mathematica software, 73–16
- Diagonal entry, 1–4, 23–5
- Diagonalization
 - eigenvalue problems, 15–10
 - eigenvalues and eigenvectors, 4–6, 4–7 to 4–8
- Diagonally dominant matrices, 9–17
- Diagonally scaled representation, 46–10
- Diagonally scaled totally unimodular (DSTU), 46–8, 46–9
- Diagonal matrices, 1–4
- DiagonalMatrix, Mathematica software
 - eigenvalues, 73–15, 73–16
 - matrices, 73–6, 73–8
- Diagonal pattern, 33–2
- Diagonal product, 27–10
- Diagonals, square matrices, 27–3
- Diagonal stability, 19–9
- Diameter, eigenvalues, 34–7
- Diao, Zijian, 62–1 to 62–19
- Dias da Silva, José A., 13–1 to 13–26
- Differentiable functions, 56–5
- Differential-algebraic equations, 55–1
- Differential-algebraic equations of order, 55–2
- Differential equations
 - constant coefficients, 55–1 to 55–5
 - eigenvalues and eigenvectors, 4–10 to 4–11
 - linear different equations, 55–1 to 55–5
 - linear differential-algebraic equations, 55–7 to 55–10, 55–14 to 55–16
 - linearization, 56–19
 - linear ordinary differential equations, 55–5 to 55–6, 55–10 to 55–14
 - stability, 55–10 to 55–16
- Differential quotient-difference (dqds) step, 45–8 to 45–9
- Digits, Maple software, 72–14
- Digraphs
 - adjacency matrix, 29–3 to 29–4
 - cycle products, 29–4 to 29–6
 - cyclic normal form, 29–9 to 29–11
 - directed graphs, 29–3 to 29–4
 - fundamentals, 29–1 to 29–3
 - irreducible, imprimitive matrices, 29–9 to 29–11
 - irreducible matrices, 29–6 to 29–8
 - matrices, 29–3 to 29–4
 - matrix completion problems, 35–2
 - max-plus algebra, 25–2
 - minimal connection, 29–12 to 29–13

- modeling and analyzing fill, 40–11
- nearly reducible matrices, 29–12 to 29–13
- nonnegative and stochastic matrices, 9–2
- P -, $P_{0,1}$ - and P_0 -matrices, 35–15 to 35–16
- primitive digraphs and matrices, 29–8 to 29–9
- sign-pattern matrices, 33–2
- strongly connected digraphs, 29–6 to 29–8
- walk products, 29–4 to 29–5
- Dihedral interior angle, 66–7
- Dilations, numerical range, 18–9 to 18–10
- Dimension
 - doubly stochastic matrices, 27–10
 - Euclidean point space, 66–2
 - Euclidean simplexes, 66–7
 - grading, 70–9
 - nonassociative algebra, 69–1
 - simultaneous similarity, 24–8
 - vector space, 2–3
- Dimensional projective spaces and subspaces, 65–6
- Dimension reduction, 49–14 to 49–15
- Dimensions, Mathematica software, 73–6, 73–7, 73–27
- Dimension theorem
 - kernel, 3–5
 - matrix range, 2–6 to 2–9
 - null space, 2–6 to 2–9
 - range, 3–5
 - rank, 2–6 to 2–9
- Dirac's bra-ket notation, 62–2
- Directed arcs, digraphs, 29–1
- Directed bigraphs, 30–4
- Directed digraphs, 29–1
- Directed edges, 29–1
- Directed graphs, 29–3 to 29–4
- Directed multigraph, 29–2
- Direction, arrival estimation, 64–15 to 64–18
- Direction space, affine spaces, 65–2
- Direct isometry, 65–5, *see also* Isometry
- Directory structure and contents, 76–3
- Direct solution, linear systems
 - fundamentals, 38–1
 - Gauss elimination, 38–7 to 38–12
 - LU decomposition, 38–7 to 38–12
 - orthogonalization, 38–13 to 38–15
 - perturbations, 38–2 to 38–5
 - QR decomposition, 38–13 to 38–15
 - symmetric factorizations, 38–15 to 38–17
 - triangular linear systems, 38–5 to 38–7
- Direct sum
 - block diagonal and triangular matrices, 10–4
 - decompositions, 2–4 to 2–6
 - direct sum decompositions, 2–5
 - group representations, 68–3
 - nonassociative algebra, 69–3
 - semisimple and simple algebras, 70–3
- Direct Toeplitz solvers, 48–4 to 48–5
- Dirichlet conditions, 59–10
- Discrete approximation, 58–13
- Discrete event systems, 25–3 to 25–4
- Discrete Fourier transform (DFT), 58–9
- Discrete invariants, 24–11
- Discrete stochastic process, 54–1
- Discrete theory, 58–2 to 58–17
- Discrete time Fourier transform, 64–2
- Discrete variables, 52–2
- Discrete Wiener filtering problem, 64–10
- Discriminant coordinates, 53–6
- Disjoint matrix multiplication, 47–8
- Dispersion, 21–7
- Dispersion matrix, 52–3
- Dissection strategy, 40–16
- Dissimilarity, 53–13
- Dissimilarity matrix, 53–14
- Distance
 - coding theory, 61–2
 - Euclidean point space, 66–2
 - Euclidean spaces, 65–4
 - graphs, 28–1
 - inner product spaces, 5–1
 - NMR protein structure determination, 60–2
 - numerical range, 18–1
- Distance bounds, 61–5 to 61–6
- Distance matrix, 60–2
- Distance-regular graph, 28–11
- Distinguished eigenvalues
 - cone invariant departure, matrices, 26–3 to 26–5
 - nonnegative and stochastic matrices, 9–2
- Distinguished face, 26–5
- Distributivity, 69–1
- Distributivity axiom, 1–1
- Divide and Conquer Bidiagonal singular value decomposition, 45–10
- Divide and conquer method, 42–12 to 42–14
- Division algebra, 69–2, 69–4
- Division Property, 6–11
- DLARGV LAPACK subroutine, 42–9
- DLARTV LAPACK subroutine, 42–9
- DLAR2V LAPACK subroutine, 42–9
- Documents, retrieved, 63–2
- Document vector, 63–1
- Domain, linear transformations, 3–1
- Domain decomposition methods, 41–12
- Dominant eigenvalue, 42–2
- Dong, Qunfeng, 60–13
- Dongarra, Jack, 74–1 to 74–7, 75–1 to 75–23, 77–1 to 77–3
- D-optimal matrices
 - balanced matrices, 32–12
 - fundamentals, 32–1
 - nonregular matrices, 32–7 to 32–9
 - nonsquare case, 32–2 to 32–12
 - regular matrices, 32–5 to 32–7
 - square case, 32–2 to 32–4
- DORGTR LAPACK subroutine, 42–8
- Dot, Mathematica software
 - matrices, 73–8
 - vectors, 73–4
- Dot product
 - Bernstein-Vazirani problem, 62–11
 - floating point numbers, 37–14
 - inner product spaces, 5–2

DotProduct, Maple software, 72–3
 Double directed tree, 29–2
 Double echelon form, 21–2
 Double generalized stars, 34–11 to 34–14
 Double path, generalized stars, 34–11
 Double precision, 37–13
 Double stars, 34–11
 Doubly nonnegative matrices, 35–10 to 35–11
 Doubly stochastic matrices
 combinatorial matrix theory, 27–10 to 27–12
 fundamentals, 9–15
 permanents, 31–3 to 31–4
 Downdating, 39–8 to 39–9
 Downer branch, 34–2
 Downer vertex, 34–2
 Drazin inverse, 55–7
 Drmac, Zlatko, 46–1 to 46–16
 Drop, Mathematica software
 fundamentals, 73–27
 matrices manipulation, 73–13
 vectors, 73–3
 Drop-off condition, 21–12
 DROT BLAS routine, 42–9
 DSBTRD LAPACK subroutine, 42–9
 DSDRV ARPACK routine, 42–21
 D-stability, 19–5 to 19–7
 DSTEBZ LAPACK subroutine, 42–15
 DSTEDC LAPACK subroutine, 42–14
 DSTEGR LAPACK subroutine, 42–17
 DSTEIN LAPACK subroutine, 42–15
 DSTU (diagonally scaled totally unimodular) matrices
 rank revealing decomposition, 46–8
 rank revealing decompositions, 46–9
 DSYEV LAPACK subroutine, 42–11
 DSYTRD LAPACK subroutine, 42–8
 Duality
 code, 61–3
 cones, 8–11, 26–2
 control theory, 57–2
 inner product spaces, 13–23
 linear functionals and annihilator, 3–8
 linear programming, 50–13 to 50–17
 optimality conditions, 51–6
 projective spaces, 65–7
 rectangular matrix multiplication, 47–5
 semidefinite programming, 51–5 to 51–7
 vector norms, 37–2
 Duality theorem, 51–7
 Dual linear program, 25–5
 Dulmage-Mendelsohn Decomposition theorem, 27–4
 Dynamical systems
 chain recurrence, 56–7 to 56–9
 eigenvalues and eigenvectors, 4–11
 flag manifolds, 56–9 to 56–11
 Floquet theory, 56–12 to 56–14
 fundamentals, 56–1 to 56–2
 Grassmannians, 56–9 to 56–11
 linear differential equations, 56–2 to 56–4
 linear dynamical systems, 56–5 to 56–7
 linearization, 56–19 to 56–21

 linear skew product flows, 56–11 to 56–12
 Morse decompositions, 56–7 to 56–9
 periodic linear differential equations, 56–12 to 56–14
 random linear dynamical systems, 56–14 to 56–16
 robust linear systems, 56–16 to 56–19
 Dynamic compensator, 57–13
 Dynamic programming, 25–3

E

Eckart-Young low rank approximation theorem, 5–11
 EDD (elementary divisor domain), 23–2
 ED (Euclidean domain), 23–2
 Edge cut, 36–3
 Edges
 digraphs, 29–1
 Euclidean simplexes, 66–7
 graphs, 28–1, 28–4
 ED-RCF (elementary divisors rational canonical form)
 matrix, 6–8 to 6–9
 Effects of reordering, 40–14 to 40–18
 Efficacy, methods comparison, 42–21
 Efficiency, error analysis, 37–16 to 37–17
 egn, Mathematica software, 73–21
 egn, Mathematica software, 73–20
 eig command, Matlab software
 eigenvalues, 15–5
 fundamentals, 71–9, 71–17, 71–18
 implicitly restarted Arnoldi method, 44–1
 Lanczos methods, 42–21
 EigenConditionNumbers, Maple software, 72–15
 Eigenpairs
 eigenvalue problems, 15–10
 matrix perturbation theory, 15–1
 Eigenproblems
 max-plus algebra, 25–6 to 25–8
 nonsymmetric, LAPACK subroutine package, 75–17 to 75–20
 Eigensharp characteristic, 30–8
 Eigenspaces
 eigenvalues and eigenvectors, 4–6
 max-plus eigenproblem, 25–6
 nonnegative and stochastic matrices, 9–2
 Eigensystem, Mathematica software, 73–14, 73–15, 73–16
 Eigentriplets
 eigenvalue problems, 15–9 to 15–10
 matrix perturbation theory, 15–1
 Eigenvalue decomposition (EVD), 42–2
 Eigenvalues, high relative accuracy
 accurate SVD, 46–2 to 46–5, 46–7 to 46–10
 fundamentals, 46–1 to 46–2
 one-sided Jacobi SVD algorithm, 46–2 to 46–5
 positive definite matrices, 46–10 to 46–14
 preconditioned Jacobi SVD algorithm, 46–5 to 46–7
 rank revealing decomposition, 46–7 to 46–10
 structured matrices, 46–7 to 46–10
 symmetric indefinite matrices, 46–14 to 46–16
 Eigenvalues, Maple software, 72–11, 72–12, 72–14

- Eigenvalues, Maple software, 72–15
- Eigenvalues, Mathematica software
 - eigenvalues, 73–14, 73–15
 - fundamentals, 73–27
 - singular values, 73–17
- Eigenvalues, numerical methods
 - high relative accuracy computation, 46–1 to 46–16
 - implicitly restarted Arnoldi method, 44–1 to 44–12
 - iterative solution methods, 41–1 to 41–17
 - singular value decomposition, 45–1 to 45–12
 - symmetric matrix techniques, 42–1 to 42–22
 - unsymmetric matrix techniques, 43–1 to 43–11
- Eigenvalues, problems
 - generalized, 15–9 to 15–11
 - perturbation theory, 15–1 to 15–6, 15–9 to 15–11
 - relative perturbation theory, 15–13 to 15–15
- Eigenvalues, symmetric matrix techniques
 - bisection method, 42–14 to 42–15
 - comparison of methods, 42–21 to 42–22
 - divide and conquer method, 42–12 to 42–14
 - fundamentals, 42–1 to 42–2
 - implicitly shifted QR method, 42–9 to 42–11
 - inverse iteration, 42–14 to 42–15
 - Jacobi method, 42–17 to 42–19
 - Lanczos method, 42–19 to 42–21
 - method comparison, 42–21 to 42–22
 - methods, 42–2 to 42–5
 - multiple relatively robust representations, 42–15 to 42–17
 - tridiagonalization, 42–5 to 42–9
- Eigenvalues, unsymmetric matrix techniques
 - dense matrix techniques, 43–3 to 43–9
 - fundamentals, 43–1
 - generalized eigenvalue problem, 43–1 to 43–3
 - sparse matrix techniques, 43–9 to 43–11
- Eigenvalues and eigenvectors
 - adjacency matrix, 28–5
 - canonical forms, 6–2 to 6–3
 - cone invariant departure, matrices, 26–3 to 26–5
 - fundamentals, 4–6 to 4–11
 - generalized eigenvalue problem, 43–1 to 43–3
 - graphs, 28–5 to 28–7
 - Hermitian matrices, 17–13 to 17–14
 - inequalities, 17–9 to 17–10
 - LAPACK subroutine package, 75–9 to 75–11, 75–11 to 75–13
 - large-scale matrix computations, 49–12
 - Maple software, 72–11 to 72–12
 - Mathematica software, 73–14 to 73–16
 - Matlab software, 71–9
 - matrix equalities and inequalities, 14–1 to 14–5, 14–8 to 14–10
 - multiplicity lists, 34–7 to 34–8
 - numerical stability and instability, 37–20
 - pseudoeigenvalues and pseudoeigenvectors, 16–1
 - reducible matrices, 9–8 to 9–9, 9–11
 - Schrödinger's equation, 59–7
 - sign-pattern matrices, 33–9 to 33–11
 - singular values and singular value inequalities, 17–13 to 17–14
 - sparse eigenvalue solvers, software, 77–2
 - spectrum and boundary points, 18–3
- Eigenvectors, Maple software, 72–11
- Eigenvectors, Mathematica software, 73–14
- eigshow command, Matlab software, 71–9
- EIGS subroutine package, 76–9 to 76–10
- eigtool, Matlab software, 16–12, 71–20
- Eijkhout, Victor, 74–1 to 74–7, 77–1 to 77–3
- Electron density distribution function, 60–7
- Electrostatics, 59–11
- Element, Mathematica software, 73–24
- Elementary analytic results, 26–12 to 26–13
- Elementary bidiagonal matrices, 21–5
- Elementary column operations, 6–11
- Elementary divisor domain (EDD), 23–2
- Elementary divisors, 6–8 to 6–11
- Elementary divisors rational canonical form (ED-RCF)
 - matrix, 6–8 to 6–9
- Elementary matrices, 1–12
- Elementary row operations
 - Gaussian and Gauss-Jordan elimination, 1–7
 - matrix equivalence, 23–5
 - Smith normal form, 6–11
- Elementary symmetric function, P-2 to P-3
- Element of volume, 13–25
- Eliminate, Mathematica software, 73–20, 73–23
- Elimination graph, 40–11
- Elimination ordering, 40–14
- Elimination sequence, 40–14
- ELMAT directory, Matlab software, 71–5
- Elsner theorem, 15–2
- Embedding graphs, 28–3
- Embre, Mark, 16–1 to 16–15
- Empty graphs, 28–2
- Empty matrix, 68–3
- Encoder
 - coding theory, 61–1
 - linear block codes, 61–3
- Encoding, 61–2
- Endpoint, 28–1
- Energy norm, 37–2
- Entry, matrices, 1–3
- Entry sign symmetric P -, $P_{0,1}$ - and P_0 -matrices, 35–19 to 35–20
- Entry weakly sign symmetric P -, $P_{0,1}$ - and P_0 -matrices, 35–19 to 35–20
- Envelope, reordering effect, 40–14
- Envelope method, 40–16
- Epsilon-pseudospectrum, square matrix, 71–20
- eqns, Mathematica software, 73–20
- Equalities and inequalities, matrices
 - determinantal relations, 14–10 to 14–12
 - eigenvalues, 14–1 to 14–5, 14–8 to 14–10
 - inversions, 14–15 to 14–17
 - nullity, 14–12 to 14–15
 - rank, 14–12 to 14–15
 - singular values, 14–8 to 14–10
 - spectrum localization, 14–5 to 14–8
- Equality-constrained least squares problems, 75–6 to 75–7

- Equal matrices, 1–3
- Equations of motion, 59–3
- Equation solving, Maple software, 72–9 to 72–11
- Equicorrelation matrix and structure, 52–9
- Equilibrium
 - linear dynamical systems, 56–5
 - matrix games, 50–18
- Equivalence
 - bilinear forms, 12–2
 - group representations, 68–1
 - Hermitian forms, 12–8
 - linear block codes, 61–3
 - linear dynamical systems, 56–5
 - linear independence, span, and bases, 2–7
 - matrix equivalence, 23–5
 - matrix representations, 68–3
 - scaling nonnegative matrices, 9–20
 - sesquilinear forms, 12–6
 - systems of linear equations, 1–9
 - trees, 34–8
 - vector norms, 37–3
- Equivalence class modulo, 25–10
- Equivalence relation, P–3
- Ergodic class matrices, 9–15
- Ergodic flow, 56–14
- Ergodicity coefficient
 - bounds, 9–5
 - irreducible matrices, 9–5
 - nonnegative and stochastic matrices, 9–2
 - reducible matrices, 9–10 to 9–11
- Error analysis
 - algorithms, 37–16 to 37–17
 - conditioning and condition numbers, 37–7 to 37–9
 - efficiency, 37–16 to 37–17
 - floating point numbers, 37–11 to 37–16
 - fundamentals, 37–1 to 37–2
 - linear systems conditioning, 37–9 to 37–11
 - matrix norms, 37–4 to 37–6
 - numerical stability and instability, 37–18 to 37–21
 - vector norms, 37–2 to 37–3
 - vector seminorms, 37–3 to 37–4
- Errors
 - estimation, 41–16 to 41–17
 - Krylov subspaces and preconditioners, 41–2
 - linear prediction, 64–7
 - LTI systems, 57–14
 - protection, coding theory, 61–1
 - vectors, 61–2
- ESPRIT algorithm, 64–17
- Estimated principal components, 53–5
- Estimation, correlations and variates, 53–7 to 53–8
- Estimation, least squares, 53–11 to 53–12
- Estimation error, 57–14
- Estimation of state, 57–11 to 57–13
- Euclidean algorithm, 6–11
- Euclidean distance
 - Euclidean point space, 66–2
 - metric multidimensional scaling, 53–14
- Euclidean distance matrices, 35–9 to 35–10
- Euclidean Distance Matrix, 51–9
- Euclidean domain (ED)
 - certain integral domains, 23–2
 - matrix equivalence, 23–6
 - matrix similarity, 24–2
- Euclidean geometry
 - fundamentals, 66–1
 - Gram matrices, 66–5 to 66–7
 - point spaces, 66–1 to 66–5
 - resistive electrical networks, 66–13 to 66–15
 - simplexes, 66–7 to 66–13
- Euclidean norm
 - matrix norms, 37–4
 - one-sided Jacobi SVD algorithm, 46–4
 - preconditioned Jacobi SVD algorithm, 46–6, 46–7
 - rank revealing decompositions, 46–8
 - symmetric indefinite matrices, 46–14
 - vector norms, 37–2
- Euclidean Parallel Postulate, Generalized, 65–3
- Euclidean properties
 - matrices, 53–14
 - plane, 65–4
 - point space, 66–1 to 66–5
 - projective spaces, 65–9
 - simplexes, 66–7 to 66–10
 - unitary similarity, 7–2
- Euclidean spaces
 - Gram matrices, 66–5
 - inner product spaces, 5–2
 - orthogonality, 5–5
 - semisimple and simple algebras, 70–4
 - vector spaces, 1–3
- Euclid's algorithm
 - certain integral domains, 23–2
 - matrix equivalence, 23–7
- Evaluation, permanents, 31–11 to 31–12
- EVD (eigenvalue decomposition), 42–2
- Even cycle, 33–2
- Exact breakdown, 49–8
- Exactly universal quantum gates, 62–7
- Exact numerical cancellation, 40–4
- Exact shifts, 44–6
- Exceptional, Jordan algebra
 - Jordan algebras, 69–12
 - nonassociative algebra, 69–3
- Exogenous input, 57–14
- Expand, Mathematica software, 73–25
- Expanders, graph parameters, 28–9
- Expansion
 - Bezout domains, 23–9
 - determinants, 4–1 to 4–2
 - reducible matrices, 9–8, 9–11 to 9–12
- exp command, Matlab software, 71–19
- Expectation, 52–2, 52–3
- Expected value
 - random vectors, 52–3
 - Schrödinger's equation, 59–7
 - state estimation, 57–12
 - statistics and random variables, 52–2
- Explicit QR iteration, 43–4 to 43–5

Explicit restarting, 44–6
 expm function, Matlab software, 11–11
 Exponentials, 56–2
 Exponential stability
 linear differential-algebraic equations, 55–14
 linear differential equations, 56–3
 linear ordinary differential equations, 55–10
 Exponent of matrix multiplication complexity, 47–2
 Exponents
 floating point numbers, 37–11
 linear skew product flows, 56–11
 primitive digraphs and matrices, 29–9
 expr, Mathematica software, 73–26
 Extended colored graphs, 66–10
 Extended graphs, 66–10
 Extended precision, 37–13
 Extended Tam-Schneider condition, 26–7
 Extendible characters, 68–9
 Extensions
 pseudospectra, 16–12 to 16–15
 sign solvability, 33–5
 Exterior algebra, 70–2
 Exterior point method, 50–23
 Exterior power, 13–12
 Exterior product, 13–13
 External direct sum
 direct sum decompositions, 2–5
 nonassociative algebra, 69–3
 External product, 13–25
 Extremal generators, 25–12
 Extreme pathways, 60–10
 Extreme point, 50–13
 Extreme ray
 flux balancing equation, 60–10
 linear independence and rank, 25–12
 Extreme values, 27–7
 Extreme vectors, 26–2
 ezcontour command, Matlab software, 71–15
 ezplot command, Matlab software, 71–14, 71–14 to 71–15, 71–17
 ezsurf command, Matlab software, 71–17

F

Faber and Manteuffel theorem, 41–8

Face

Collatz-Wielandt sets, 26–5
 Euclidean simplexes, 66–7
 geometry, 51–5
 Perron-Frobenius theorem, 26–2
 reducible matrices, 26–8

Facially exposed face, 51–5

Factorizations, *see also* QR factorization

Arnoldi factorization, 44–2 to 44–4
 bipartite graphs, 30–8 to 30–10
 function computation methods, 11–9 to 11–10
 information retrieval, 63–5 to 63–8
 LU factorizations, 1–13
 nonnegative matrix factorization, 63–5 to 63–8

orthogonal, least squares solutions, 39–5 to 39–6
 sparse matrix methods, 40–4 to 40–10
 total positive and total negative matrices, 21–5 to 21–6
 Factorizations, direct solution of linear solutions

fundamentals, 38–1
 Gauss elimination, 38–7 to 38–12
 LU decomposition, 38–7 to 38–12
 orthogonalization, 38–13 to 38–15
 perturbations, 38–2 to 38–5
 QR decomposition, 38–13 to 38–15
 symmetric factorizations, 38–15 to 38–17
 triangular linear systems, 38–5 to 38–7

Faithful characteristics

characters, 68–5
 group representations, 68–2
 matrix representations, 68–3

Fallat, Shaun M., 21–1 to 21–12

Fast algorithms, 47–2 to 47–4, *see also* Algorithms

Fast Fourier transform (FFT), 58–17 to 58–20, 60–8

Fast large-scale matrix computations, 49–2

Fast matrix inversion, 47–9 to 47–10

Fast matrix multiplication

advanced techniques, 47–7 to 47–9
 algorithms, 47–2 to 47–5
 applications, 47–9 to 47–10
 approximation algorithms, 47–6 to 47–7
 fundamentals, 47–1 to 47–2

F-distribution, 53–10

Feasible region, 50–1

Feasible solutions, 50–1

Feasible values, 50–1

Feedback controller, 57–13

Fejer's theorem, 8–10

Fekete's Criterion, 21–7

Fermions, 59–10

FFT, *see* Fast Fourier transform (FFT)

Fiedler, Miroslav, 66–1 to 66–15

Fiedler vectors, 36–1, 36–4, 36–7 to 36–9

Field of relational functions, 23–2

Field of values

convergence rates, 41–16
 numerical range, 18–1

Fields, P–3

Fill element, 40–4

Fill graphs, 40–14

Fill-in, sparse matrix factorizations, 49–3

Fill matrix, 40–4

Filter polynomial, 44–6

Final class, 54–5

Final subset, 9–2

Fine, Morse decomposition, 56–7

Finite dimensional

direct sum decompositions, 2–5
 nonassociative algebra, 69–2
 vector space, 2–3

Finite energy, 64–2

Finite impulse response (FIR)

adaptive filtering, 64–12
 signal processing, 64–2, 64–3
 Wiener filter, 64–11

- Finitely generated elements
 - Bezout domains, 23–8
 - max-plus algebra, 25–2
- Finitely generated ideals, 23–2
- Finite Markov chain, 54–2
- Finite Markov chains, 54–9 to 54–11
- Finite power, 64–5
- Finite precision arithmetic, 41–16 to 41–17
- Finite time exponential growth rate, 56–16
- FIR (finite impulse response)
 - adaptive filtering, 64–12
 - signal processing, 64–2, 64–3
 - Wiener filter, 64–11
- First, Mathematica software
 - matrices manipulation, 73–13
 - singular values, 73–18
 - vectors, 73–3
- First level radix 2 FFT, 58–17 to 58–18
- First (population) canonical correlations and variates, 53–7
- FIR Wiener filtering problem, 64–10
- Fischer's Determinantal Inequality, 8–10
- Fischer's inequality, 14–11
- Fixed point
 - linear dynamical systems, 56–5
 - linearization, 56–19
- Fixed spaces, 3–6
- Flag manifolds, 56–7, 56–9 to 56–11
- Flatten, Mathematica software
 - fundamentals, 73–27
 - linear programming, 73–24
 - matrices manipulation, 73–14
- Flexible algebra, 69–10
- Flip map, 22–2
- Floating point numbers, 37–11 to 37–16
- Floating point operation (flop)
 - algorithms and efficiency, 37–16
 - large-scale matrix computations, 49–2
- Floquet exponents, 56–17
- Floquet theory
 - dynamical systems, 56–12 to 56–14
 - random linear dynamical systems, 56–15 to 56–16
- Flow lattice, 30–2
- Flux balancing equation, 60–10 to 60–13
- fname command, Matlab software, 71–10, 71–11
- FOM (Full Orthogonalization Method), 41–7
- Ford-Fulkerson theorem, 27–7
- Forest, graphs, 28–2
- for loops, Matlab software, 71–11
- format short command, Matlab software, 71–8
- Formed space, 67–5
- Formulating linear programs, 50–3 to 50–7
- Formulation, 50–3 to 50–7
- Forward errors, 37–18, 37–20 to 37–21
- Forward stability, 37–18
- Four (4)-cockades, 30–4
- Fourier analysis
 - discrete theory, 58–8 to 58–17
 - fast Fourier transform, 58–17 to 58–20
 - function/functional theory, 58–2 to 58–8, 58–12 to 58–17
 - fundamentals, 58–1
- Fourier coefficients, 5–4
- Fourier expansion, 5–4
- Fourier transforms
 - Green's functions, 59–10
 - Karle-Hauptman matrix, 60–8
- Frame, 56–9
- Frameticks, Mathematica software, 73–27
- F-ratio, 53–3
- Free algebras, 69–18
- Free distance, 61–12
- Free Lie algebra, 70–2
- Free variables, 1–10
- Freeware (software), 77–1 to 77–3
- Frequency-domain analysis, 57–5 to 57–6
- Frequency response, 64–2
- Freund, Roland W., 49–1 to 49–15
- Friedland, Shmuel, 23–1 to 23–10, 24–1 to 24–11
- Friendship theorem, 28–7
- Frobenius inequality, 14–13
- Frobenius-König theorem, 27–4
- Frobenius norm
 - eigenvalues, 15–4
 - elementary analytic results, 26–12
 - irreducible matrices, 29–7
 - matrices function behavior, 16–10
 - matrix norms, 37–4
 - protein structure comparison, 60–4
 - semidefinite programming, 51–3
 - singular value decomposition, 5–11
 - square matrices, 27–6, 29–11
 - unitarily invariant norms, 17–6
 - unitary similarity, 7–2
 - weak combinatorial invariants, 27–6
- Frobenius normal form, 27–5
- Frobenius reciprocity, 68–9, 68–10
- Frobenius-Victory theorem, 26–8, 26–9
- Frontal/multifrontal methods, 40–10
- Front end, Mathematica software, 73–1
- Frucht-Kantorovich Inequality, 52–10
- Full cones, 8–10
- Full level radix 2 FFT, 58–18 to 58–19
- Full Orthogonalization Method (FOM), 41–7
- Full rank least squares problem, 5–14
- Full-rank model, 52–8
- Full reorthogonalization procedure, 42–20 to 42–21
- Fully indecomposable, 27–3 to 27–4
- Fulton studies, 17–13
- Functional inequalities
 - irreducible matrices, 9–5
 - reducible matrices, 9–10
- Function evaluation operator, 62–5 to 62–6
- Function/functional theory, 58–2 to 58–8, 58–12 to 58–17
- Functions, Matlab software, 71–11
- Functions of matrices
 - computational methods, 11–9 to 11–12
 - cosine, 11–7 to 11–8

- exponential, 11–5 to 11–6
- fundamentals, 11–1
- logarithm, 11–6 to 11–7
- sign function, 11–8
- sine, 11–7 to 11–8
- square root, 11–4 to 11–5
- theory, 11–1 to 11–4

Fundamental period, 58–2

Fundamental subspaces, 39–4

Fundamental tensor, 13–25

G

Galerkin condition

- Arnoldi factorization, 44–3
- Krylov subspace projection, 44–2

Gale-Ryser theorem, 27–7

gallery command, Matlab software, 71–5

Gantmacher-Lyapunov theorem, 26–14

Gantmacher studies, 19–4

Gaubert, Stéphane, 25–1 to 25–14

Gaussian elimination, *see also* LU factorizations

- algorithm efficiency, 37–17
- bipartite graphs, 30–7
- bisection and inverse iteration, 42–15
- direct solution of linear systems, 38–7 to 38–12
- fundamentals, 1–7 to 1–9
- Karle-Hauptman matrix, 60–8
- modeling and analyzing fill, 40–10
- numerical stability and instability, 37–20, 37–21
- reordering effect, 40–16, 40–18
- sparse matrix factorizations, 40–5, 40–9

Gaussian Network Model, 60–10

Gaussian properties, 8–9

Gauss-Jordan elimination, 1–7 to 1–9

Gauss-Markov model

- least squares estimation, 53–11
- linear statistical models, 52–8

Gauss-Markov theorem, 52–11

Gauss multipliers, 38–7

Gauss-Newton method, 51–8

Gauss-Seidel algorithm, 41–14

Gauss-Seidel methods, 41–3 to 41–4

Gauss transformations

- Gauss elimination, 38–7
- sparse matrix factorizations, 40–4

Gauss vectors, 38–7

GCDD (greatest common divisor domain), 23–2

GBAL LAPACK subroutine, 43–3

General Inverse Eigenvalue Problem (GIEP), 34–8

Generalized cycle, digraphs, 29–2

Generalized cycle products, 29–5 to 29–6

Generalized eigenvalue problem, 43–1 to 43–3

Generalized eigenvalues and eigenvectors, 59–7

Generalized Euclidean Parallel Postulate, 65–3

Generalized inverse, 52–4

Generalized Laplacian, 36–10 to 36–11

Generalized least squares estimator, 52–8

Generalized least squares problem, 39–1

Generalized line graphs, 28–4

Generalized Minimal Residual (GMRES)

- convergence rates, 41–15 to 41–16
- Krylov space methods, 41–7, 41–9 to 41–11
- linear systems of equations, 49–13
- matrices function behavior, 16–11
- preconditioners, 41–12

Generalized octonions, 69–4

Generalized quaternions, 69–4

Generalized Schur complement, 52–4

Generalized sign pattern, 33–2

Generalized Singleton bound, 61–12

Generalized singular value decomposition (GSVD), 15–12

Generalized stars, 34–10 to 34–14

Generalized variance, 52–3

General linear group

- group representations, 68–1
- matrix group, 67–1

General Schur Theorem, 43–2

Generator matrix

- convolutional codes, 61–11
- linear block codes, 61–3

Generators

- certain integral domains, 23–2
- linear independence and rank, 25–12
- polynomial, 61–7

Geometry and geometric aspects

- affine spaces, 65–1 to 65–4
- eigenvalues and eigenvectors, 4–6
- Euclidean geometry, 66–1 to 66–15
- Euclidean spaces, 65–4 to 65–6
- fundamentals, 65–1
- least squares solutions, 39–4 to 39–5
- linear programming, 50–13
- matrix power asymptotics, 25–8
- max-plus eigenproblem, 25–6
- nonnegative and stochastic matrices, 9–2
- projective spaces, 65–6 to 65–9
- semidefinite programming, 51–5

Geometry'Rotations' package, Mathematica software, 73–4

Gerhard, Jürgen, 72–21

Gersgorin discs, 14–5 to 14–7

Gersgorin theorem

- generalized cycle products, 29–5
- pseudospectra, 16–3

GGBAK LAPACK subroutine, 43–7

GBBAL LAPACK subroutine, 43–7

GGHRD LAPACK subroutine, 43–7

GIEP (General Inverse Eigenvalue Problem), 34–8

Gilbert Varshamov bound

- linear code classes, 61–10
- main linear coding problem, 61–6

Givens algorithm, 46–5 to 46–6

givens function, Matlab software, 42–9

Givens QR factorization, 38–14

Givens rotations

- orthogonalization, 38–13 to 38–14
- QR factorization, 39–9
- tridiagonalization, 42–5, 42–7 to 42–8

- Givens transformation, 38–13
- Glide reflection, 65–5
- Global invariant manifolds, 56–19
- Glossary, G–1 to G40
- GMRES (Generalized Minimal Residual)
 - convergence rates, 41–15 to 41–16
 - Krylov space methods, 41–7, 41–9 to 41–11
 - linear systems of equations, 49–13
 - matrices function behavior, 16–11
 - preconditioners, 41–12
- Golay codes, 61–8
- Goldman-Tucker theorem, 51–6
- Golub-Kahan singular value decomposition, 45–6
- Golub-Reinsch singular value decomposition, 45–6
- Gondran-Minoux properties, 25–12, 25–13
- Google (search engine)
 - information retrieval, 63–10 to 63–14
 - Markov chains, 54–4 to 54–5
 - PageRank, 63–11
 - Web search, 63–9
- Goppa code, algebraic geometric, 61–10
- grad, Mathematica software, 73–15, 73–16
- Graded algebras, 70–8 to 70–10
- Grading
 - eigenvalue problems, 15–13
 - Krylov subspaces, 49–6
 - polar decomposition, 15–8
 - singular value problems, 15–15
 - tensor algebras, 13–20 to 13–22
- Gradual underflow, 37–11 to 37–12
- Gragg studies, 44–4
- Graham-Pollak theorem, 30–9
- Gramian, Euclidean simplexes, 66–8
- Gram matrices
 - Euclidean geometry, 66–5 to 66–7
 - Hermitian matrices, 8–1, 8–2
- Gram-Schmidt calculation, Maple software, 72–6
- Gram-Schmidt methods
 - Arnoldi factorization, 44–3 to 44–4
 - unitary similarity, 7–2, 7–4
- Gram-Schmidt orthogonalization, 5–8 to 5–10
- GramSchmidt, Mathematica software, 73–4
- Graphical user interfaces (GUIs), 71–19 to 71–22
- Graphics, Matlab software, 71–14 to 71–17
- Graphics' Arrow', Mathematica software, 73–5
- Graphs, *see also* Algebraic connectivity; Euclidean geometry; *specific type of graph*
 - adjacency matrix, 28–5 to 28–7
 - association schemes, 28–11 to 28–12
 - doubly stochastic matrices, 27–10
 - eigenvalues, 28–5 to 28–7
 - fundamentals, 28–1 to 28–3
 - matrix completion problems, 35–2
 - matrix representations, 28–7 to 28–9
 - modeling and analyzing fill, 40–11
 - multiplicative D -stability, 19–6
 - multiplicities and parter vertices, 34–2
 - parameters, 28–9 to 28–11
 - simplexes, 66–10
 - special types, 28–3 to 28–5
- Grassmann, Taksar, Heyman (GTH) trick, 54–13
- Grassmann characteristics
 - dynamical systems, 56–9 to 56–11
 - Floquet theory, 56–13
 - manifolds, dynamical systems, 56–7
 - matrix pair, 15–12
 - tensor algebras, 13–21
 - tensors, 13–12 to 13–17
- Gray Code order, 31–12
- Greatest common divisor domain (GCDD), 23–2
- Greatest common divisors, 23–2
- Greatest integer function, P–3
- Greenbaum, Anne, 41–1 to 41–17
- Green's functions, 59–10 to 59–11
- Griesmer bound, 61–5
- Grobman-Hartman theorem, 56–20
- Group, P–3 to P–4
- Group inverse, 9–2
- Group of invertible linear operators, 67–1
- Group representations
 - characters, 68–5 to 68–6
 - character table, 68–6 to 68–8
 - fundamentals, 68–1 to 68–3
 - induction of characters, 68–8 to 68–10
 - matrix representations, 68–3 to 68–5
 - orthogonality relations, 68–6 to 68–8
 - restriction of characters, 68–8 to 68–10
 - symmetric group representations, 68–10 to 68–11
- Group ring, 68–2
- Grover's search algorithm, 62–15 to 62–17
- GSVD, *see* Generalized singular value decomposition (GSVD)
- GTH (Grassmann, Taksar, Heyman) trick, 54–13
- GUI, *see* Graphical user interfaces (GUIs)
- Gunaratne, Ajith, 60–13

H

- Hacjan studies, 50–23
- Hadamard-Fischer inequality, 8–10
- Hadamard inequalities
 - determinantal relations, 14–11
 - inequalities, 17–11
- Hadamard matrix
 - nonsquare case, 32–5
 - permanents, 31–8
 - square case, 32–2
- Hadamard product
 - complex sign and ray patterns, 33–14
 - positive definite matrices, 8–9
 - rank and nullity, 14–14
 - square matrices, 27–4
 - totally positive and negative matrices, 21–10, 21–11
- Hadamard's determinantal inequality, 8–10, 8–11
- Haemers, Willem H., 28–1 to 28–12
- Half-line subset, 13–24
- Halfspaces, 25–11
- Hall, Frank J., 33–1 to 33–17
- Hall matrices, 27–3, 27–4
- Hamilton cycle, 28–1

- Hamiltonian, minimally chordal symmetric, 35–15
- Hamiltonian operator, 59–2
- Hamiltonian system, 56–13
- Hamilton-Jacobi partial differential equations, 25–6
- Hamming association scheme, 28–12
- Hamming properties
 - code, 61–6, 61–9
 - distance, 61–2
 - weight, 61–2
- Han, Lixing, 5–1 to 5–16
- Hankel matrices
 - Maple software, 72–18
 - structured matrices, 48–2, 48–3
 - totally positive and negative matrices, 21–12
- HankelMatrix, Mathematica software, 73–6
- Hansen, Per Christian, 39–1 to 39–12
- Hard constraints, 51–1
- Hardware floats, Maple software, 72–14
- Hardy, Littlewood, Pólya theorem, 27–11
- Hardy space, 57–5
- Hare, Dave, 72–21
- Hartman-Grobman theorem, 56–20
- Hat matrix, 52–9
- Hautus-Popov test, 57–8
- Heat equation, 59–11
- Heat kernel, 59–11
- Height characteristic, 9–7, 26–8
- Hentzel, Irvin, 69–25
- Hereditary, 35–2
- Hermite normal form, 23–5 to 23–7, 23–6
- Hermitian characteristics
 - angular momentum and representations, 59–9
 - Arnoldi factorization, 44–3
 - classical groups, 67–5
 - differential-algebraic equations, 55–15
 - differential equations, 55–11
 - extensions, 16–13
 - iterative solution methods, 41–4 to 41–7
 - Jordan algebras, 69–13
 - Kronecker products, 10–9
 - Schrödinger's equation, 59–7, 59–8
 - Schur complements, 10–7
 - splitting theorems and stability, 26–14
- Hermitian forms, 12–7 to 12–9
- Hermitian matrices
 - adjoint, 1–4
 - adjoint operators, 5–6
 - Arnoldi process, 49–10
 - bilinear forms, 12–7 to 12–9
 - eigenvalues, 8–3 to 8–5, 15–5 to 15–6
 - fundamentals, 1–4, 8–1 to 8–2
 - inertia, 19–2
 - multiplicities and Parter vertices, 34–2
 - numerical range, 18–6
 - order properties, eigenvalues, 8–3 to 8–5
 - positive definite and semidefinite matrices, 35–8
 - positive definite characteristics, 5–2
 - pseudo-inverse, 5–12
 - relative perturbation theory, 15–14
 - singular value decomposition, 5–11
 - singular values, 17–2, 17–13 to 17–14
 - sparse matrices, 49–4 to 49–5
 - spectral theory, 7–5, 7–8
 - standard linear preserver problems, 22–5
 - submatrices and block matrices, 10–2
 - symmetric factorizations, 38–15
- Hermitian positive definite and semidefinite, ARPACK, 76–7 to 76–8
- Hermitian preconditioning, 41–3
- Hermitian properties
 - linear operators, 5–5
 - pencils, 24–6
 - property L , 24–6
- HermitianTranspose, Maple software, 72–3, 72–5
- Hershkowitz, Daniel, 19–1 to 19–10
- hes, Mathematica software, 73–16
- HessenbergDecomposition, Mathematica software, 73–19, 73–27
- Hessenberg pattern, 33–3, 33–4
- Hessian properties
 - Hermitian matrices, 8–2
 - semidefinite programming, 51–10
- Hestenes studies, 46–2
- Hidden constraint, 51–10
- Higham, Nicholas J., 11–1 to 11–12
- Higham and Tisseur studies, 16–12
- Highest weight vector, 70–7
- High relative accuracy, eigenvalues and singular values
 - accuracy, 46–2 to 46–5, 46–7 to 46–10
 - fundamentals, 46–1 to 46–2
 - one-sided Jacobi SVD algorithm, 46–2 to 46–5
 - positive definite matrices, 46–10 to 46–14
 - preconditioned Jacobi SVD algorithm, 46–5 to 46–7
 - rank revealing decomposition, 46–7 to 46–10
 - structured matrices, 46–7 to 46–10
 - symmetric indefinite matrices, 46–14 to 46–16
- High relative accuracy bidiagonal singular value decomposition, 45–7, 45–8
- hilb command, Matlab software, 71–18
- Hilbert matrix
 - linear systems conditioning, 37–11
 - preconditioned Jacobi SVD algorithm, 46–7
 - rank revealing decomposition, 46–10
- HilbertMatrix, Mathematica software
 - matrices, 73–6
 - singular values, 73–18
- Hilbert-Schmidt inner product, 13–23, 13–24
- Hilbert spaces
 - quantum computation, 62–2
 - random signals, 64–4
 - Schrödinger's equation, 59–7
- Hill's equation, 56–14
- Hirsch and Bendixson inequalities, 14–2
- HITS (Hypertext Induced Topic Search), 63–9
- HKM method, 51–8
- H -matrices, 19–9
- Hodge star operator, 13–24 to 13–26

- Hoffman polynomial, 28–5, 28–6
 - Hoffman-Wielandt inequality, 7–7
 - Hoffman-Wielandt theorem, 15–2
 - Hogben, Leslie, 6–1 to 6–14, 35–1 to 35–20
 - Hölder inequality, 37–3
 - Hölder norm, 37–2
 - Holmes, Randall R., 68–1 to 68–11
 - Homogeneity
 - cone programming, 51–2
 - coordinates, 65–7
 - differential equation, 2–3, 2–4
 - Euclidean simplexes, 66–8
 - function field, linear code classes, 61–10
 - linear differential equations, 55–2
 - line coordinates, 65–7
 - Markov chains, 54–1
 - partial inverse M -matrices, 35–14
 - pencil strict equivalence, 23–9
 - polynomials, 23–2, 23–9
 - projective spaces, 65–7
 - systems of linear equations, 1–9, 1–10
 - tensor algebras, 13–21
 - vector norms, 37–2
 - vector seminorms, 37–3
 - Homogeneous of degree, 13–21
 - Homomorphism
 - bimodules, 69–6
 - modules, 70–7
 - nonassociative algebra, 69–3
 - Homotopy approach, 20–12
 - Hopf algebra, 69–18
 - Horn inequalities, 14–9
 - Hotelling's distribution, 53–9, 53–10
 - Hotelling studies, 32–1
 - Householder method
 - algorithm efficiency, 37–17
 - singular value decomposition, 45–5
 - Householder properties
 - algorithm, 46–5 to 46–6
 - QR factorization, 38–13
 - reduction, bidiagonal form, 45–5
 - transformation, 38–13
 - vectors, 38–13
 - Householder reflections
 - orthogonalization, 38–13, 38–15
 - tridiagonalization, 42–6
 - Howell, Kenneth, 58–1 to 58–20
 - lhs, Mathematica software, 73–20
 - HSEQR LAPACK subroutine, 43–6
 - Hunacek, Mark, 65–1 to 65–9
 - Hurwitz characteristics, 57–9
 - Hurwitz command, Maple software, 72–20
 - Hurwitz matrix, 57–7
 - Hyperacute simplexes, 66–10
 - Hyperbolic characteristics
 - linear dynamical systems, 56–5
 - linearization, 56–19
 - symmetric indefinite matrices, 46–14
 - Hyperbolic Jacobi, 46–15
 - Hyperlink matrix, 63–10
 - Hyperplanes, 66–2, 66–2 to 66–4
 - Hypertext Induced Topic Search (HITS), 63–9
 - Hypotenuse, simplexes, 66–10
- ## I
-
- IAP (inertially arbitrary pattern), 33–11
 - Ideal characteristics
 - algebras, 69–18
 - Lie algebras, 70–1
 - nonassociative algebra, 69–3
 - Idempotence
 - general properties, 69–5
 - invariant subspaces, 3–6
 - nilpotence, 2–12
 - Identity, linear transformations, 3–1
 - Identity matrix
 - fundamentals, 1–4, 1–6
 - matrices, 1–4
 - max-plus algebra, 25–1
 - systems of linear equations, 1–13
 - IdentityMatrix, Mathematica software
 - decomposition, 73–19
 - eigenvalues, 73–14
 - matrices, 73–6, 73–8
 - matrix algebra, 73–11
 - Identity pattern, 33–2
 - IEPs, *see* Inverse eigenvalue problems (IEPs)
 - If, Mathematica software, 73–8
 - IF-RCF (invariant factors rational canonical form)
 - matrix, 6–12
 - if statements, Matlab software, 71–11
 - IIR (infinite impulse response), 64–2, 64–3, 64–6 to 64–7
 - Ill-conditioned properties
 - conditioning and condition numbers, 37–7
 - linear systems, 37–10 to 37–11
 - symmetric indefinite matrices, 46–16
 - Image, kernel and range, 3–5
 - Immanant, 21–10, 31–13
 - Implicitly restarted Arnoldi method (IRAM)
 - Arnoldi factorization, 44–2 to 44–4
 - convergence, 44–9 to 44–10
 - fundamentals, 44–1
 - generalized eigenproblem, 44–11
 - implicit restarting, 44–6 to 44–8
 - Krylov methods, 44–11 to 44–12
 - Krylov subspace projection, 44–1 to 44–2
 - polynomial restarting, 44–5 to 44–6
 - restarting process, 44–4 to 44–5
 - sparse matrices, 43–10
 - spectral transformations, 44–11 to 44–12
 - subspaces, 44–9 to 44–10
 - Implicitly shifted QR method, 42–9 to 42–11
 - Implicit QR iteration, 43–5
 - Implicit restarting, 44–6 to 44–8
 - Imprimitive digraphs, 29–9
 - Imprimitive gate, 62–7
 - Improper divisors, 23–2
 - Improper hydroplane, 66–8
 - Improper point, 66–8
 - Impulse response, 64–2

- imread command, Matlab software, 71–15
- Incidence matrix
 - binary matrices, 31–5
 - matrix representations, 28–7
 - square case, 32–2
- Incomplete Cholesky decomposition, 41–11
- Incomplete LU decomposition, 41–11
- Inconsistency, systems of linear equations, 1–9
- Increment, 74–2
- Indecomposition, 70–4
- Indefinite matrices, 8–6
- Independence
 - direct sum decompositions, 2–5
 - graph parameters, 28–9
- Independent and identical distribution, 53–3
- Index
 - Bezout domains, 23–9
 - Collatz-Wielandt sets, 26–5
 - doubly stochastic matrices, 27–10
 - eigenvectors, 6–2
 - linear differential-algebraic equations, 55–7
 - polynomial interpolation, 11–2
 - reducible matrices, 9–11
- Indexing module, 63–9
- Index of imprimitivity
 - imprimitive matrices, 29–10
 - irreducible matrices, 9–3
 - reducible matrices, 9–7
- Index of primitivity, 29–9
- Index period, 58–3
- Indices, Web search, 63–9
- Indirect isometry, 65–5
- Individual condition number, 15–1, 15–10
- Induced bases, 2–5, 13–15
- Induced inner product, 13–23
- Induced subdigraphs, 29–2
- Induced subgraphs, 28–2
- Induction of characters, 68–8 to 68–10
- Inductive structure, 27–4 to 27–6
- Inequalities, *see also* Equalities and inequalities, matrices
 - irreducible matrices, 9–5
 - reducible matrices, 9–10
 - singular values and singular value inequalities, 17–7 to 17–12
- Inertia
 - congruence, 8–5
 - matrix stability and inertia, 19–2 to 19–3
 - minimum rank, 33–11 to 33–12
 - preservation, 19–5
- Inertia and stability
 - additive D -stability, 19–7 to 19–8
 - fundamentals, 19–1 to 19–2
 - inertia, 19–2 to 19–3
 - Lyapunov diagonal stability, 19–9 to 19–10
 - multiplicative D -stability, 19–5 to 19–7
 - stability, 19–3 to 19–5
- Inertially arbitrary pattern (IAP), 33–11
- Inertia set, 33–11
- Inexact preconditioners, 41–16 to 41–17
- Infeasibility, 50–1
- Inference
 - multivariate normal, 53–4 to 53–5
 - statistical, 53–12 to 53–13
- Infinite dimensional, 2–3
- Infinite impulse response (IIR), 64–2, 64–3, 64–6 to 64–7
- Infinite Markov chain, 54–2
- Infinitesimal element, 21–11
- infinity, Maple software, 72–5
- Infinity, Mathematica software, 73–26
- Information retrieval
 - fundamentals, 63–1
 - Google's page rank, 63–10 to 63–14
 - latent semantic indexing, 63–3 to 63–5
 - nonnegative matrix factorization, 63–5 to 63–8
 - vector space method, 63–1 to 63–3
 - Web search, 63–8 to 63–10
- Inheritance principle, 8–7
- Inhomogeneity
 - Euclidean simplexes, 66–8
 - linear differential equations, 55–2
- Initial minor data sets, 21–7
- Initial parameter settings, ARPACK, 76–6
- Initial state vector, 4–10
- Initial value problem, 55–2, 56–2
- Injective, kernel and range, 3–5
- inline command, Matlab software, 71–11 to 71–12
- Inner distribution, 28–11
- Inner product and inner product spaces
 - adjoints of linear operators, 5–5 to 5–6
 - floating point numbers, 37–14
 - fundamentals, 5–1 to 5–3
 - multilinear algebra, 13–22 to 13–24
 - orthogonality relations, 68–7
- Innovations process, 64–6
- Innovations representation, 64–6
- Input, algorithms and efficiency, 37–16
- Input space, 57–2
- Insert, Mathematica software, 73–13
- Instability, *see also* Stability
 - error analysis, 37–18 to 37–21
 - numerical methods, 37–18 to 37–21
- Integers, Mathematica software, 73–24
- Integral domains, matrices over
 - Bezout domains, 23–8 to 23–9
 - certain integral domains, 23–1 to 23–4
 - fundamentals, 23–1
 - linear equations, 23–8 to 23–9
 - matrix equivalence, 23–4 to 23–8
 - strict equivalence, pencils, 23–9 to 23–10
- Integral kernels, 59–10
- Integrate, Mathematica software, 73–1
- Interactive software
 - freeware, 77–1 to 77–3
 - Maple, 72–1 to 72–21
 - Mathematica, 73–1 to 73–27
 - Matlab, 71–1 to 71–22
- Interchange
 - classes of $(0,1)$ -matrices, 27–7
 - tournament matrices, 27–9
- interface, Maple software, 72–7

- Interior point methods, 50–23 to 50–24
- Interior solution, 50–23
- Interlaces, P–4
- Interlacing inequalities, 34–2
- Interlacing inequalities, 8–3 to 8–4
- Internal direct sum, 69–3
- Internal stability, 57–15, *see also* Stability
- International Standard Book Number (ISBN) code, 61–4
- Intersection, graphs, 28–2
- Intersection number, 28–11
- Intraclass correlation, 52–9
- Invariant factors
 - canonical forms, 6–12 to 6–14
 - inverse eigenvalue problems, 20–3
 - matrix equivalence, 23–5
- Invariant factors rational canonical form (IF-RCF)
 - matrix, 6–12
- Invariant manifold theorem, 56–20, 56–21
- Invariant polynomials
 - inverse eigenvalue problems, 20–3
 - matrix equivalence, 23–5
- Invariants
 - face, 26–2
 - group representations, 68–1
 - probability vectors, 54–2
 - random linear dynamical systems, 56–14
 - simultaneous similarity, 24–8
 - subspaces, projections, 3–6 to 3–7
 - vectors, 54–2
- inv command, Matlab software, 71–17
- Inverse, Maple software, 72–13
- Inverse, Mathematica software
 - eigenvalues, 73–16
 - linear systems, 73–20, 73–22
 - matrices, 73–6, 73–8
 - matrix algebra, 73–10, 73–11
- Inverse, random vectors, 52–4
- Inverse discrete Fourier transform (inverse DFT), 58–9
- Inverse eigenvalue problems (IEPs)
 - affine parameterized IEPs, 20–10 to 20–12
 - fundamentals, 20–1
 - nonnegative matrices, 9–22, 20–5 to 20–10
 - nonzero spectra, nonnegative matrices, 20–7 to 20–8
 - numerical methods, 20–11 to 20–12
 - prescribed entries, 20–1 to 20–3
 - spectra, nonnegative matrices, 20–6 to 20–10
 - trees, 34–8
 - 2x2 block type, 20–3 to 20–5
- Inverse Fourier transform
 - Fourier analysis, 58–2
 - Karle-Hauptman matrix, 60–8
- Inverse iteration, 42–2, 42–14 to 42–15
- Inverse Lanczos methods, 16–11
- Inverse matrices, 1–12
- InverseMatrixNorm, Mathematica software, 73–10
- Inverse M -matrices
 - completion problems, 35–14 to 35–15
 - fundamentals, 9–17
- Inverse nonnegatives
 - inverse patterns, 33–12
 - M -matrices, 9–17
 - splitting theorems and stability, 26–13
- Inverse patterns, 33–12 to 33–14
- Inverse-positivity matrices, 9–17
- Inverse sign-pattern matrices, 33–12 to 33–14
- Inverse tridiagonal matrices, 21–3
- Inversion
 - fast matrix multiplication, 47–9, 47–9 to 47–10
 - isomorphism, 3–7
 - matrix equalities and inequalities, 14–15 to 14–17
- Inverted file, Web search, 63–9
- Invertibility
 - fundamentals
 - isomorphism, 3–7
 - matrix equivalence, 23–5
- Invertible linear operators, 67–1
- Invertible matrices
 - decomposable tensors, 13–7
 - fundamentals, 1–12
 - LU factorizations, 1–15
- Invertible Matrix Theorem, 1–12
- Invert spectral transformation mode, ARPACK, 76–7
- inv function, Matlab software, 71–3
- Involution, 69–2
- IRAM, *see* Implicitly restarted Arnoldi method (IRAM)
- Irreducibility
 - bimodules, 69–6
 - certain integral domains, 23–2
 - characterizing, 9–3
 - characters, 68–5, 68–6
 - group representations, 68–1
 - imprimitive matrices, 29–9 to 29–11
 - matrix group, 67–1
 - matrix representations, 68–3
 - max-plus algebra, 25–2
 - modules, 70–7
 - Perron-Frobenius theorem, 26–2
 - simultaneous similarity, 24–8
- Irreducible classes, 54–5 to 54–7
- Irreducible components, 27–5 to 27–6
- Irreducible matrices
 - balancing, 43–4
 - digraphs, 29–6 to 29–8
 - fundamentals, 9–2 to 9–7
 - nonnegative matrices, 9–2 to 9–7
- ISBN (International Standard Book Number) code, 61–4
- Isolated invariant, 56–7
- Isolated vertex, 28–2
- Isometry
 - classical groups, 67–5
 - Euclidean spaces, 65–4
- Isomorphism
 - graphs, 28–1
 - Grassmannian manifolds, 56–10
 - group representations, 68–1
 - matrix representations, 68–3
 - nonassociative algebra, 69–3
 - nonsingularity characterization, 3–7 to 3–8
- Isoperimetric number, 28–9

Isotalo, Jarkko, 53–14

Iterative methods

function computation methods, 11–10

Krylov subspaces and preconditioners, 41–2

Toeplitz solvers, 48–5

Iterative solution methods, linear solutions

CG convergence rates, 41–14 to 41–15

convergence rates, 41–14 to 41–16

error estimation, 41–16 to 41–17

finite precision arithmetic, 41–16 to 41–17

fundamentals, 41–1 to 41–2

GMRES convergence rates, 41–15 to 41–16

Hermitian problems, 41–4 to 41–7

inexact preconditioners, 41–16 to 41–17

Krylov subspaces, 41–2 to 41–4

MINRES convergence rates, 41–14 to 41–15

non-Hermitian problems, 41–7 to 41–11

non-optimal Krylov space methods, 41–7 to 41–11

optimal Krylov space methods, 41–4 to 41–11

preconditioned algorithms, 41–12 to 41–14

preconditioners, 41–2 to 41–4, 41–11 to 41–12

stopping criteria, 41–16 to 41–17

matrix function, 11–4

matrix product, 18–9

Perron-Schaefer condition, 26–7

simultaneous similarity, 24–10

Jordan canonical form, *see also* real-Jordan canonical form

canonical forms, 6–3 to 6–6

differential equations, 55–6, 55–8

matrix function, 11–1 to 11–2, 11–4

reducible matrices, 9–8

unitary similarity, 7–3

Jordan characteristics

chain, 6–4

form matrices, 56–3

identity, nonassociative algebra, 69–3

invariants, 6–3, 6–4

product, nonassociative algebra, 69–3

structures, simultaneous similarity, 24–9

jordan command, Matlab software, 71–9

JordanDecomposition, Mathematica software, 73–18

JordanForm, Maple software, 72–11, 72–12

Jordan-Wielandt matrix, 17–2

J

Jacobian characteristics

linearization, 56–20

nonassociative algebra, 69–2

Jacobi-Davidson methods, 43–10 to 43–11

Jacobi identity

Akivis identity, 69–19

Lie algebras, 70–2

nonassociative algebra, 69–2

Jacobi matrices, 21–4

Jacobi methods

Krylov subspaces and preconditioners, 41–3 to 41–4

symmetric matrix eigenvalue techniques, 42–17 to 42–19

Jacobi rotation

Jacobi method, 42–17

singular value decomposition, 45–11 to 45–12

Jacobi's Theorem, 4–5

Jeffrey, David J., 72–1 to 72–21

Johnson, Charles R., 34–1 to 34–15

Johnson association scheme, 28–12

Join, graphs, 28–2

Join, Mathematica software

matrices manipulation, 73–13

vectors, 73–3

Jordan algebras

computational methods, 69–21

nonassociative algebra, 69–3, 69–12 to 69–14

power associative algebras, 69–15

Jordan basis, 6–3, 6–4, 6–5

Jordan blocks

Jordan canonical form, 6–3

K -reducible matrices, 26–9

linear differential equations, 56–4

K

Kahan-Cao-Xie-Li theorem, 15–3

Kahan matrix, 17–4

Kalman-Bucy filter

LTI systems, 57–17

state estimation, 57–11, 57–12, 57–13

Kalman decomposition, 57–7 to 57–8

Kalman filter, 57–13

Kamm and Nagy studies, 48–9

Kantorovich Inequality, 52–10

Kapranov rank, 25–13

Karle-Hauptman matrix, 60–7 to 60–9

Karp's formula

maximal cycle mean, 25–5

max-plus eigenproblem, 25–7

Kaufman studies, 44–4

Kernel

Bezout domains, 23–8

characters, 68–5

group representations, 68–2

least squares solution, 39–4

linear independence, span, and bases, 2–6

linear inequalities and projections, 25–10

matrix representations, 68–3

range, 3–5 to 3–6

kernel, Mathematica software, 73–1

Killing form

Malcev algebras, 69–16, 69–17

semisimple and simple algebras, 70–3

Kinetic energy

Lagrangian mechanics, 59–5

oscillation modes, 59–2

Kingman's inequality, 25–5

Kleene star, 25–2, 25–3

Kliemann, Wolfgang, 6–14, 56–1 to 56–21

Klienbergl, Jon, 63–9
 Klyachko studies
 eigenvalues, 17–13
 Hermitian matrices, 8–4
 Knott Inequality, 52–10
 Knutson and Tao studies
 eigenvalues, 17–13
 Hermitian matrices, 8–4
 Koteljanskii’s Determinantal Inequality, 8–10
 Kravcuk polynomials, 28–12
 K -reducible matrices, 26–8 to 26–13
 Krein condition, 28–11
 Kreiss Matrix theorem, 16–10
 Kronecker canonical form, 55–7 to 55–10
 Kronecker product
 linear maps, 13–9
 Matlab software, 71–4
 matrix similarities, 24–1
 partitioned matrices, 10–8 to 10–9
 positive definite matrices, 8–10
 rank and nullity, 14–13
 semidefinite programming, 51–3, 51–4
 structured matrices, 48–4
 tensor products, 13–8
 Kronecker symbol, 28–11
 kron function, Matlab software, 71–4
 Krylov characteristics
 implicitly restarted Arnoldi method, 44–11 to 44–12
 matrix, Lanczos method, 42–19
 sequence, 49–5
 spaces, 41–2
 vectors, 42–20
 Krylov-Schur algorithm, 43–9
 Krylov subspaces
 eigenvalue computations, 49–12
 iterative solution methods, 41–2 to 41–4
 large-scale matrix computations, 49–5 to 49–6
 projection, 44–1 to 44–2
 k -step, 44–3
 Ky-Fan norms
 polar decomposition, 15–8
 unitarily invariant norms, 17–5, 17–6

L

Lagrangian function, 51–5
 Lagrangian mechanics, 59–5 to 59–6
 Lagrangian properties
 multiplier, differential equations, 55–3
 primal-dual interior point algorithm, 51–8
 relaxation, 51–10
 Laguerre polynomial, 31–10
 Lanczos algorithm
 Krylov space methods, 41–4 to 41–5, 41–10
 symmetric matrix eigenvalue techniques, 42–19 to 42–21
 Lanczos matrices
 eigenvalue computations, 49–12
 linear systems of equations, 49–13
 symmetric Lanczos process, 49–7

Lanczos methods
 implicit restarting, 44–6
 pseudospectra computation, 16–11
 total least squares problem, 48–9
 Lanczos process, 49–10
 Lanczos vectors
 Arnoldi factorization, 44–3
 nonsymmetric Lanczos process, 49–8
 symmetric Lanczos process, 49–7
 Landau’s theorem, 27–9
 Langou, Julien, 74–1 to 74–7, 75–1 to 75–23, 77–1 to 77–3
 Langville, Amy N., 63–1 to 63–14
 LAPACK subroutine package
 dense matrices, 43–3
 DLARGV subroutine, 42–9
 DLARTV subroutine, 42–9
 DLAR2V subroutine, 42–9
 DORGTR, 42–8
 DORGTR subroutine, 42–8
 DSBTRD subroutine, 42–9
 DSTEBZ subroutine, 42–15
 DSTEDC subroutine, 42–14
 DSTEGR subroutine, 42–17
 DSTEIN subroutine, 42–15
 DSYEV subroutine, 42–11
 DSYTRD, 42–8
 DSYTRD subroutine, 42–8
 equality-constrained least squares problems, 75–6 to 75–7
 execution times, 42–22
 fundamentals, 75–1 to 75–2
 GEBAL subroutine, 43–3
 GGBAK subroutine, 43–7
 GGBAL subroutine, 43–7
 GGHRD subroutine, 43–7
 HSEQR subroutine, 43–6
 least squares problems, 75–4 to 75–7
 linear equality-constrained least squares problems, 75–6 to 75–7
 linear least squares problems, 75–4 to 75–6
 linear model problem, 75–8 to 75–9
 linear system of equations, 75–2 to 75–4
 models, 75–8 to 75–9
 nonsymmetric eigenproblems, 75–17 to 75–20
 nonsymmetric eigenvalue problem, 75–11 to 75–13
 singular value decomposition, 75–13 to 75–15, 75–20 to 75–23
 symmetric definite eigenproblems, 75–15 to 75–17
 symmetric eigenvalue problem, 75–9 to 75–11
 TGEVC subroutine, 43–7
 TGSEN subroutine, 43–7
 TGSNA subroutine, 43–7
 TREVC subroutine, 43–6
 TREXC subroutine, 43–7
 tridiagonalization, 42–8
 TRSEN subroutine, 43–7
 TRSNA subroutine, 43–7
 Laplace expansion
 determinantal relations, 14–10
 determinants, 4–1, 4–5

- multiplication, 13–18 to 13–19
 - permanents, 31–2
 - permanents evaluation, 31–12
- Laplace transforms
 - frequency-domain analysis, 57–6
 - LTI systems, 57–14
- Laplacian matrices
 - algebraic connectivity, 36–1, 36–2
 - Fiedler vectors, 36–7, 36–8 to 36–9
 - Hermitian matrices, 8–5
- Laplacian properties
 - algebraic connectivity, 36–10 to 36–11
 - eigenvalues, 28–7
 - graph parameters, 28–9, 28–10
 - graphs, 28–8 to 28–9
 - matrix representations, 28–7
- Large-scale matrix computations
 - Arnoldi process, 49–10 to 49–11
 - dimension reduction, 49–14 to 49–15
 - eigenvalue computations, 49–12
 - fundamentals, 49–1 to 49–2
 - Krylov subspaces, 49–5 to 49–6
 - linear dynamical systems, 49–14 to 49–15
 - linear systems, equations, 49–12 to 49–14
 - nonsymmetric Lanczos process, 49–8 to 49–10
 - sparse matrix factorizations, 49–2 to 49–5
 - symmetric Lanczos process, 49–6 to 49–7
- Laser method, 47–8, 47–9
- Last, Mathematica software
 - matrices manipulation, 73–13
 - singular values, 73–18
 - vectors, 73–3
- Latent semantic indexing (LSI), 63–3 to 63–5
- Latin rectangle, 31–6
- Lawley-Hotelling trace statistic, 53–13
- LDPC (low density parity check) codes, 61–11
- LDU factorization, 1–14 to 1–15
- Leading diagonal, 15–12
- Leading dimension, 74–2
- Leading entry, 1–7
- Leading principle matrices, 1–6, 1–15
- Leading principle minors
 - determinants, 4–3
 - recognition and testing, 21–7
 - stability, 19–3
- Leading principle submatrices, 1–4
- Leal Duarte, António, 34–1 to 34–15
- Least squares algorithms, 39–6 to 39–7
- Least squares estimation
 - linear statistical models, 52–8
 - multivariate statistical analysis, 53–11 to 53–12
- Least squares problems
 - fundamentals, 5–14 to 5–16
 - LAPACK subroutine package, 75–4 to 75–7
 - Matlab software, 71–7 to 71–9
 - numerical stability and instability, 37–20
- Least squares solutions, linear systems
 - algebraic aspects, 39–4 to 39–5
 - algorithms, 39–6 to 39–7
 - damped least squares, 39–9 to 39–10
 - data fitting, 39–3 to 39–4
 - downdating, 39–8 to 39–9
 - fundamentals, 39–1 to 39–3
 - geometric aspects, 39–4 to 39–5
 - orthogonal factorizations, 39–5 to 39–6
 - QR factorization, 39–8 to 39–9
 - rank revealing decompositions, 39–11 to 39–12
 - sensitivity, 39–7 to 39–8
 - updating, 39–8 to 39–9
- Lebesgue spaces, 57–5
- Lee and Seung algorithm, 63–6, 63–7
- Left alternative algebra, 69–10
- Left alternative identities, 69–2
- Left deflating subspaces, 55–7
- left divide operator, Matlab software, 71–7
- Left eigenvector, 4–6
- Left Kronecker indices, 55–7
- Left Krylov subspace, 49–8
- Left Lanczos vectors, 49–8
- Left-looking methods, 40–10
- Left Moufang identity, 69–10
- Left multiplication operators, 69–5
- Left-normalized product, 69–16
- Left preconditioning
 - BiCGSTAB algorithm, 41–12
 - Krylov subspaces and preconditioners, 41–3
- Left reducing subspaces, 55–7
- Left regular representation, 68–2
- Left singular space, 45–1
- Left singular vectors, 5–10, 45–1
- Legs, simplexes, 66–10
- Length, Mathematica software
 - fundamentals, 73–27
 - matrices, 73–6, 73–7
 - matrix algebra, 73–12
 - vectors, 73–3, 73–4
- Length characteristics
 - Euclidean point space, 66–2
 - graphs, 28–1
 - max-plus algebra, 25–2
 - sign-pattern matrices, 33–2
 - stars, 34–10
- length command, Matlab software, 71–2, 71–9
- Length of a walk
 - digraphs, 29–2
 - graphs, 28–1
- Leon, Steven J., 71–1 to 71–22
- Level characteristic, 26–8
- Levinson-Durbin algorithm, 64–8, 64–9
- Lévy-Desplanques theorem, 14–6
- Li, Chi-Kwong, 18–1 to 18–11
- Li, Ren-Cang, 15–1 to 15–16
- Lie algebras
 - angular momentum and representations, 59–10
 - fundamentals, 70–1 to 70–3
 - graded algebras, 70–8 to 70–10
 - modules, 70–7 to 70–10
 - nonassociative algebra, 69–2, 69–3
 - semisimple algebras, 70–3 to 70–7
 - simple algebras, 70–3 to 70–7
- Liénard-Chipart Stability Criterion, 19–4
- Likelihood function, 53–4

- Likelihood ratio statistic, 53–13
- Limits, nonnegative and stochastic matrices, 9–2
- Limit set, chain recurrence, 56–7
- linalg package, Maple, 72–1
- Linder, David, 72–21
- Line, Mathematica software, 73–5
- Linear algebra
 - adjoints of linear operators, 5–5 to 5–6
 - annihilator, 3–8 to 3–9
 - bases, 2–10 to 2–12, 3–4
 - change of basis, 2–10 to 2–12, 3–4
 - coordinates, 2–10 to 2–12
 - determinants, 4–1 to 4–6
 - dimension theorem, 2–6 to 2–9
 - direct sum decompositions, 2–4 to 2–6
 - eigenvalues and eigenvectors, 4–6 to 4–11
 - Gaussian elimination, 1–7 to 1–9
 - Gauss-Jordan elimination, 1–7 to 1–9
 - Gram-Schmidt orthogonalization, 5–8 to 5–10
 - idempotence, 2–12
 - inner product spaces, 5–1 to 5–3, 5–5 to 5–6
 - invariant subspaces, 3–6 to 3–7
 - isomorphism, 3–7 to 3–8
 - kernel, 3–5 to 3–6
 - least-squares problems, 5–14 to 5–16
 - linear functionals, 3–8 to 3–9
 - linear independence, 2–1 to 2–3
 - linear transformations, 3–1 to 3–9
 - LU factorization, 1–13 to 1–15
 - Maple software, 72–12 to 72–13
 - matrices, 1–3 to 1–6, 1–11 to 1–13, 2–6 to 2–9, 3–3 to 3–4
 - nilpotence, 2–12
 - nonsingularity characterizations, 2–9 to 2–10, 3–7 to 3–8
 - null space, 2–6 to 2–9
 - orthogonality, 5–3 to 5–10
 - orthogonal projection, 5–6 to 5–8
 - projections, 3–6 to 3–7
 - pseudo-inverse, 5–12 to 5–14
 - QR factorization, 5–8 to 5–10
 - range, 3–5 to 3–6
 - rank, 2–6 to 2–9
 - similarity, 3–4
 - singular value decomposition, 5–10 to 5–12
 - span, 2–1 to 2–3
 - systems of linear equations, 1–9 to 1–11
 - transformations, linear, 3–1 to 3–9
 - vectors, 1–1 to 1–3, 2–3 to 2–4, 3–2 to 3–3
- LinearAlgebra'Cholesky', Mathematica software, 73–27
- LinearAlgebra'MatrixManipulation', Mathematica software
 - decomposition, 73–19
 - fundamentals, 73–26
 - linear systems, 73–22
- LinearAlgebra'MatrixManipulation' package, Mathematica software, 73–2
- LinearAlgebra[Modular], Maple software, 72–12 to 72–13
- LinearAlgebra'Orthogonalization' package, Mathematica software, 73–4, 73–5
- LinearAlgebra package, Maple, 72–1, 72–2
- LinearAlgebra'Tridiagonal', Mathematica software, 73–20
- Linear block codes, 61–3 to 61–4
- Linear characters, 68–5
- Linear code classes, 61–6 to 61–11
- Linear combination
 - Euclidean point space, 66–2
 - span and linear independence, 2–1
- Linear constraints, 50–1
- Linear dependence, 2–1, 66–2
- Linear differential-algebraic equations, 55–7 to 55–10, 55–14 to 55–16
- Linear differential equation of order, 55–1
- Linear differential equations
 - differential equations, 55–1 to 55–5
 - dynamical systems, 56–2 to 56–4
 - numerical methods, 54–12
- Linear dynamical systems, 49–14 to 49–15, 56–5 to 56–7, *see also* Dynamical systems; Linear systems
- Linear equality-constrained least squares problems, 75–6 to 75–7
- Linear equations
 - cone invariant departure, matrices, 26–11 to 26–12
 - matrices over integral domains, 23–8 to 23–9
 - systems of linear equations, 1–9
- LinearEquationsToMatrices, Mathematica, 73–20, 73–22
- Linear form
 - linear functionals and annihilator, 3–8
 - linear programming, 50–1
- Linear functionals, annihilator, 3–8 to 3–9
- Linear hull, 66–2
- Linear independence and rank
 - Euclidean point space, 66–2
 - max-plus algebra, 25–12 to 25–14
 - span, 2–1 to 2–3
- Linear inequalities and projections, 25–10 to 25–12
- Linearization
 - alternative algebras, 69–11
 - dynamical systems, 56–19 to 56–21
 - Jordan algebras, 69–12
- Linear least squares problems, 75–4 to 75–6
- Linear maps, 3–1, 13–8 to 13–10, *see also* Linear transformations
- Linear matrix groups, 67–3 to 67–5
- Linear matrix similarities, 24–1
- Linear model problem, 75–8 to 75–9
- Linear operators
 - linear transformations, 3–1
 - nonassociative algebra, 69–3
- Linear ordinary differential equations
 - fundamentals, 55–5 to 55–6
 - stability, 55–10 to 55–14
- Linear output feedback, 57–7
- Linear prediction, 64–7 to 64–9
- Linear preserver problems
 - additive preservers, 22–7 to 22–8
 - fundamentals, 22–1 to 22–2
 - multiplicative preservers, 22–7 to 22–8
 - nonlinear preservers, 22–7 to 22–8

- problems, 22–4 to 22–7
- standard forms, 22–2 to 22–4
- Linear programming
 - canonical forms, 50–7 to 50–8
 - duality, 50–13 to 50–17
 - formulation, 50–3 to 50–7
 - fundamentals, 50–1 to 50–2
 - geometric interpretation, phase 2, 50–13
 - interior point methods, 50–23 to 50–24
 - linear approximation, 50–20 to 50–23
 - Mathematica software, 73–23 to 73–24
 - matrix games, 50–18 to 50–20
 - parametric programming, 50–17 to 50–18
 - phase 2 geometric interpretation, 50–13
 - pivoting, 50–10 to 50–11
 - sensitivity analysis, 50–17 to 50–18
 - simplex method, 50–11 to 50–13
 - standard forms, 50–7 to 50–8
 - standard row tableaux, 50–8 to 50–10
- LinearProgramming, Mathematica software, 73–24
- Linear programs, 50–2
- Linear-quadratic Gaussian (LQG) problem, 57–14
- Linear-quadratic regulator (LQR) problem, 57–14
- Linear representation, 68–1
- Linear semidefinite programming, 51–3
- Linear skew product flows, 56–11 to 56–12
- LinearSolve, Maple software, 72–9
- LinearSolve, Mathematica software
 - decomposition, 73–19
 - linear systems, 73–20, 73–21, 73–22
- Linear state feedback, 57–7
- Linear statistical models
 - fundamentals, 52–8 to 52–15
 - random vectors, 52–1 to 52–8
- Linear subspaces, 66–2
- Linear system perturbations, 38–2 to 38–5
- Linear systems
 - conditioning, error analysis, 37–9 to 37–11
 - dynamical, 49–14 to 49–15
 - equations, 49–12 to 49–15
 - fundamentals, 1–9
 - large-scale matrix computations, 49–12 to 49–15
 - Mathematica software, 73–20 to 73–23
 - Matlab software, 71–7 to 71–9
 - structured matrices computations, 48–5 to 48–8
- Linear systems, direct solution
 - fundamentals, 38–1
 - Gauss elimination, 38–7 to 38–12
 - LU decomposition, 38–7 to 38–12
 - orthogonalization, 38–13 to 38–15
 - perturbations, 38–2 to 38–5
 - QR decomposition, 38–13 to 38–15
 - symmetric factorizations, 38–15 to 38–17
 - triangular linear systems, 38–5 to 38–7
- Linear systems, iterative solution methods
 - CG convergence rates, 41–14 to 41–15
 - convergence rates, 41–14 to 41–16
 - error estimation, 41–16 to 41–17
 - finite precision arithmetic, 41–16 to 41–17
 - fundamentals, 41–1 to 41–2
 - GMRES convergence rates, 41–15 to 41–16
 - Hermitian problems, 41–4 to 41–7
 - inexact preconditioners, 41–16 to 41–17
 - Krylov subspaces, 41–2 to 41–4
 - MINRES convergence rates, 41–14 to 41–15
 - non-Hermitian problems, 41–7 to 41–11
 - non-optimal Krylov space methods, 41–7 to 41–11
 - optimal Krylov space methods, 41–4 to 41–11
 - preconditioned algorithms, 41–12 to 41–14
 - preconditioners, 41–2 to 41–4, 41–11 to 41–12
 - stopping criteria, 41–16 to 41–17
- Linear systems, least squares solutions
 - algebraic aspects, 39–4 to 39–5
 - algorithms, 39–6 to 39–7
 - damped least squares, 39–9 to 39–10
 - data fitting, 39–3 to 39–4
 - downdating, 39–8 to 39–9
 - fundamentals, 39–1 to 39–3
 - geometric aspects, 39–4 to 39–5
 - orthogonal factorizations, 39–5 to 39–6
 - QR factorization, 39–8 to 39–9
 - rank revealing decompositions, 39–11 to 39–12
 - sensitivity, 39–7 to 39–8
 - updating, 39–8 to 39–9
- Linear systems of equations
 - LAPACK subroutine package, 75–2 to 75–4
 - large-scale matrix computations, 49–12 to 49–15
- Linear time-invariant (LTI) systems
 - analysis, 57–7 to 57–10
 - control theory, 57–2
 - signal processing, 64–1
- Linear transformations
 - Euclidean spaces, 65–4
 - linear algebra, 3–1 to 3–9
- Linear unbiased estimator, 52–9
- Linear zero function, 52–9
- Line graphs, 28–4
- Lines
 - affine spaces, 65–2
 - Euclidean point space, 66–2
 - matrices, 27–2
 - projective spaces, 65–6
 - segments, 50–13, 65–2
- Linklessly embedded graphs, 28–4
- ListDensityPlot, Mathematica software, 73–27
- L*-matrices
 - sign-pattern matrices, 33–5 to 33–7
 - sign solvability, 33–5
- L*-module, 70–7
- Local invariant polynomials, 23–9
- Local Perron-Schaefer conditions, 26–6
- Local similarity, 22–7, 24–1
- Local spectral radius, 26–2
- Local stable/unstable manifold, 56–19
- Location, numerical range, 18–4 to 18–6
- Loewner ordering, 8–10
- Loewner partial ordering, 52–4
- Loewy, Raphael, 12–1 to 12–9
- Log, Mathematica software, 73–26
- Logarithms, 11–6

Log-barrier problem, 51–8
 Logical operators, Matlab software, 71–12
 Look-ahead strategies, 40–17, 41–10
 Loop digraphs, 29–2
 Loop graphs, 28–1
 Lorentz cone, 51–2
 Lovász parameter, 28–10
 Low density parity check (LDPC) codes, 61–11
 Lower bounds, 47–8
 Lower Collatz-Wielandt numbers, 26–4
 LowerDiagonalMatrix, Mathematica software, 73–6
 Lower Hessenberg matrices, 10–4
 Lowering operator, 59–8, 59–9
 Lower shift matrices, 48–1
 Lower triangular linear matrix, 38–5
 Lower triangular matrices, 1–4, 10–4
 Löwner partial ordering, 52–4
 LQG (linear-quadratic Gaussian) problem, 57–14
 LQR (linear-quadratic regulator) problem, 57–14
 LSI, *see* Latent semantic indexing (LSI)
 LTI (linear time-invariant) systems
 analysis, 57–7 to 57–10
 control theory, 57–2, 57–7 to 57–10
 signal processing, 64–1
 LUBackSubstitute, Mathematica software, 73–19
 LUBackSubstitution, Mathematica software, 73–20, 73–22
 LU decomposition
 direct solution of linear systems, 38–7 to 38–12
 incomplete LU decomposition, 41–11
 LUDecomposition, Maple software, 72–9, 72–10
 LUDecomposition, Mathematica software
 decomposition, 73–18, 73–19
 linear systems, 73–20, 73–22
 Luenberger observer, 57–11, 57–12, 57–13
 LU factorizations
 Gaussian Eliminations, 1–13 to 1–15
 sparse matrix factorizations, 49–3
 LUMatrices, Mathematica software, 73–19
 Lyapunov equation
 differential-algebraic equations, 55–15 to 55–16
 differential equations, 55–11
 matrix equations, 57–10
 semidefinite programming, 51–9
 Lyapunov exponents
 Floquet theory, 56–13
 linear differential equations, 56–2
 linearization, 56–21
 linear skew product flows, 56–11
 random linear dynamical systems, 56–15, 56–16
 Lyapunov properties
 diagonal stability, 19–9 to 19–10
 scaling factor, 19–9
 spectrum, 56–16, 56–17
 stability, 55–10, 55–14
 Lyapunov spaces
 Grassmannian and flag manifolds, 56–10, 56–11
 linear differential equations, 56–2
 random linear dynamical systems, 56–15
 Lyapunov Stability Criterion, 19–4

M

Machado, Armando, 13–1 to 13–26
 Machine epsilon, 37–12
 Mackey's Subgroup theorem, 68–9
 Main angles, adjacency matrix, 28–5
 Main Coding Theory Problem, 61–5
 Main diagonal, matrices, 1–4
 Main Linear Coding Problem, 61–5 to 61–6, 61–12
 Majorization, P–4
 Malcev algebras, 69–2, 69–3, 69–16 to 69–17
 Malcev module, 69–16
 Malcev theorem, 67–2
 M - and M_0 - matrices, 35–12 to 35–13
 Manipulation of matrices, 73–13 to 73–14
 Mantissa, 37–11
 Map, Mathematica software
 decomposition, 73–19
 eigenvalues, 73–16
 fundamentals, 73–26
 matrices, 73–7
 matrix algebra, 73–10
 singular values, 73–17
 Maple software
 arrays, 72–8 to 72–9
 canonical forms, 72–15 to 72–16
 eigenvalues and eigenvectors, 72–11 to 72–12
 equation solving, 72–9 to 72–11
 functions of matrices, 72–19 to 72–20
 fundamentals, 72–1 to 72–2
 linear algebra, 72–12 to 72–13
 matrices, 72–4 to 72–7
 matrix factoring, 72–9 to 72–11
 modular arithmetic, 72–12 to 72–13
 numerical linear algebra, 72–13 to 72–15
 stability of matrices, 72–20 to 72–21
 structured matrices, 72–16 to 72–18
 vectors, 72–2 to 72–4
 Mapping theorems, 16–2
 MapThread, Mathematica software, 73–13
 Markov chains
 censoring, 54–11
 doubly stochastic matrices, 27–10
 eigenvalues and eigenvectors, 4–10
 finite, 54–9 to 54–11
 fundamentals, 54–1 to 54–5
 irreducible classes, 54–5 to 54–7
 numerical methods, 54–12 to 54–14
 Perron-Schaefer condition, 26–7
 states classification, 54–7 to 54–9
 Markov decision process, deterministic, 25–3
 Markowitz cost, 40–17
 Markowitz scheme, 40–18
 Maschke's theorem
 group representations, 68–2
 matrix groups, 67–2
 matrix representations, 68–4
 Matching, bipartite graphs, 30–2
 Mathematical physics
 angular momentum, 59–9 to 59–10
 fundamentals, 59–1 to 59–2

- Green's functions, 59–10 to 59–11
- Lagrangian mechanics, 59–5 to 59–6
- oscillation modes, 59–2 to 59–5
- rotation group, representations, 59–9 to 59–10
- Schrödinger's equation, 59–6 to 59–9
- Mathematical programming, 50–1
- Mathematica software
 - appendix, 73–25 to 73–27
 - decompositions, 73–18 to 73–19
 - eigenvalues, 73–14 to 73–16
 - fundamentals, 73–1 to 73–2, 73–25 to 73–27
 - linear programming, 73–23 to 73–24
 - linear systems, 73–20 to 73–23
 - manipulation of matrices, 73–13 to 73–14
 - matrices, 73–5 to 73–9
 - matrix algebra, 73–9 to 73–12
 - singular values, 73–16 to 73–18
 - vectors, 73–3 to 73–5
- Mathias, Roy, 17–1 to 17–15
- Mathieu equation, 56–14
- Matlab software
 - bucky command, Matlab, 71–11
 - built-in functions, 71–4 to 71–5
 - built-in functions, Matlab software, 71–4 to 71–5
 - cat command, Matlab, 71–2
 - circul function, Matlab, 71–6
 - colormap command, Matlab, 71–15
 - colspace command, Matlab, 71–17
 - contour command, Matlab, 71–15
 - det command, Matlab, 71–17
 - det function, Matlab, 71–3
 - eig command, Matlab, 71–9, 71–17, 71–18
 - eigenvalues, 15–5
 - eigenvalues and eigenvectors, 71–9
 - eigs function, 76–9 to 76–10
 - eigshow command, Matlab, 71–9
 - eigtool, Matlab, 16–12, 71–20
 - ELMAT directory, Matlab, 71–5
 - exp command, Matlab, 71–19
 - expm function, Matlab, 11–11
 - ezcontour command, Matlab, 71–15
 - ezplot command, Matlab, 71–14, 71–14 to 71–15, 71–17
 - ezsurf command, Matlab, 71–17
 - fname command, Matlab, 71–10, 71–11
 - for loops, Matlab, 71–11
 - format short command, Matlab, 71–8
 - functions, Matlab, 71–11
 - fundamentals, 71–1
 - gallery command, Matlab, 71–5
 - givens function, Matlab, 42–9
 - graphical user interfaces, 71–19 to 71–22
 - graphics, 71–14 to 71–17
 - graphics, Matlab software, 71–14 to 71–17
 - hilb command, Matlab, 71–18
 - if statements, Matlab, 71–11
 - implicitly restarted Arnoldi method, 44–1
 - imread command, Matlab, 71–15
 - inline command, Matlab, 71–11 to 71–12
 - inv command, Matlab, 71–17
 - inv function, Matlab, 71–3
 - jordan command, Matlab, 71–9
 - kron function, Matlab, 71–4
 - Lanczos methods, 42–21
 - least squares solutions, 71–7 to 71–9
 - left divide operator, Matlab, 71–7
 - length command, Matlab, 71–2, 71–9
 - linear systems, 71–7 to 71–9
 - logical operators, Matlab, 71–12
 - matrices, 71–1 to 71–3
 - matrix arithmetic, 71–3 to 71–4
 - max(size) command, Matlab, 71–2
 - mesh command, Matlab, 71–15
 - meshgrid command, Matlab, 71–14
 - M-files*, Matlab, 71–11, 71–20
 - multidimensional arrays, 71–1 to 71–3
 - narin command, Matlab, 71–12
 - narout command, Matlab, 71–12
 - norm command, Matlab, 71–17
 - null command, Matlab, 71–8, 71–17
 - numnull command, Matlab, 71–13
 - orth command, Matlab, 71–17
 - peaks function, Matlab, 71–17
 - pi function, Matlab, 71–4
 - planerot function, Matlab, 42–9
 - plot command, Matlab, 71–14
 - programming, 71–11 to 71–14
 - rand command, Matlab, 71–6
 - rank command, Matlab, 71–17
 - relational operators, Matlab, 71–12
 - right divide operator, Matlab, 71–7
 - roots function, Matlab, 72–16
 - rref command, Matlab, 71–17
 - size command, Matlab, 71–2
 - spare matrices, 71–9 to 71–11
 - SPARFUN directory, Matlab, 71–10
 - special matrices, 71–5 to 71–7
 - spy command, Matlab, 71–10
 - submatrices, 71–1 to 71–3
 - subs command, Matlab, 71–17, 71–18
 - sum command, Matlab, 71–17
 - surfc command, Matlab, 71–15
 - symbolic mathematics, 71–17 to 71–19
 - sym command, Matlab, 71–17
 - syms command, Matlab, 71–17
 - toeplitz function, Matlab, 71–6
 - trace command, Matlab, 71–17
 - vpa command, Matlab, 71–17, 71–19
 - while loops, Matlab, 71–11
- Matlab software's EIGS subroutine package, 76–9 to 76–10
- Matrices
 - balanced, 32–5, 32–12
 - bipartite graphs, 30–4 to 30–7
 - classes of $(0,1)$ -matrices, 27–7 to 27–10
 - defective, 4–6
 - derogatory, 4–6
 - digraphs, 29–3 to 29–4
 - dimension theorem, 2–6 to 2–9
 - D-optimal matrices, 32–5 to 32–9, 32–12
 - elementary, 1–11 to 1–13

- empty matrix, 68–3
- equal, 1–3
- factoring, 72–9 to 72–11
- function behaviors, 16–8 to 16–11
- fundamentals, 1–3, 1–3 to 1–6
- group representations, 68–3 to 68–5
- inverses, 1–11 to 1–13
- inversion, 47–9
- linear transformations, 3–3 to 3–4
- logarithm, 11–11
- Lyapunov diagonal stability, 19–9
- Maple software, 72–4 to 72–7, 72–16 to 72–18, 72–19 to 72–20
- mappings, 18–11
- Mathematica software, 73–5 to 73–9, 73–13 to 73–14
- Matlab software, 71–1 to 71–3, 71–5 to 71–7
- max-plus algebra, 25–1
- nonregular, 32–7 to 32–9
- nonsquare case, 32–5
- norms, 37–4 to 37–6
- norms, error analysis, 37–4 to 37–6
- null space, 2–6 to 2–9
- rank, 2–6 to 2–9
- representations, 28–7 to 28–9
- singular values and singular value inequalities, 17–3 to 17–5, 17–13 to 17–14
- spectral theory, 7–5 to 7–9
- systems of linear equations, 1–9
- (± 1) -matrices, 31–8
- Matrices, combinatorial theory
 - classes of $(0,1)$ -matrices, 27–7 to 27–10
 - convex polytopes, 27–10 to 27–12
 - doubly stochastic matrices, 27–10 to 27–12
 - monotone class, 27–7 to 27–8
 - square matrices, 27–3 to 27–6
 - strong combinatorial invariants, 27–3 to 27–5
 - structure and invariants, 27–1 to 27–3
 - tournament matrices, 27–8 to 27–10
 - weak combinatorial invariants, 27–5 to 27–6
- Matrices, completion problems
 - completely positive matrices, 35–10 to 35–11
 - copositive matrices, 35–11 to 35–12
 - doubly nonnegative matrices, 35–10 to 35–11
 - entry sign symmetric P -, $P_{0,1}$ - and P_0 -matrices, 35–19 to 35–20
 - entry weakly sign symmetric P -, $P_{0,1}$ - and P_0 -matrices, 35–19 to 35–20
 - Euclidean distance matrices, 35–9 to 35–10
 - fundamentals, 35–1 to 35–8
 - inverse M -matrices, 35–14 to 35–15
 - M - and M_0 -matrices, 35–12 to 35–13
 - nonnegative P -, $P_{0,1}$ - and P_0 -matrices, 35–17 to 35–18
 - P -, $P_{0,1}$ - and P_0 -matrices, 35–15 to 35–17
 - positive definite matrices, 35–8 to 35–9
 - positive P -matrices, 35–17 to 35–18
 - positive semidefinite matrices, 35–8 to 35–9
 - strictly copositive matrices, 35–11 to 35–12
- Matrices, equalities and inequalities
 - determinantal relations, 14–10 to 14–12
 - eigenvalues, 14–1 to 14–5, 14–8 to 14–10
 - inversions, 14–15 to 14–17
 - nullity, 14–12 to 14–15
 - rank, 14–12 to 14–15
 - singular values, 14–8 to 14–10
 - spectrum localization, 14–5 to 14–8
- Matrices, fast multiplication
 - advanced techniques, 47–7 to 47–9
 - algorithms, 47–2 to 47–5
 - applications, 47–9 to 47–10
 - approximation algorithms, 47–6 to 47–7
 - fundamentals, 47–1 to 47–2
- Matrices, functions
 - computational methods, 11–9 to 11–12
 - cosine, 11–7 to 11–8
 - exponential, 11–5 to 11–6
 - fundamentals, 11–1
 - logarithm, 11–6 to 11–7
 - sign function, 11–8
 - sine, 11–7 to 11–8
 - square root, 11–4 to 11–5
 - theory, 11–1 to 11–4
- Matrices, perturbation theory
 - eigenvalue problems, 15–1 to 15–6, 15–9 to 15–11, 15–13 to 15–15
 - polar decomposition, 15–7 to 15–9
 - relative distances, 15–13 to 15–16
 - singular value problems, 15–6 to 15–7, 15–12 to 15–13, 15–15 to 15–16
- Matrices, sign-pattern
 - allowing properties, 33–9 to 33–11
 - complex sign patterns, 33–14 to 33–15
 - eigenvalue characterizations, 33–9 to 33–11
 - fundamentals, 33–1 to 33–3
 - inertia, minimum rank, 33–11 to 33–12
 - inverses, 33–12 to 33–14
 - L -matrices, 33–5 to 33–7
 - orthogonality, 33–16 to 33–17
 - powers, 33–15 to 33–16
 - ray patterns, 33–14 to 33–16
 - sign-central patterns, 33–17
 - sign nonsingularity, 33–3 to 33–5
 - sign solvability, 33–5 to 33–7
 - S -matrices, 33–5 to 33–7
 - stability, 33–7 to 33–9
- Matrices, sparse
 - analyzing fill, 40–10 to 40–13
 - effect of reorderings, 40–14 to 40–18
 - factorizations, 40–4 to 40–10
 - fundamentals, 40–1 to 40–2
 - modeling, 40–10 to 40–13
 - reordering effect, 40–14 to 40–18
 - sparse matrices, 40–2 to 40–4
- Matrices, special properties
 - canonical forms, 6–1 to 6–14
 - Hermitian, 8–1 to 8–5
 - nonnegative matrices, 9–1 to 9–15, 9–20 to 9–23
 - partitioned matrices, 10–1 to 10–9
 - positive definite matrices, 8–6 to 8–12
 - spectral theory, 7–5 to 7–9
 - stochastic matrices, 9–15 to 9–17
 - unitary similarity, 7–1 to 7–5

- Matrices, stability and inertia
 - additive D -stability, 19–7 to 19–8
 - fundamentals, 19–1 to 19–2
 - inertia, 19–2 to 19–3
 - Lyapunov diagonal stability, 19–9 to 19–10
 - multiplicative D -stability, 19–5 to 19–7
 - stability, 19–3 to 19–5
- Matrices computations, large-scale
 - Arnoldi process, 49–10 to 49–11
 - dimension reduction, 49–14 to 49–15
 - eigenvalue computations, 49–12
 - fundamentals, 49–1 to 49–2
 - Krylov subspaces, 49–5 to 49–6
 - linear dynamical systems, 49–14 to 49–15
 - linear systems, equations, 49–12 to 49–14
 - nonsymmetric Lanczos process, 49–8 to 49–10
 - sparse matrix factorizations, 49–2 to 49–5
 - symmetric Lanczos process, 49–6 to 49–7
- Matrices generation, Maple software, 72–4 to 72–5
- Matrices over $\mathbb{R}$, 31–8 to 31–9
- Matrices sign function, 11–12
- Matrix, Maple software, 72–1, 72–4 to 72–5, 72–13 to 72–14, 72–15
- matrix, Maple software, 72–1
- Matrix algebra, 73–9 to 73–12
- Matrix approximation, 17–12 to 17–13
- Matrix arithmetic, 71–3 to 71–4
- Matrix condition number, 38–2
- MatrixConditionNumber, Mathematica software
 - matrix algebra, 73–10
 - singular values, 73–18
- Matrix cosine and sine, 11–11
- Matrix direct sum, 2–7
- Matrix equations, control theory, 57–10 to 57–11
- Matrix equivalence, 23–4 to 23–8
- MatrixExp, Mathematica software, 73–9, 73–11
- Matrix exponential, 11–10
- MatrixExponential, Maple software, 72–19
- Matrix factoring, 72–9 to 72–11
- MatrixForm, Mathematica software
 - decomposition, 73–19
 - matrices, 73–6, 73–7, 73–8
 - matrix algebra, 73–10, 73–11
 - vectors, 73–2, 73–4, 73–5
- MatrixFunction, Maple software, 72–19 to 72–20
- Matrix function behaviors, 16–8 to 16–11
- Matrix games, 50–18 to 50–20
- Matrix groups
 - BN structure, 67–4 to 67–5
 - classical groups, 67–5 to 67–7
 - fundamentals, 67–1 to 67–3
 - linear groups, 67–3 to 67–5
- Matrix in real-Jordan canonical form, 6–7
- MatrixInverse, Maple software, 72–5
- Matrix inversion, 47–9
- Matrix logarithm, 11–11
- Matrix mappings, 18–11
- Matrix multiplication complexity, 47–8
- MatrixNorm, Mathematica software, 73–27
- Matrix pencils
 - eigenvalue problems, 15–9
 - extensions, 16–12
 - generalized eigenvalue problem, 43–2
 - linear differential-algebraic equations, 55–7
- Matrix perturbation theory
 - eigenvalue problems, 15–1 to 15–6, 15–9 to 15–11, 15–13 to 15–15
 - polar decomposition, 15–7 to 15–9
 - relative distances, 15–13 to 15–16
 - singular value problems, 15–6 to 15–7, 15–12 to 15–13, 15–15 to 15–16
- Matrix polynomial extension, 16–13
- Matrix polynomial of degree, 9–8
- MatrixPower, Mathematica software
 - matrices, 73–6, 73–8
 - matrix algebra, 73–9, 73–10, 73–11, 73–12
- Matrix power asymptotics, 25–8 to 25–9
- Matrix products
 - fundamentals, 1–4
 - Maple software, 72–6
 - numerical range, 18–8 to 18–9
- MatrixQ, Mathematica software, 73–6
- Matrix quadratic forms, 53–8 to 53–11
- MatrixRank, Mathematica software
 - fundamentals, 73–27
 - matrix algebra, 73–10, 73–12
- Matrix representations, 28–7 to 28–9
- Matrix sign function, 11–12
- Matrix square root, 11–4 to 11–5, 11–11
- Matrix stability and inertia
 - additive D -stability, 19–7 to 19–8
 - fundamentals, 19–1 to 19–2
 - inertia, 19–2 to 19–3
 - Lyapunov diagonal stability, 19–9 to 19–10
 - multiplicative D -stability, 19–5 to 19–7
 - stability, 19–3 to 19–5
- Matrix-tree theorem, 28–8
- Matrix-vector multiplication, 37–8
- Matrix-vector product, 1–4
- Matrix-Vector products, Maple software, 72–6 to 72–7
- Max algebra, 9–23
- Max-cut problem, 51–1, 51–2, 51–10
- Maximal cycle mean, 25–4 to 25–6
- Maximal ideal, 23–2
- Maximal rank, 33–11
- Maximal sign nonsingularity, 33–3
- Maximize, Mathematica software, 73–23, 73–27
- Maximum absolute column sum norm, 37–4
- Maximum absolute row sum norm, 37–4
- Maximum distance separable (MDS), 61–5
- Maximum distance separable (MDS) convolutional codes, 61–12
- Maximum entropy method, 64–14, 64–15
- Maximum likelihood estimates, 53–4
- Maximum multiplicity, 34–4 to 34–6
- Maximum rank deficiency, 34–4
- Max-plus algebra
 - asymptotics, matrix powers, 25–8 to 25–9
 - eigenproblem, 25–6 to 25–8

- fundamentals, 25–1 to 25–4
 - linear independence and rank, 25–12 to 25–14
 - linear inequalities and projections, 25–10 to 25–12
 - maximal cycle mean, 25–4 to 25–6
 - permanent, 25–9 to 25–10
 - permanents, 25–9 to 25–10
- Max-plus Collatz-Wielandt formulas, 25–4 to 25–5
- Max-plus Cramer's formula, 25–13
- Max-plus diagonal scaling, 25–6
- Max-plus eigenproblem, 25–6 to 25–8
- Max-plus permanent, 25–9 to 25–10
- Max-plus polynomial function, 25–9
- Max-plus semiring, 25–1
- Maxplus toolbox, 25–6
- max(size) command, Matlab software, 71–2
- Maxwell's equations, 59–2
- McDonald studies, 26–7
- McLaurin expansion, 23–9
- McMillan degree, 57–6
- MDS (maximum distance separable), 61–5
- MDS (maximum distance separable) convolutional codes, 61–12
- Mean
 - multivariate normal inference, 53–4
 - statistics and random variables, 52–2
- Mean square prediction error, 64–7
- Mean value, 52–2
- Mean vectors, 52–3
- Measured outputs, 57–14
- Measure of relative separation, 17–7
- Mehrmann, Volker, 55–1 to 55–16
- Meini, Beatrice, 54–1 to 54–14
- Meini iteration, 11–11
- Menagé number, 31–6
- Menger matrix
 - Euclidean simplexes, 66–8, 66–9
 - fundamentals, 66–12
 - resistive electrical networks, 66–15
- Merging
 - matrix power asymptotics, 25–8
 - nonnegative IEPs, 20–8
- Mesh, Mathematica software, 73–27
- mesh command, Matlab software, 71–15
- meshgrid command, Matlab software, 71–14
- Message blocks of length, 61–1
- Metabolites, 60–10
- Method of Condensation, 4–4
- Metric dynamical systems, 56–12
- Metric multidimensional scaling, 53–13 to 53–14
- Metric net, simplexes, 66–10
- Metrics, P–4
- Meyer, Carl D., 63–1 to 63–14
- M-files*, Matlab software, 71–11, 71–20
- MGS (modified Gram-Schmidt) process, 44–4
- Midpoint
 - affine spaces, 65–2
 - Euclidean point space, 66–2
- MIEPs (multiplicative IEPs), 20–10
- Mills, Mark, 2–1 to 2–12
- Minimal connection, 29–12 to 29–13
- Minimally chordal symmetric Hamiltonian, 35–15
- Minimally potentially stable, 33–7
- Minimal matrix norms, 37–4
- Minimal polynomials
 - convergence in gap, 44–9
 - eigenvalues and eigenvectors, 4–6
 - Krylov subspaces, 49–6
- Minimal rank, 33–11
- Minimal realization, 57–6
- Minimal Residual (MINRES) algorithm
 - convergence rates, 41–14 to 41–15
 - Krylov space methods, 41–4, 41–6, 41–10
- Minimal sign-central matrices, 33–17
- Minimax theorem, 50–18
- Minimize, Mathematica software
 - fundamentals, 73–27
 - linear programming, 73–23, 73–24
- Minimum co-cover, 27–2
- Minimum cover, 27–2
- Minimum deficiency algorithm, 40–17
- Minimum entropy controller, 57–15
- Minimum-norm least squares solution, 39–1
- Minimum phase, 64–2
- Minimum rank, 34–4 to 34–6
- Minimum rank inertia, 33–11 to 33–12
- Minors, *see also* Principal minors
 - determinants, 4–1
 - graphs, 28–4
- Minors, Mathematica software, 73–10, 73–11
- Min-plus semiring, 25–1
- MINRES (Minimal Residual) algorithm
 - convergence rates, 41–14 to 41–15
 - Krylov space methods, 41–4, 41–6, 41–10
- Minus algebra, 69–2
- Mirsky theorem, 15–6
- Mixed strategies, matrix games, 50–18
- M-matrices*
 - fundamentals, 9–17 to 9–20
 - matrix completion problems, 35–12 to 35–13
 - stability, 19–4
- M_0 -matrices, 35–12 to 35–13
- M-norm*, 37–2
- Möbius function, 20–7
- Model matrix, 52–8
- Models, *see also specific model*
 - fill, sparse matrix methods, 40–10 to 40–13
 - full-rank, 52–8
 - Gauss-Markov model, 52–8, 53–11
 - LAPACK subroutine package, 75–8 to 75–9
 - linear statistical, 52–1 to 52–15
 - multivariate linear model, 53–11
 - multivariate statistical analysis, 53–11 to 53–13
 - Padé, 49–14, 49–15
 - Padé model, 49–14
 - reduced-order model, 49–14
 - signal model, 64–16
 - univariate linear model, 53–11
- Modified Gram-Schmidt (MGS) process, 44–4
- Modified incomplete Cholesky decomposition, 41–11
- Modular arithmetic, 72–12 to 72–13
- Module, Mathematica software, 73–26

- Modules
 - group representations, 68–2
 - Lie algebras, 70–7 to 70–10
 - matrix representations, 68–4
- Molecular distance geometry problem, 60–2
- Moment generating function, 53–3
- Monic polynomials, 23–2
- Monotone class, 27–7 to 27–8
- Monotone vector norm, 37–2
- Mood studies, 32–1
- Moore-Penrose inverse
 - inverse patterns, 33–12, 33–13
 - least squares solutions, 39–2
 - linear statistical models, 52–12
 - Maple software, 72–7
- Moore-Penrose pseudo-inverse
 - extensions, 16–13
 - pseudo-inverse, 5–12
- Morgan studies, 44–6
- Morse decompositions
 - dynamical systems, 56–7 to 56–9
 - Grassmannian and flag manifolds, 56–9 to 56–10
 - robust linear systems, 56–17
- Morse sets, 56–7
- Morse spectrum, 56–16
- Most, Mathematica software
 - fundamentals, 73–27
 - matrices manipulation, 73–13
 - vectors, 73–3
- Motzkin and Taussky studies, 7–8
- Moufang identities, 69–10
- Moulton Plane, 65–9
- (MRRR) multiple relatively robust representations, 42–15 to 42–17
- Mukhopadhyay, Kriti, 60–13
- Multi-bigraph, 30–4
- Multidimensional arrays, 71–1 to 71–3
- Multigrid method, 41–3, 41–11
- Multilinear algebra
 - alt multiplication, 13–17 to 13–19
 - antisymmetric maps, 13–10 to 13–12
 - associated maps, 13–19 to 13–20
 - decomposable tensors, 13–7
 - Grassmann tensors, 13–12 to 13–17
 - Hodge star operator, 13–24 to 13–26
 - inner product spaces, 13–22 to 13–24
 - linear maps, 13–8 to 13–10
 - multilinear maps, 13–1 to 13–3
 - orientation, 13–24 to 13–26
 - symmetric maps, 13–10 to 13–12
 - symmetric tensors, 13–12 to 13–17
 - sym multiplication, 13–17 to 13–19
 - tensor algebras, 13–20 to 13–22
 - tensor multiplication, 13–17 to 13–19
 - tensor products, 13–3 to 13–7, 13–8 to 13–10, 13–22 to 13–24
- Multilinear maps, 13–1 to 13–3
- Multinomial distribution, 52–4
- Multiple linear regression, 52–8
- Multiple relatively robust representations (MRRR), 42–15 to 42–17
- Multiplication, 69–1, *see also* Fast matrix multiplication
- Multiplication algebra, 69–5
- Multiplicative D -stability, 19–5 to 19–7
- Multiplicative Ergodic theorem, 56–14 to 56–15
- Multiplicative IEPs (MIEPs), 20–10
- Multiplicative perturbation
 - eigenvalue problems, 15–13
 - polar decomposition, 15–8
 - singular value problems, 15–15
- Multiplicative preservers, 22–7 to 22–8
- Multiplicity
 - algebraic connectivity, 36–10 to 36–11
 - characters, 68–6
 - composition, 13–13
 - max-plus permanent, 25–9
 - singular value decomposition, 45–1
- Multiplicity lists, *see also* Symmetric matrices
 - double generalized stars, 34–11 to 34–14
 - eigenvalues, 34–7 to 34–8
 - fundamentals, 34–1 to 34–2
 - generalized stars, 34–10 to 34–11
 - maximum multiplicity, 34–4 to 34–6
 - minimum rank, 34–4 to 34–6
 - parter vertices, 34–2 to 34–4
 - stars, generalized, 34–10 to 34–14
 - trees, 34–8 to 34–10
 - vines, 34–15
- Multiset, P–4 to P–5
- Multivariate Gauss-Markov theorem, 53–12
- Multivariate linear model, 53–11
- Multivariate normal distribution
 - multivariate statistical analysis, 53–3 to 53–5
 - positive definite matrices, 8–9
- Multivariate statistical analysis
 - canonical correlations and variates, 53–7
 - correlations and variates, 53–7 to 53–8
 - data matrix, 53–2 to 53–3
 - discriminant coordinates, 53–6
 - estimation, correlations and variates, 53–7 to 53–8
 - fundamentals, 53–1 to 53–2
 - inference, multivariate normal, 53–4 to 53–5
 - least squares estimation, 53–11 to 53–12
 - matrix quadratic forms, 53–8 to 53–11
 - metric multidimensional scaling, 53–13 to 53–14
 - models, 53–11 to 53–13
 - multivariate normal distribution, 53–3 to 53–5
 - principal component analysis, 53–5 to 53–6
 - statistical inference, 53–12 to 53–13
- Murakami, Lúcia I., 69–1 to 69–25
- MUSIC algorithm, 64–17

N

- Nagy, Kamm and, studies, 48–9
- Naming conventions, 76–3 to 76–4
- narin command, Matlab software, 71–12
- narout command, Matlab software, 71–12
- Narrow-band signals, 64–16
- Narrow sense Bose-Chadhuri-Hocquenghem (BCH) code, 61–8

- Natural norms, 37–4
- Natural ordering, 41–2
- N -cycle matrices, 48–2
- Near breakdown, 49–8
- Nearby floating point numbers, 37–13
- Nearest-neighbor decoding, 61–2
- Nearly decomposable, 27–3
- Nearly reducible matrices, 29–12 to 29–13
- Nearly sign-central matrices, 33–17
- Nearly sign nonsingularity, 33–3
- Negative definite properties, 12–3, 12–8
- Negative half-life, 13–24
- Negative orientation, 13–24
- Negative semidefinite properties
 - Hermitian forms, 12–8
 - symmetric bilinear forms, 12–3
- Negative semistability, 33–7
- Negative stability
 - sign pattern matrices, 33–7
 - stability, 19–3
- Negative subdivision, 20–8
- Negative vertices, 36–7
- Neighbors, graphs, 28–2
- Nested basis, 49–5
- Nested dissection ordering, 40–16 to 40–17
- Net trace, 20–7
- Neubauer, Michael G., 32–1 to 32–12
- Neumann, Michael, 5–1 to 5–16
- Neumann boundary conditions, 59–10
- Neumann series, 14–16
- Newton iteration, 11–11
- Newton-Schultz iteration, 11–12
- Newton's Law, 59–1
- Newton's method
 - interior point methods, 50–24
 - numerical methods, PIEPs, 20–11
 - primal-dual interior point algorithm, 51–8
 - total least squares problem, 48–9
- Ng, Esmond G., 40–1 to 40–18
- Ng, Michael, 48–1 to 48–9
- Nielsen, Hans Bruun, 39–1 to 39–12
- Nil algebra, 69–5
- Nil ideal properties, 69–5
- Nilpotence
 - alternative algebras, 69–10
 - general properties, 69–5
 - idempotence, 2–12
 - invariant subspaces, 3–6
 - reducible matrices, 9–11 to 9–12
- Nilpotency index, 69–5
- Nilpotent radical algebras, 69–6
- Nil radical algebras, 69–5
- Nil-semisimple algebras, 69–5
- Nodes, digraphs, 29–1
- Noise subspace, 64–16
- Noisy channel, 61–2
- Noisy transmission, 61–2
- Nonassociative algebra, 69–3, *see also* Algebra
 - applications
 - Akivis algebra, 69–16 to 69–17
 - alternative algebras, 69–10 to 69–12
 - composition algebras, 69–8 to 69–10
 - computational methods, 69–20 to 69–25
 - fundamentals, 69–1 to 69–4
 - Jordan algebras, 69–12 to 69–14
 - Malcev algebras, 69–16 to 69–17
 - noncommutative Jordan algebras, 69–14 to 69–16
 - power associative algebras, 69–14 to 69–16
 - properties, 69–4 to 69–8
 - right alternative algebras, 69–14 to 69–16
 - Sabinin algebra, 69–16 to 69–17
- Nonbasic variables, 50–10
- Noncentral distribution, 53–3
- Noncentral F-distribution, 53–3
- Noncollinear points, 65–2
- Noncommutative algorithms, 47–2
- Noncommutative Jordan algebra, 69–14
- Noncommutative Jordan algebras, 69–14 to 69–16
- Nondefective matrices, 4–6
- Nondegenerate properties
 - bilinear forms, 12–2
 - sesquilinear forms, 12–6
- Nonderogatory matrices, 4–6
- Nondifferentiation, 18–3
- Nonempty sets, row and column indices, 1–4
- Non-Hermitian case, 49–12
- Non-Hermitian Lanczos algorithm, 41–7
- Non-Hermitian problems, 41–7 to 41–11
- Nonhomogenous products, 9–22
- Nonlinear preservers, 22–7 to 22–8
- Nonnegative IEPs (NIEPs)
 - fundamentals, 20–5
 - merging results, 20–8
 - nonzero spectra, 20–7 to 20–8
 - spectra, 20–6 to 20–7
 - sufficient conditions, 20–8 to 20–10
- Nonnegatives
 - constraints, 50–3
 - factorization, 9–22
 - fundamentals, 9–1
 - integer rank, 30–8
 - matrix factorization, 63–5 to 63–8
 - sign pattern matrices, 33–12
 - stable matrices, 9–17
 - vectors, 26–2
- Nonnegatives, matrices
 - fundamentals, 9–1 to 9–2
 - inequalities, 17–11
 - inverse eigenvalue problem, 9–22
 - irreducible matrices, 9–2 to 9–7
 - max algebra, 9–23
 - nonhomogenous products, 9–22
 - nonnegative factorization, 9–22
 - P -, $P_{0,1}$ - and P_0 -matrices, completion problems, 35–17 to 35–18
 - permanents, 31–7
 - Perron-Frobenius theorem, 26–2
 - product form, 9–23
 - reducible matrices, 9–7 to 9–15
 - scaling, 9–20 to 9–23
 - sets, 9–23
- Nonnormality constant, 44–10